

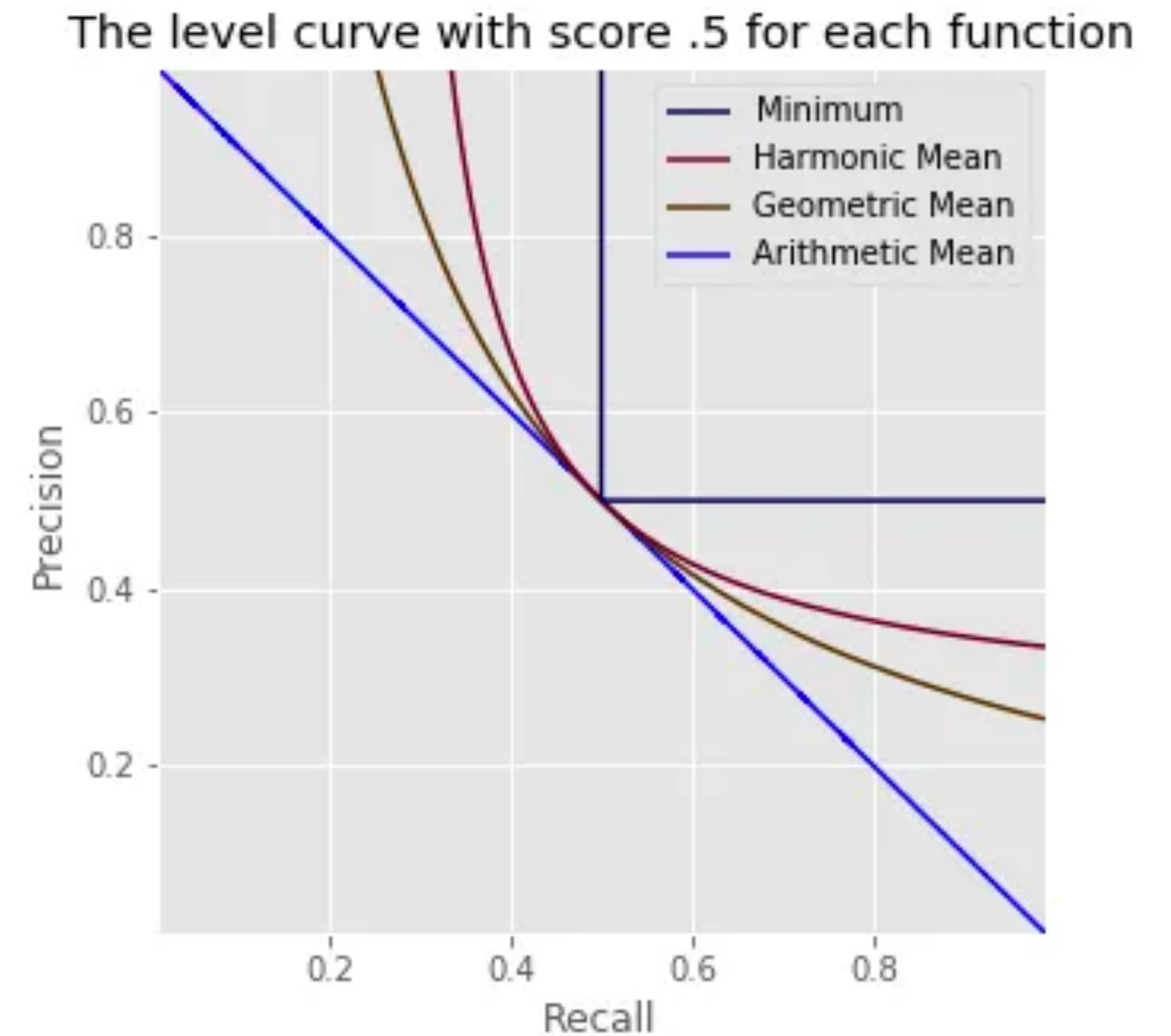
Sequence Models II

Wei Xu

(many slides from Greg Durrett)

Harmonic Mean (Math Review)

- ▶ Arithmetic Mean = $(P + R) / 2$
- ▶ Geometric Mean = $\sqrt{P \times R}$
- ▶ Harmonic Mean = $2 \times P \times R / (P + R)$



HMMs (Recap)

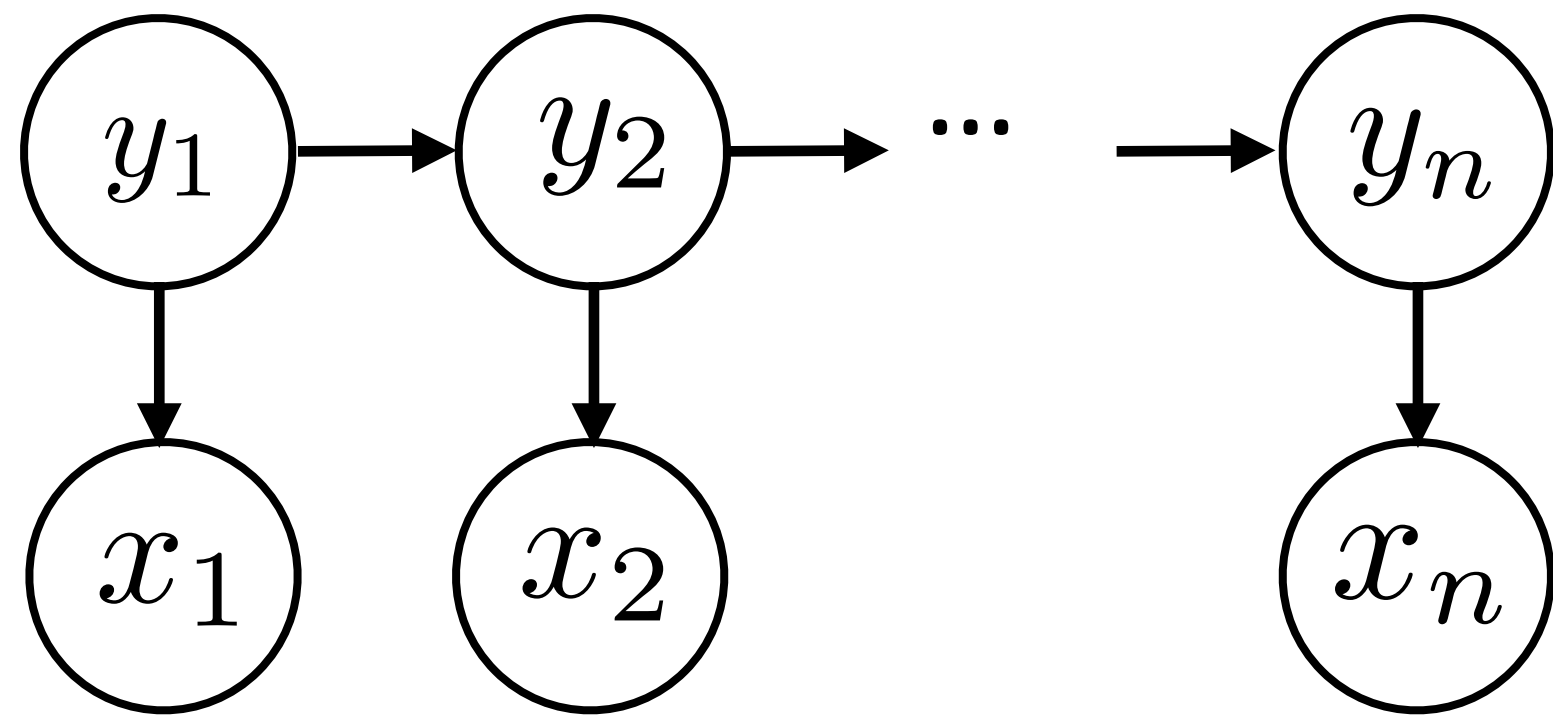
► Input $\mathbf{x} = (x_1, \dots, x_n)$ Output $\mathbf{y} = (y_1, \dots, y_n)$



Andrew Viterbi, 1967

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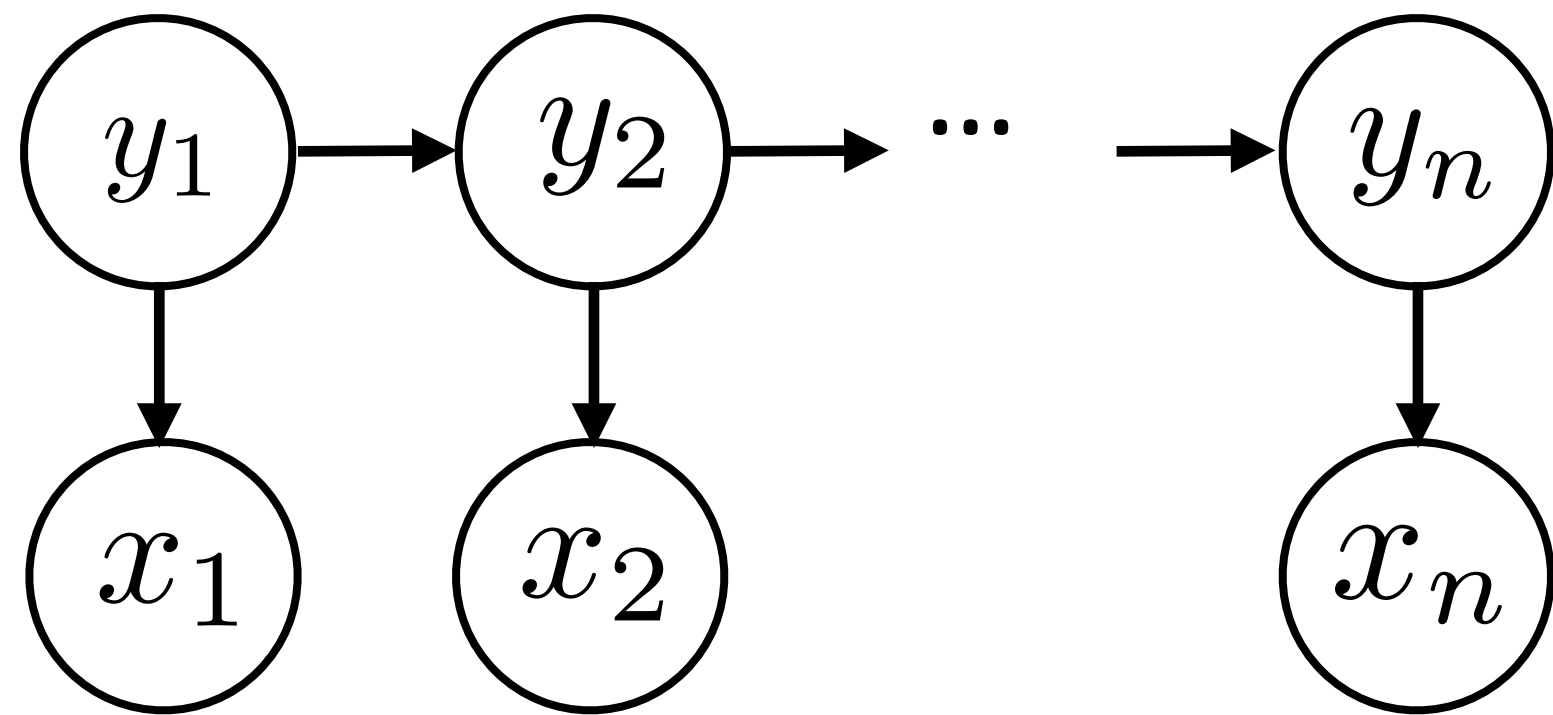
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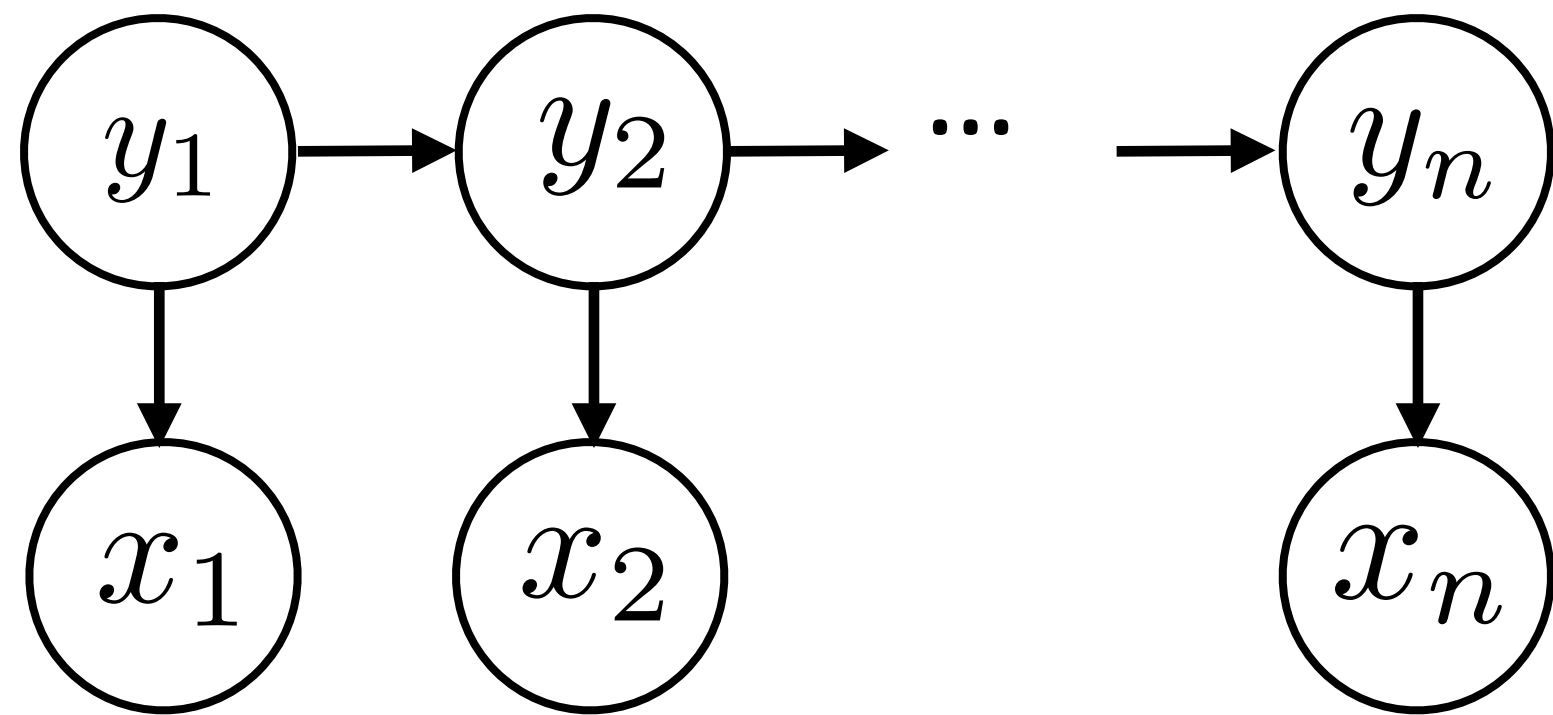
$$P(\mathbf{y}, \mathbf{x}) = P(y_1) \prod_{i=2}^n P(y_i | y_{i-1}) \prod_{i=1}^n P(x_i | y_i)$$



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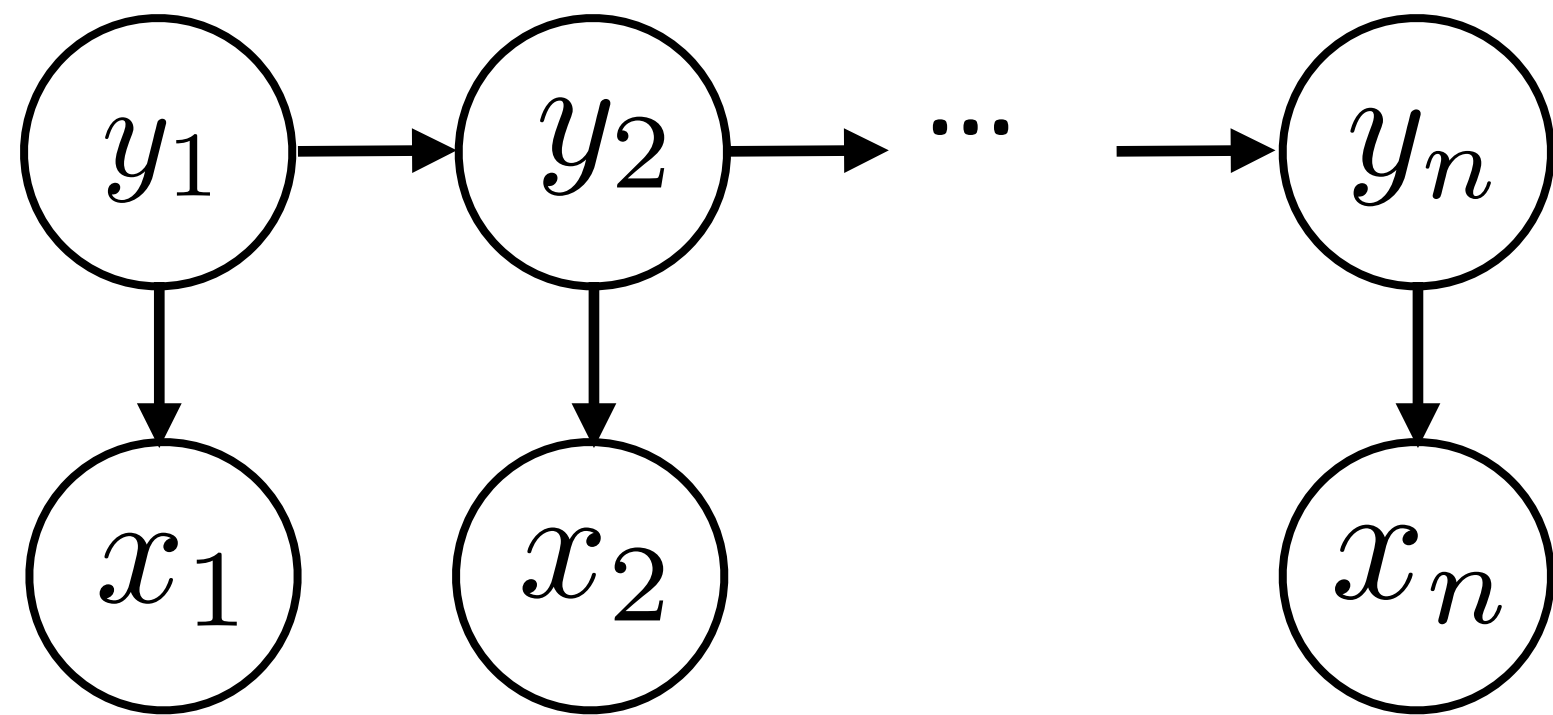
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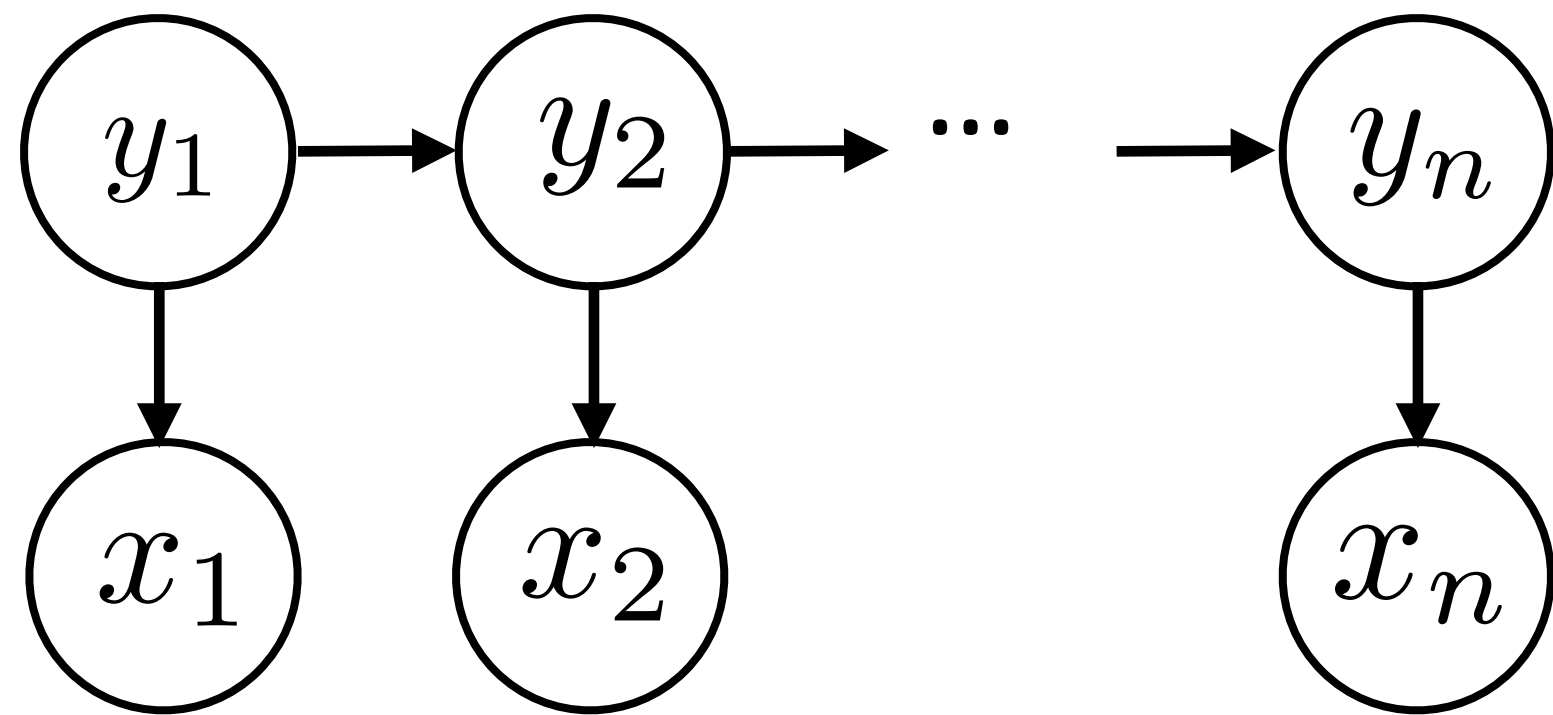
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- Viterbi: $\operatorname{score}_i(s) = \max_{y_{i-1}} P(s | y_{i-1}) P(x_i | s) \operatorname{score}_{i-1}(y_{i-1})$



Andrew Viterbi, 1967

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- ▶ What did Viterbi compute? $P(\mathbf{y}_{\max}|\mathbf{x}) = \max_{y_1, \dots, y_n} P(\mathbf{y}|\mathbf{x})$

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- ▶ In addition to finding the best path, we may want to compute marginal probabilities of paths $P(y_i = s|\mathbf{x})$

$$P(y_i = s|\mathbf{x}) = \sum_{y_1, \dots, y_{i-1}, y_{i+1}, \dots, y_n} P(\mathbf{y}|\mathbf{x})$$

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- ▶ Can compute marginals with dynamic programming as well using an algorithm called forward-backward

This Lecture

- ▶ Named entity recognition (NER)
- ▶ CRFs: model (+features for NER), inference, learning
- ▶ Reading: Eisenstein Chapter 7 & 8.3

Named Entity Recognition

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Barack Obama will travel to Hangzhou today for the G20 meeting .

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PERSON

LOC

ORG

Named Entity Recognition

B-PER I-PER O O O B-LOC O O O B-ORG O O

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- ▶ BIO tagset: begin, inside, outside

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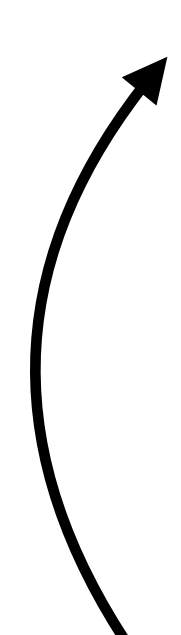
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- ▶ BIO tagset: begin, inside, outside
- ▶ Sequence of tags — should we use an HMM?
- ▶ Why might an HMM not do so well here?
 - ▶ Lots of O's, so tags aren't as informative about context
 - ▶ Insufficient features/capacity with multinomials (especially for unks)

Where we're going

- ▶ Flexible discriminative model for tagging tasks that can use arbitrary features of the input. Similar to logistic regression, but *structured*

B-PER I-PER
 *Barack Obama* will travel to *Hangzhou* today for the *G20* meeting .

Curr_word=Barack & **Label=B-PER**

Next_word=Obama & **Label=B-PER**

Curr_word_starts_with_capital=True & **Label=B-PER**

Posn_in_sentence=1st & **Label=B-PER**

Label=B-PER & Next-Label = I-PER

...

Evaluating NER

B-PER	I-PER	O	O	O	B-LOC	O	O	O	B-ORG	O	O
<i>Barack Obama will travel to Hangzhou today for the G20 meeting .</i>											
PERSON					LOC		ORG				

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- Prediction of all Os still gets 66% accuracy on this example!

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 - ▶ Recall: of the gold named entities, how many did we find?

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 - ▶ F-measure: harmonic mean of these two

Personal Disclosure Detection

Category	#Spans	Avg Len	Example
<i>Demographic Attributes</i>			
LOCATION	525	5.70±3.85	I live in the UK and a diagnosis is really expensive, even with health insurance
AGE	308	2.93±1.72	I am a 23-year-old who is currently going through the last leg of undergraduate school
RELATIONSHIP STATUS	287	6.72±5.97	My partner has not helped at all, and I'm bed ridden now
AGE/GENDER	248	1.42±0.71	For some context, I (20F), still live with my parents
PET	192	6.93±7.31	Hi, I have two musk turtles and have never had any health problems before at all
APPEARANCE	173	6.96±6.25	Same here. I am 6'2. No one can sit behind me.
HUSBAND/BF	148	6.89 ±7.24	My husband and I vote for different parties
WIFE/GF	144	5.24±4.42	My gf and I applied, we're new but fairly active!
GENDER	110	3.28±3.10	Am I insane? Eh. I'm just a girl who wants to look on the outside how I feel on the inside.
RACE/NATIONALITY	99	3.63±2.37	As Italian I hope tonight you will won the world cup
SEXUAL ORIENTATION	58	6.52±7.47	I'm a straight man but I do wanna say this
NAME	21	3.81±3.48	Hello guys, my name is xxx and I love travelling
CONTACT	14	5.69±3.56	xxx is my ig
<i>Personal Experiences</i>			
HEALTH	783	10.36±9.78	I am pretty sure I have autism, but I don't want to get an official diagnosis.
FAMILY	543	9.27±8.73	My little brother (9M) is my pride and joy
OCCUPATION	428	8.90±6.60	I'm a motorcycle tourer (by profession), but when I'm off the saddle I'm mostly bored
MENTAL HEALTH	285	16.86±16.28	I get asked this pretty regularly.. but I struggle with depression and ADHD
EDUCATION	229	9.92±7.71	Hi there, I got accepted to UCLA (IS), which I'm pumped about.
FINANCE	153	12.00±9.19	Yes. I was making \$68k a year and had around \$19k in debt

Table 1: Statistics and examples for each self-disclosure category in our dataset, sorted by decreasing frequency. Personal identifiable information are redacted as ‘xxx’ to be shown here.

Personal Disclosure Detection

Class (#Spans)	RoBERTa-large			DeBERTaV3-large			GPT-4		
	Span F ₁	Partial F ₁	Token F ₁	Span F ₁	Partial F ₁	Token F ₁	Span F ₁	Partial F ₁	Token F ₁
AGE (35)	60.87	72.46	<u>87.04</u>	64.62	70.77	84.97	<i>65.71</i>	80.0	80.54
AGE&GENDER (17)	73.68	84.21	<u>81.63</u>	54.05	70.27	75.47	69.77	74.42	70.97
RACE/NATIONALITY (8)	60.0	60.0	64.94	82.35	82.35	<u>81.25</u>	70.59	70.59	71.64
GENDER (17)	61.11	61.11	56.47	72.73	72.73	61.73	57.14	57.14	<u>62.50</u>
LOCATION (41)	52.87	71.26	76.60	57.78	73.33	<u>83.61</u>	41.30	54.35	69.63
APPEARANCE (31)	<i>44.07</i>	64.41	<u>78.14</u>	35.48	67.74	76.27	12.77	42.55	54.71
WIFE/GF (30)	53.33	66.67	74.19	65.52	75.86	<u>78.36</u>	48.39	64.52	74.32
FINANCE (33)	35.82	68.66	76.57	<i>40.0</i>	71.43	<u>77.68</u>	31.17	54.55	69.49
OCCUPATION (44)	40.0	64.44	72.39	45.65	65.22	<u>75.08</u>	39.56	52.75	68.21
FAMILY (44)	<i>50.0</i>	58.70	66.52	35.29	49.02	<u>69.23</u>	46.60	58.25	64.68
HEALTH (40)	44.21	56.84	<u>76.54</u>	<i>45.10</i>	58.82	75.94	26.45	38.02	50.17
MENTAL HEALTH (46)	39.22	64.71	<u>81.18</u>	<i>40.0</i>	63.16	75.75	36.36	52.73	66.51
HUSBAND/BF (14)	68.75	75.0	<u>71.91</u>	64.71	70.59	71.11	55.17	68.97	68.49
EDUCATION (21)	55.32	68.09	85.23	65.38	69.23	<u>86.73</u>	42.55	51.06	67.37
PET (15)	46.15	46.15	56.88	48.28	55.17	<u>61.54</u>	48.28	48.28	56.25
RELATIONSHIP STATUS (31)	38.36	41.10	58.80	36.92	43.08	<u>60.11</u>	34.29	42.86	57.58
SEXUAL ORIENTATION (12)	28.57	76.19	68.0	25.0	58.33	66.0	<i>52.17</i>	69.57	<u>71.29</u>
AVERAGE	48.32	64.71	72.53	48.88	65.71	<u>74.17</u>	41.93	57.68	66.14

Table 8: Test performance per class for fine-tuned models and prompted GPT-4-0125-preview. *Italic* highlights the best model for each class on span F₁. **Bold** on partial span F₁. Underline on token F₁.

CRFs

Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data

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Abstract

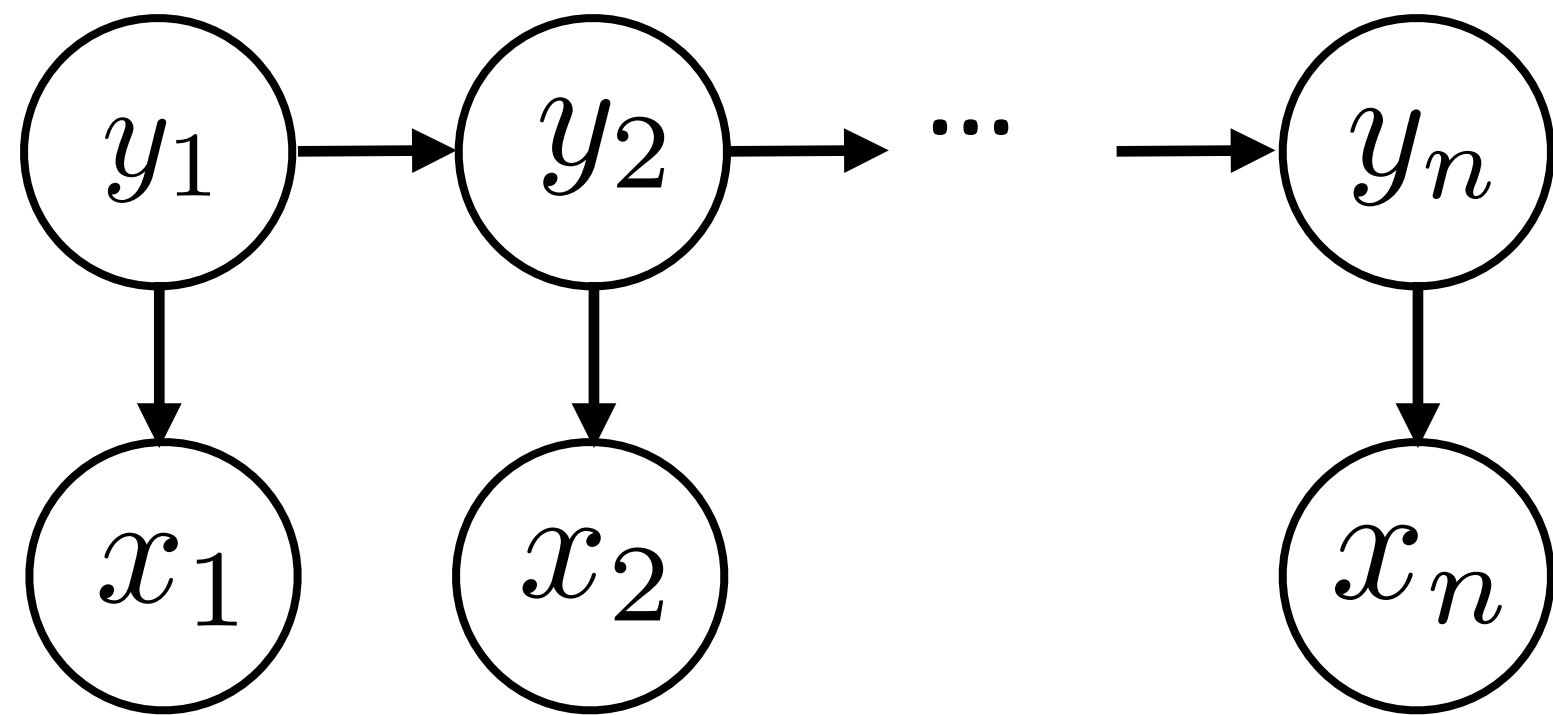
We present *conditional random fields*, a framework for building probabilistic models to segment and label sequence data. Conditional random fields offer several advantages over hidden Markov models and stochastic grammars for such tasks, including the ability to relax strong independence assumptions made in those models. Conditional random fields also avoid a fundamental limitation of maximum entropy Markov models (MEMMs) and other discriminative Markov models based on directed graphical models which can be biased towards states

minimize the joint likelihood of training examples. To define a joint probability over observation and label sequences, a generative model needs to enumerate all possible observation sequences, typically requiring a representation in which observations are task-appropriate atomic entities, such as words or nucleotides. In particular, it is not practical to represent multiple interacting features or long-range dependencies of the observations, since the inference problem for such models is intractable.

This difficulty is one of the main motivations for looking at conditional models as an alternative. A conditional model specifies the probabilities of possible label sequences given an observation sequence. Therefore, it does not expend

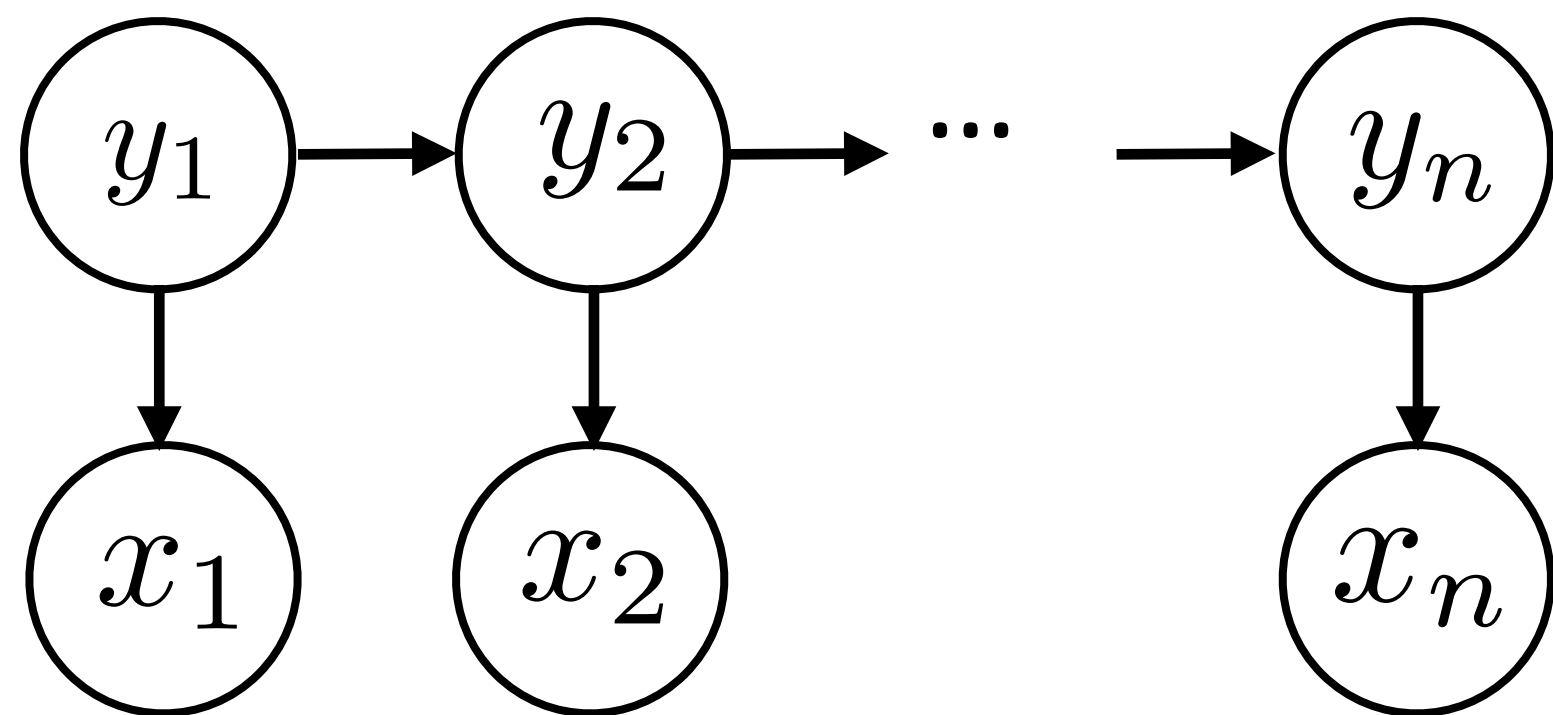
HMMs, Formally

- ▶ HMMs are expressible as Bayes nets (factor graphs)



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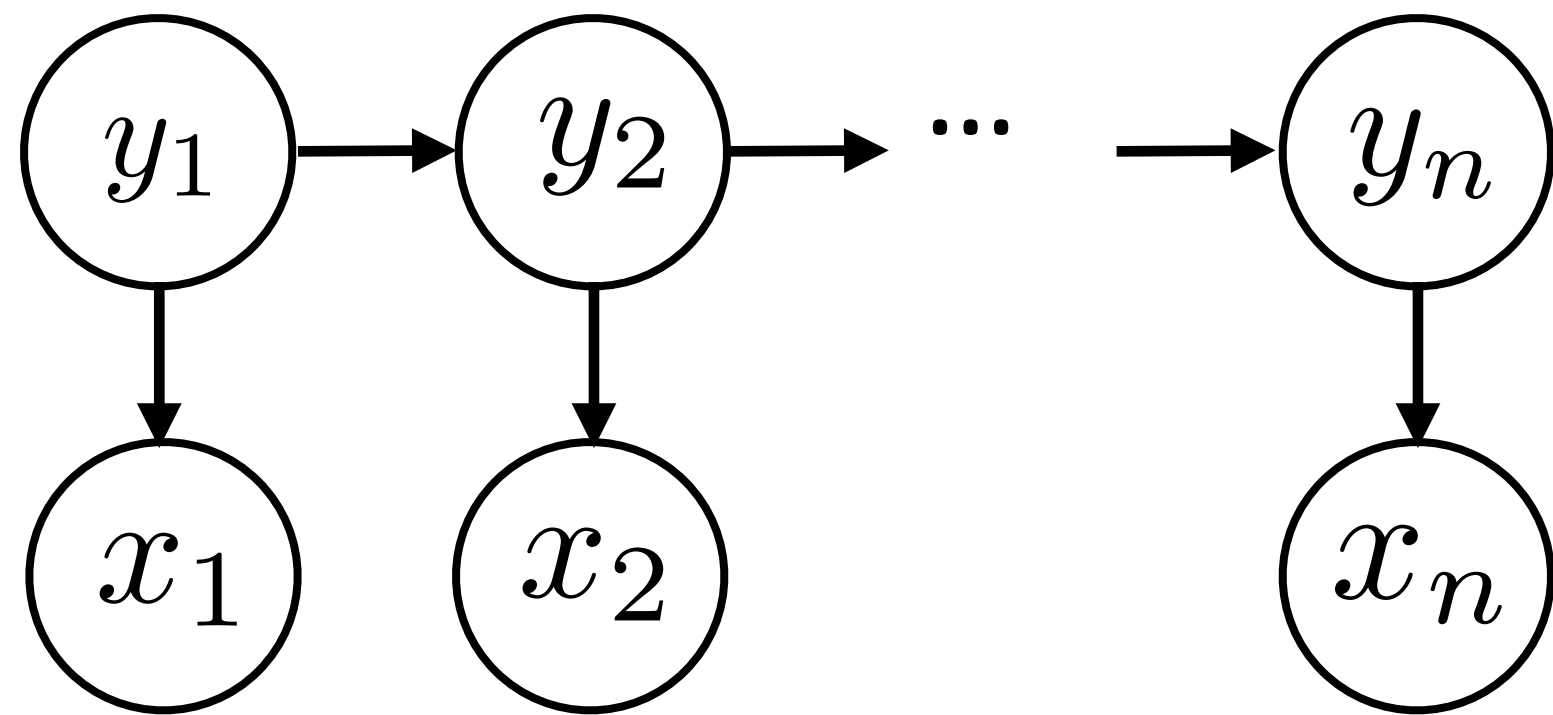


- ▶ This reflects the following decomposition:

$$P(\mathbf{y}, \mathbf{x}) = P(y_1)P(x_1|y_1)P(y_2|y_1)P(x_2|y_2) \dots$$

HMMs, Formally

- ▶ HMMs are expressible as Bayes nets (factor graphs)



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$$P(\mathbf{y}, \mathbf{x}) = P(y_1)P(x_1|y_1)P(y_2|y_1)P(x_2|y_2) \dots$$

- ▶ Locally normalized model: each factor is a probability distribution that normalizes

Conditional Random Fields

- ▶ HMMs: $P(\mathbf{y}, \mathbf{x}) = P(y_1)P(x_1|y_1)P(y_2|y_1)P(x_2|y_2) \dots$
- ▶ CRFs: discriminative models with the following globally-normalized form:

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_k \exp(\phi_k(\mathbf{x}, \mathbf{y}))$$

normalizer $\uparrow$ any real-valued scoring function of its arguments

Conditional Random Fields

▶ HMMs: $P(\mathbf{y}, \mathbf{x}) = P(y_1)P(x_1|y_1)P(y_2|y_1)P(x_2|y_2) \dots$

▶ CRFs: discriminative models with the following globally-normalized form:

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▶ Special case: linear feature-based potentials $\phi_k(\mathbf{x}, \mathbf{y}) = w^\top f_k(\mathbf{x}, \mathbf{y})$

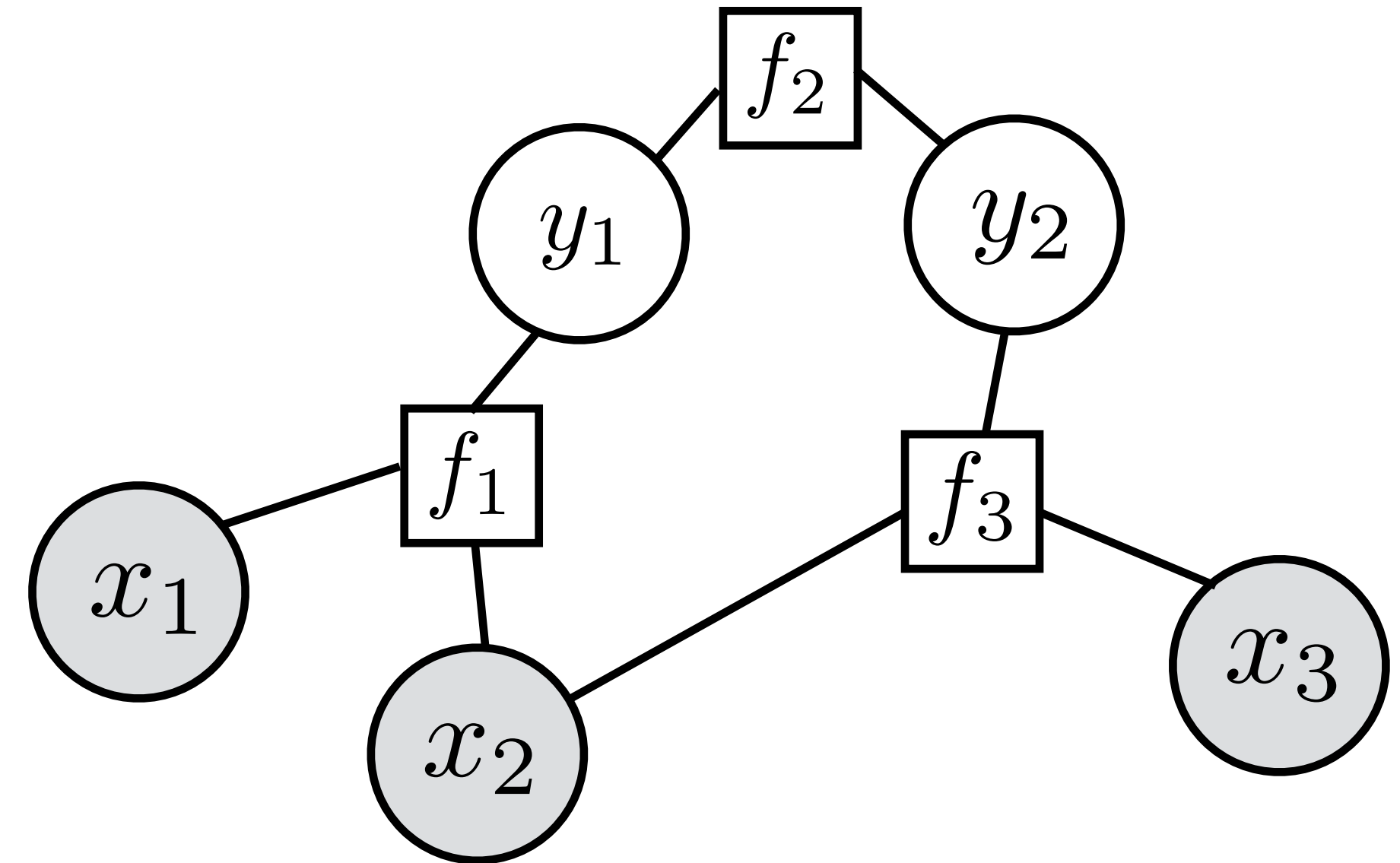
$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \exp \left(\sum_{k=1}^n w^\top f_k(\mathbf{x}, \mathbf{y}) \right)$$

▶ Looks like our single weight vector multiclass logistic regression model

HMMs vs. CRFs

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \exp \left(\sum_{k=1}^n w^\top f_k(\mathbf{x}, \mathbf{y}) \right)$$

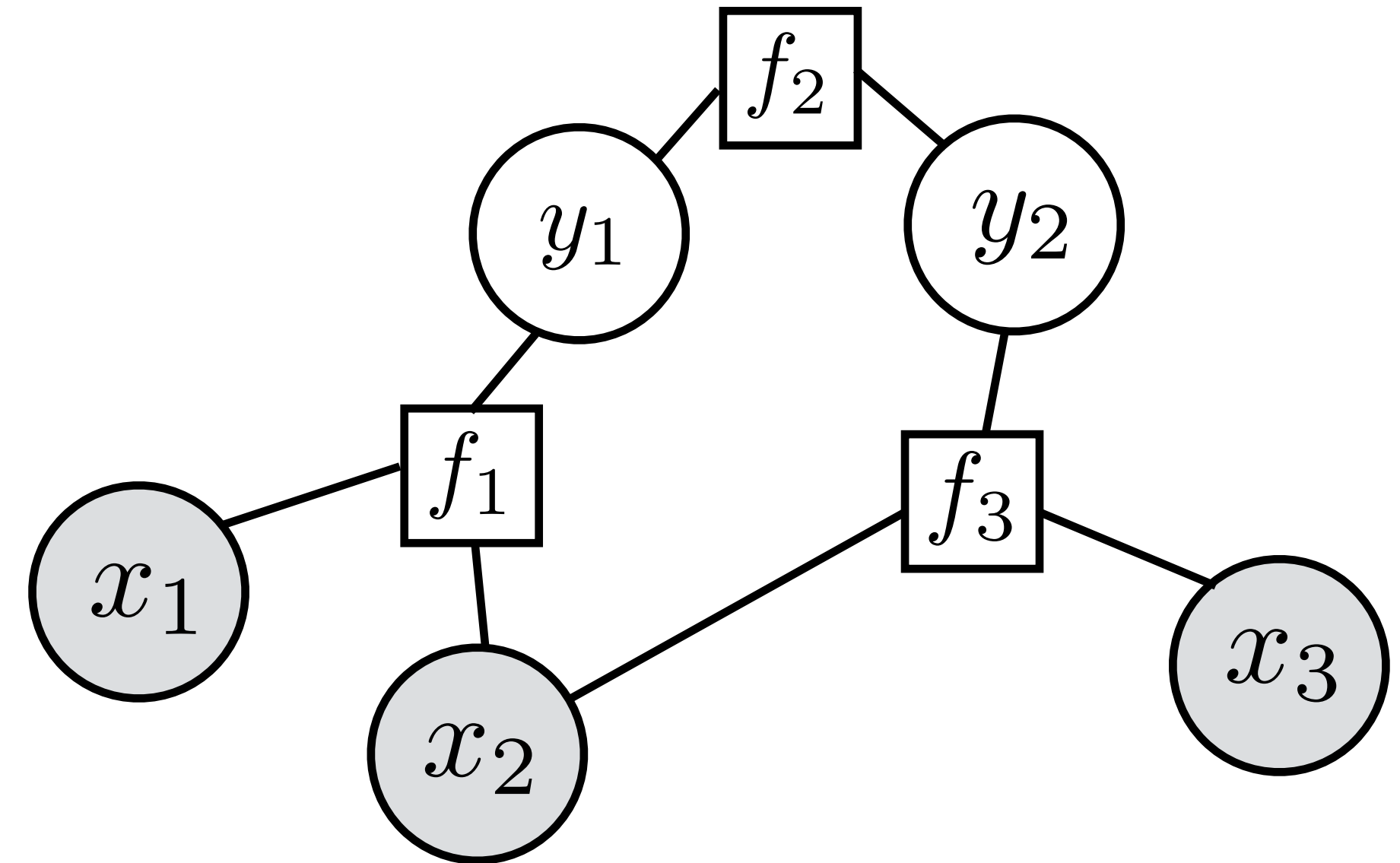
- Conditional model: x's are observed



- Each square represent a potential (feature function) and touches the y's and x's it depends on

HMMs vs. CRFs

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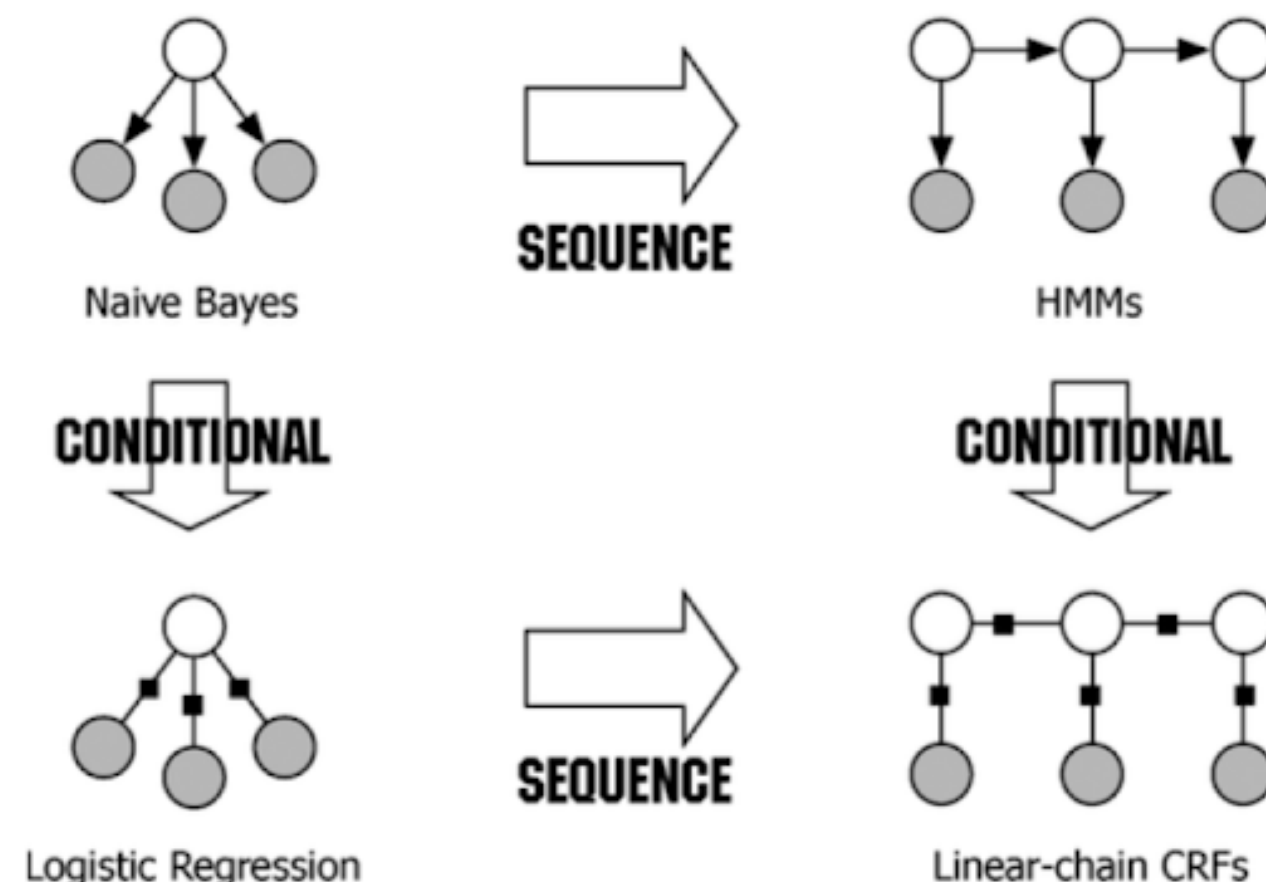


- Conditional model: x's are observed

- Naive Bayes : logistic regression :: HMMs : CRFs

local vs. global normalization <-> generative vs. discriminative

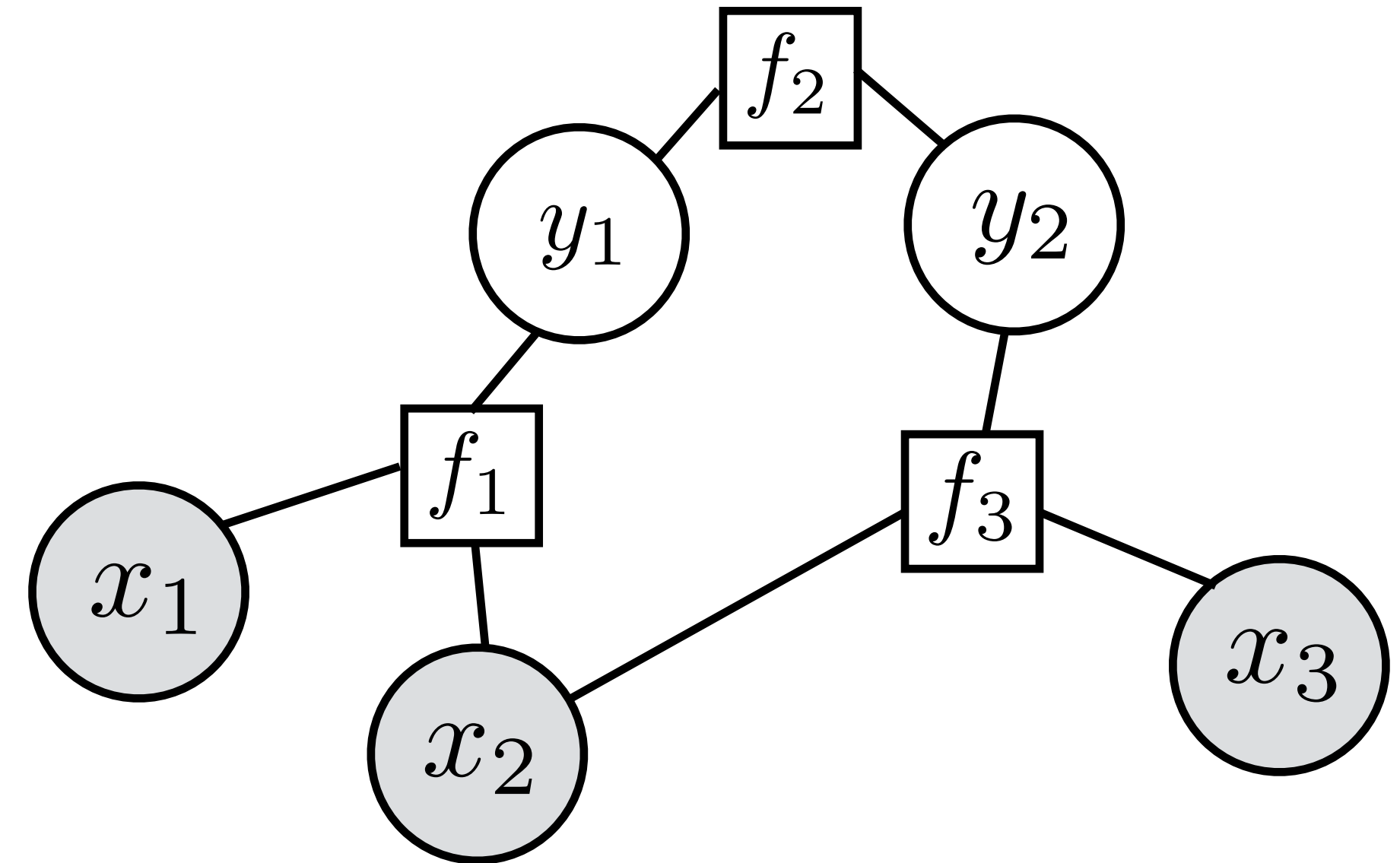
(locally normalized discriminative models do exist (MEMMs))



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HMMs vs. CRFs

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \exp \left(\sum_{k=1}^n w^\top f_k(\mathbf{x}, \mathbf{y}) \right)$$



- ▶ Conditional model: $\mathbf{x}$'s are observed
- ▶ Naive Bayes : logistic regression :: HMMs : CRFs
local vs. global normalization $\leftrightarrow$ generative vs. discriminative
(locally normalized discriminative models do exist (MEMMs))
- ▶ HMMs: in the standard setup, emissions consider one word at a time
- ▶ CRFs: features over many words simultaneously, non-independent features (e.g., suffixes and prefixes), doesn't have to be a generative model

Problem with CRFs

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \exp \left(\sum_{k=1}^n w^\top f_k(\mathbf{x}, \mathbf{y}) \right)$$

- ▶ Normalizing constant

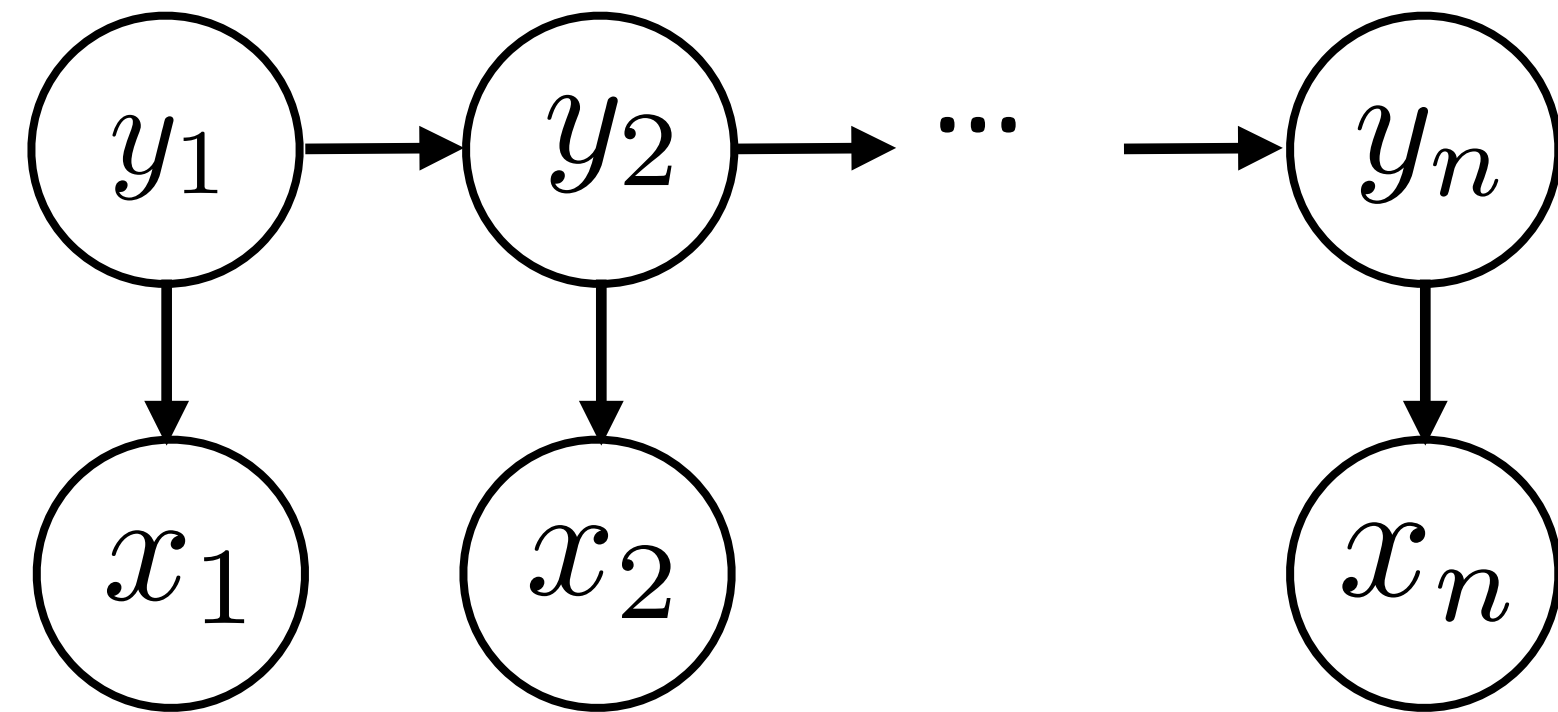
$$Z = \sum_{\mathbf{y}'} \exp \left(\sum_{k=1}^n w^\top f_k(\mathbf{x}, \mathbf{y}') \right)$$

- ▶ Inference: $\mathbf{y}_{\text{best}} = \operatorname{argmax}_{\mathbf{y}'} \exp \left(\sum_{k=1}^n w^\top f_k(\mathbf{x}, \mathbf{y}') \right)$

- ▶ If $\mathbf{y}$ consists of 5 variables with 30 values each, how expensive are these?
- ▶ Need to constrain the form of our CRFs to make it tractable

Sequential CRFs

- ▶ HMMs: $P(\mathbf{y}, \mathbf{x}) = P(y_1)P(x_1|y_1)P(y_2|y_1)P(x_2|y_2) \dots$

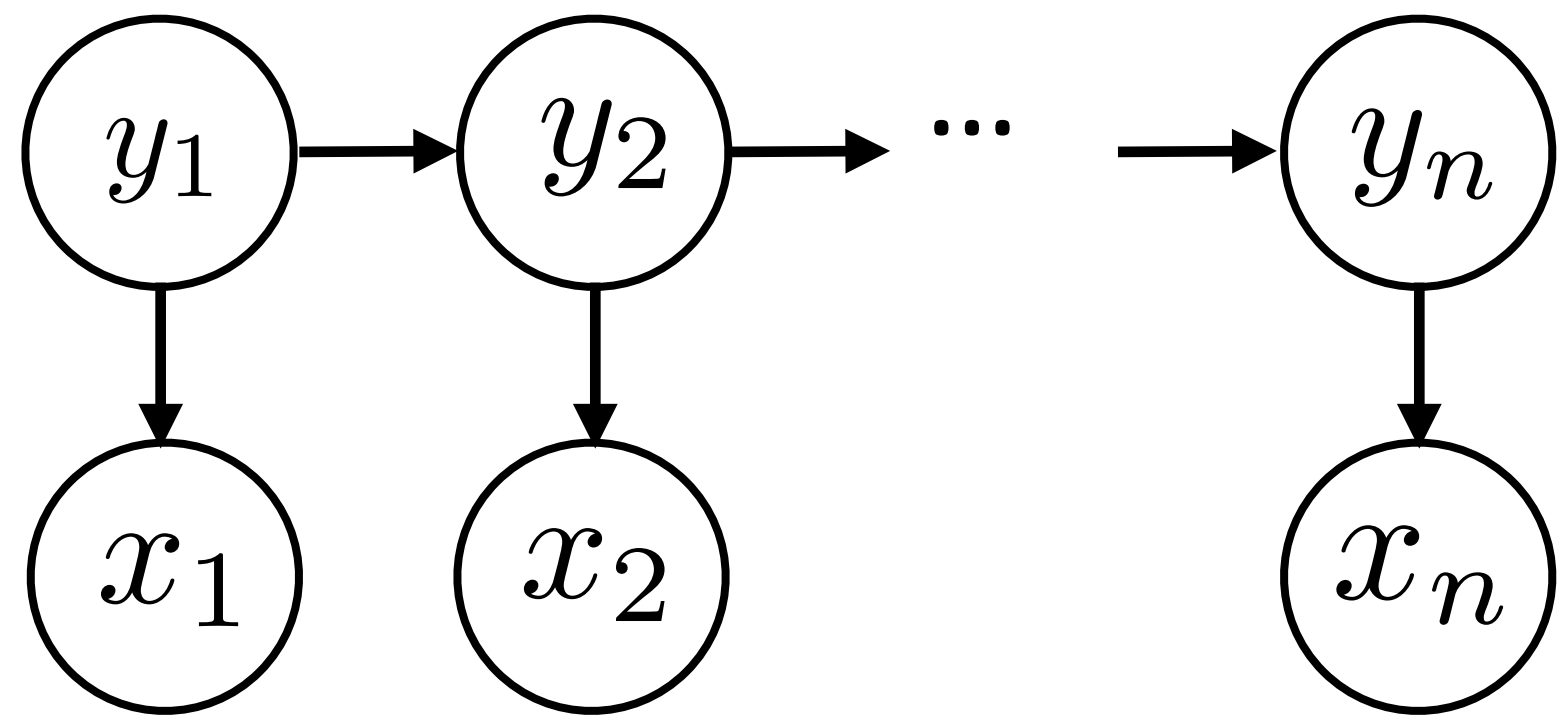


- ▶ CRFs:

$$P(\mathbf{y}|\mathbf{x}) \propto \prod_k \exp(\phi_k(\mathbf{x}, \mathbf{y}))$$

Sequential CRFs

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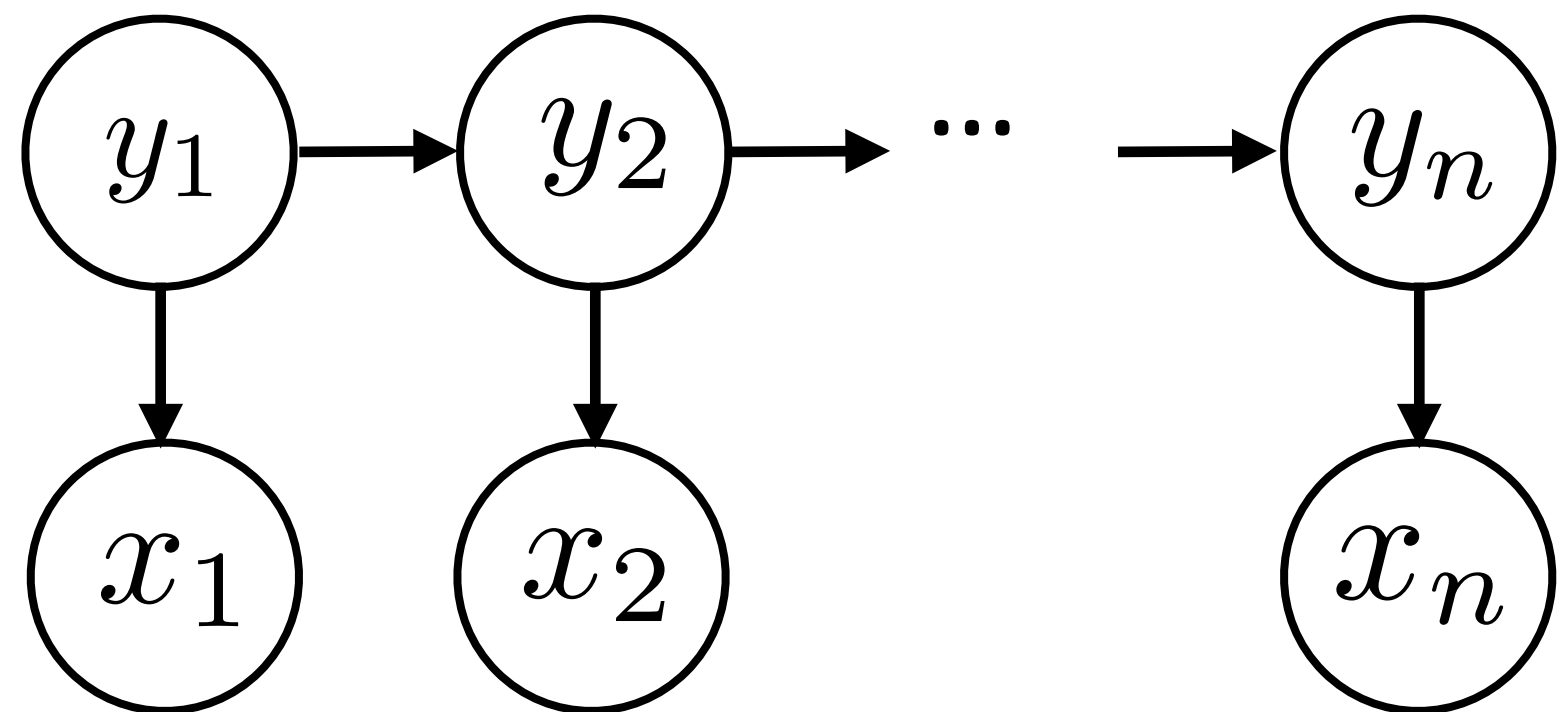
- ▶ CRFs:

$$P(\mathbf{y}|\mathbf{x}) \propto \prod_k \exp(\phi_k(\mathbf{x}, \mathbf{y}))$$

$$P(\mathbf{y}|\mathbf{x}) \propto \exp(\phi_o(y_1)) \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(x_i, y_i))$$

Sequential CRFs

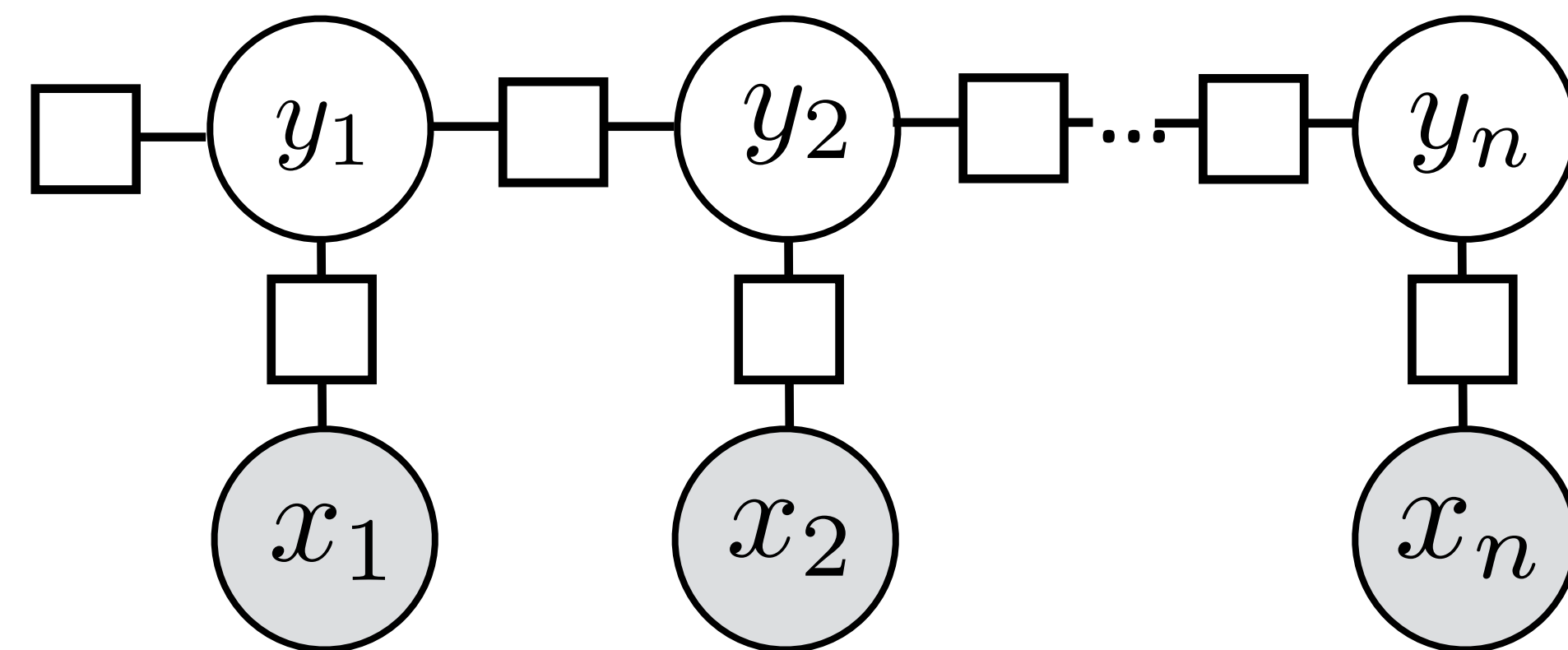
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- ▶ CRFs:

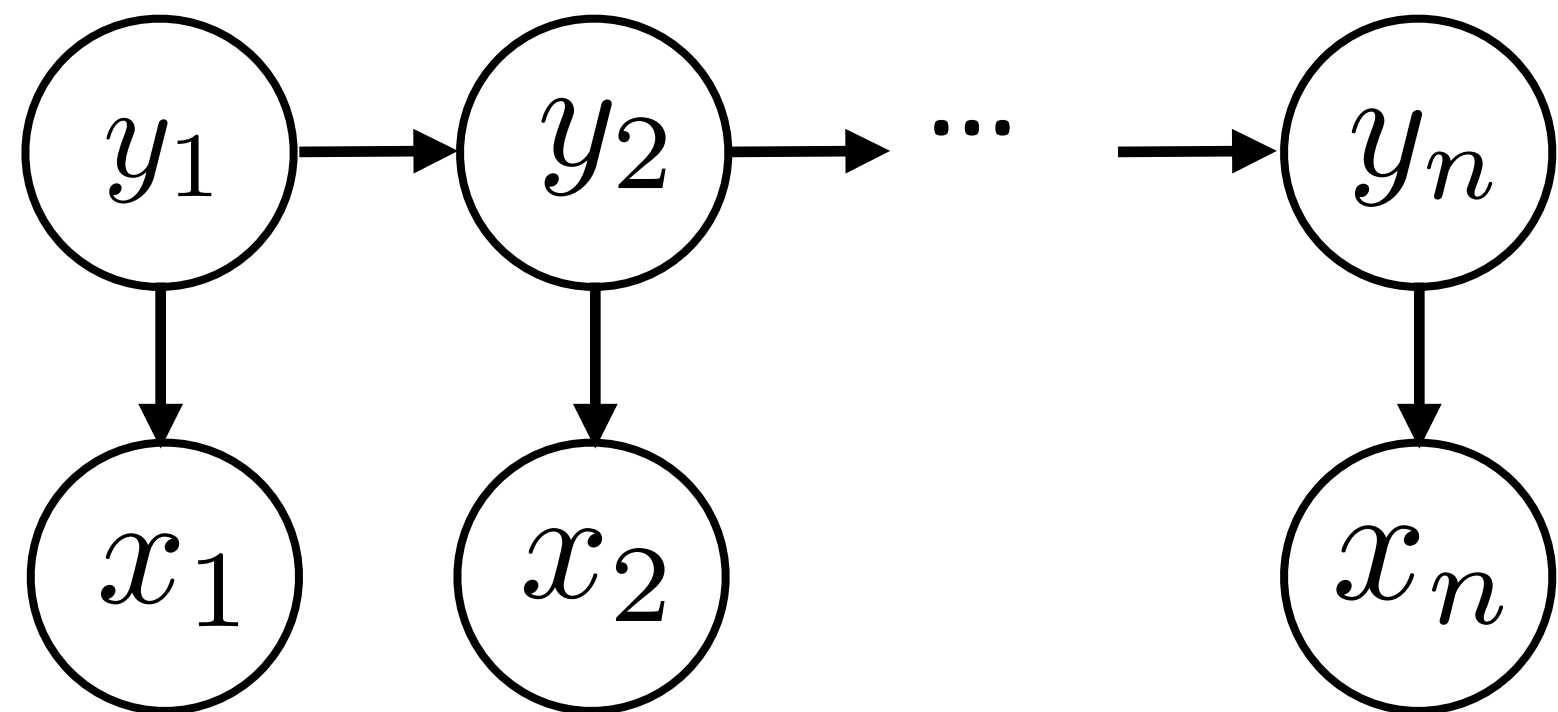
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$$P(\mathbf{y}|\mathbf{x}) \propto \exp(\phi_o(y_1)) \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(x_i, y_i))$$



Sequential CRFs

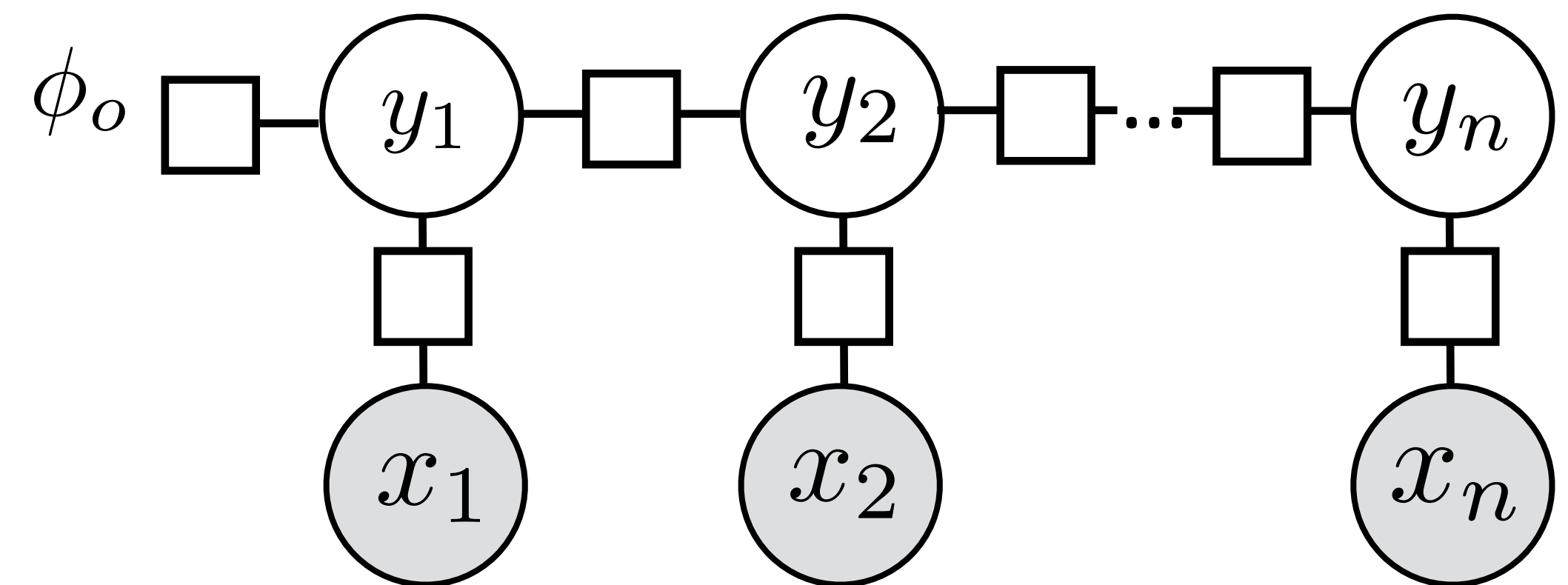
- ▶ HMMs: $P(\mathbf{y}, \mathbf{x}) = P(y_1)P(x_1|y_1)P(y_2|y_1)P(x_2|y_2) \dots$



- ▶ CRFs:

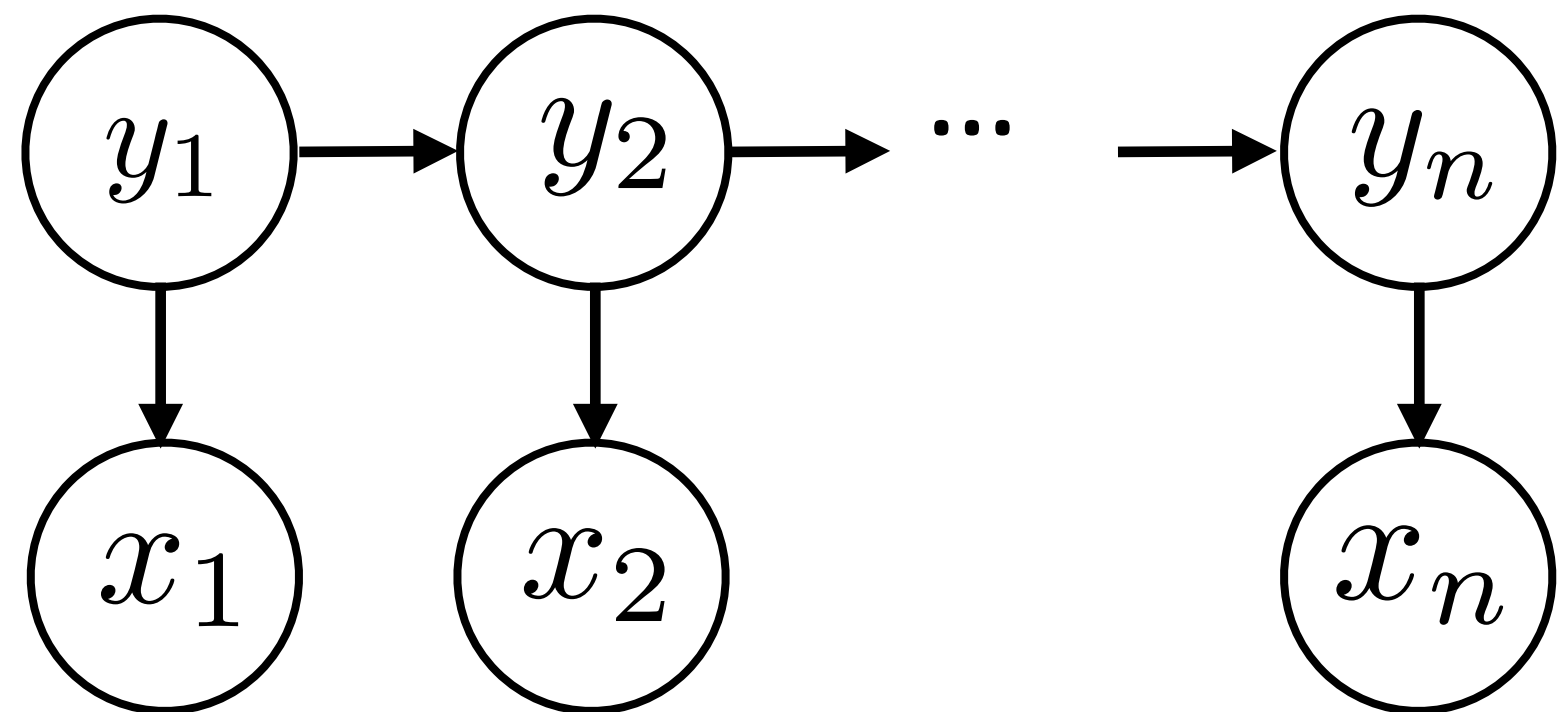
$$P(\mathbf{y}|\mathbf{x}) \propto \prod_k \exp(\phi_k(\mathbf{x}, \mathbf{y}))$$

$$P(\mathbf{y}|\mathbf{x}) \propto \exp(\phi_o(y_1)) \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(x_i, y_i))$$



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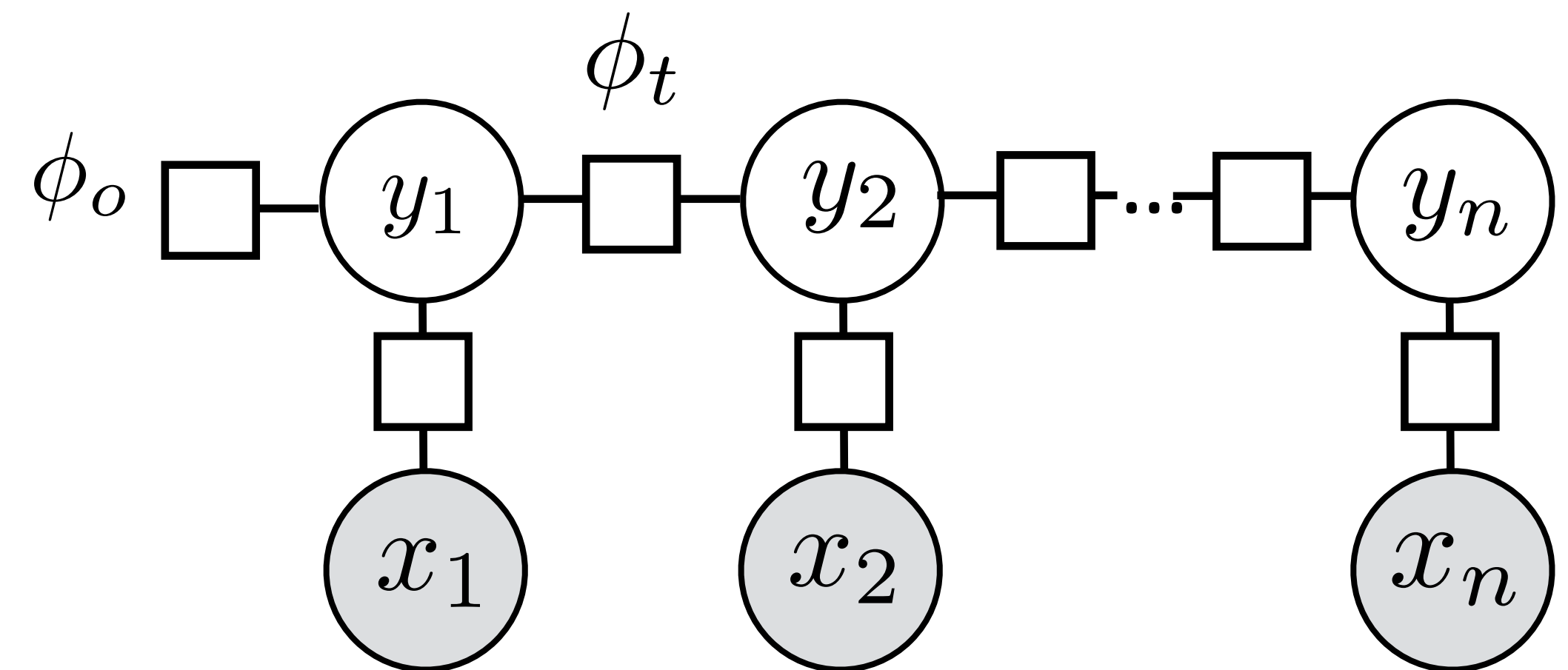
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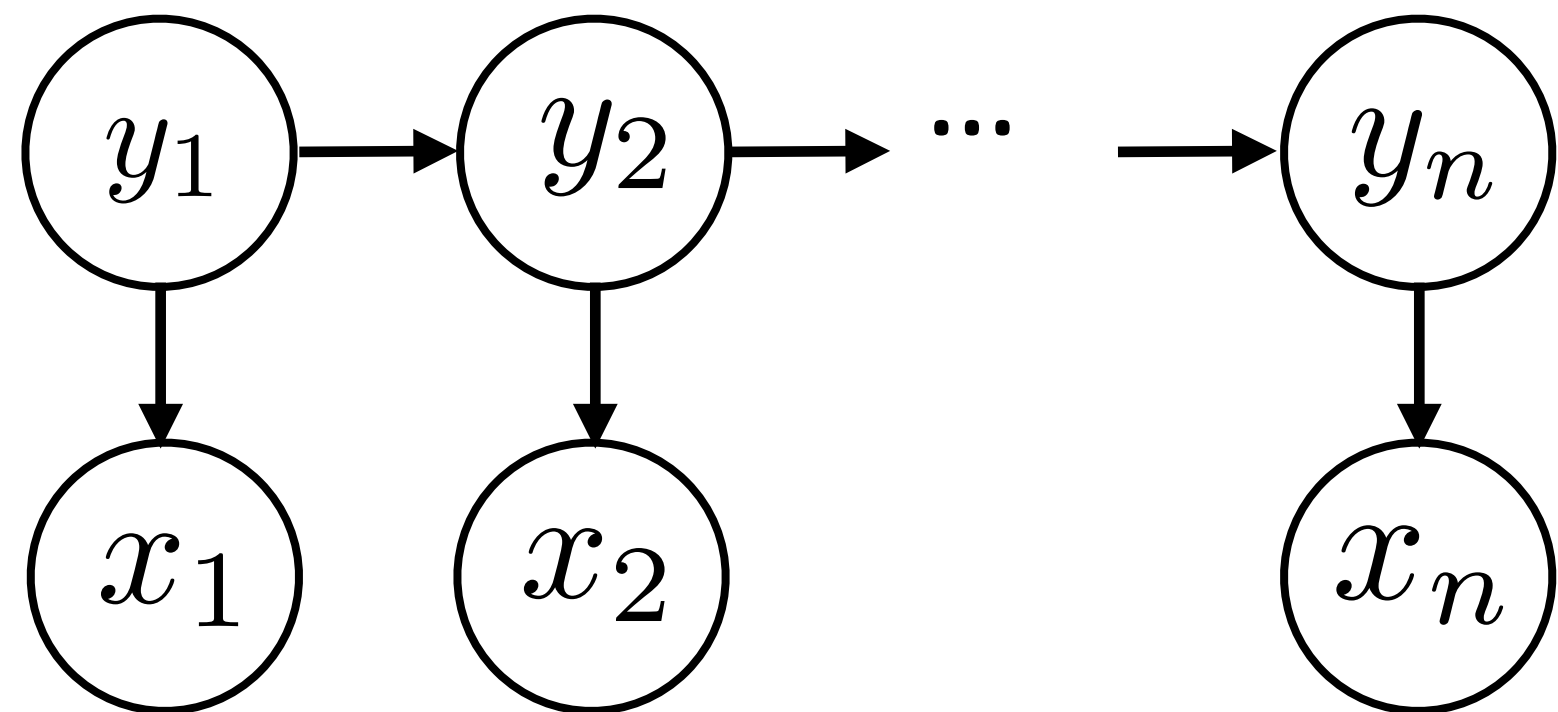
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Sequential CRFs

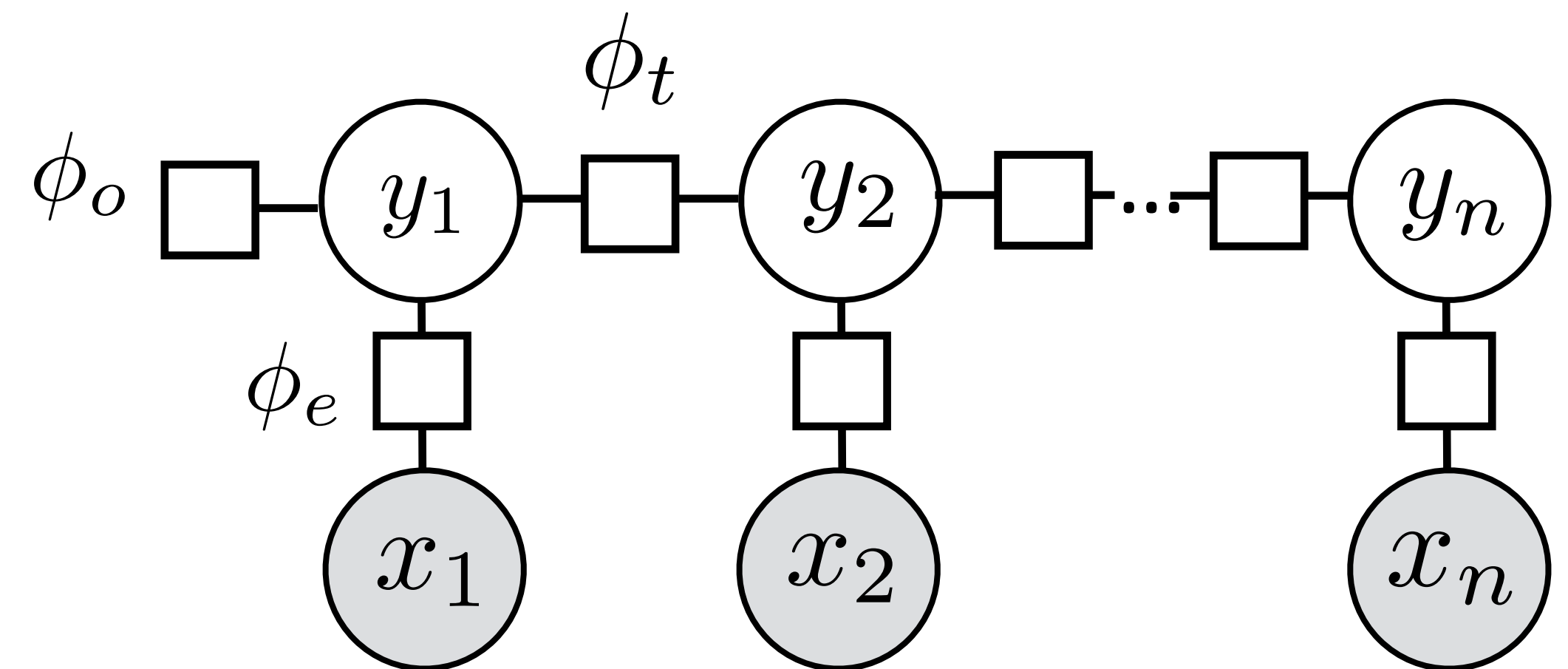
- ▶ HMMs: $P(\mathbf{y}, \mathbf{x}) = P(y_1)P(x_1|y_1)P(y_2|y_1)P(x_2|y_2) \dots$



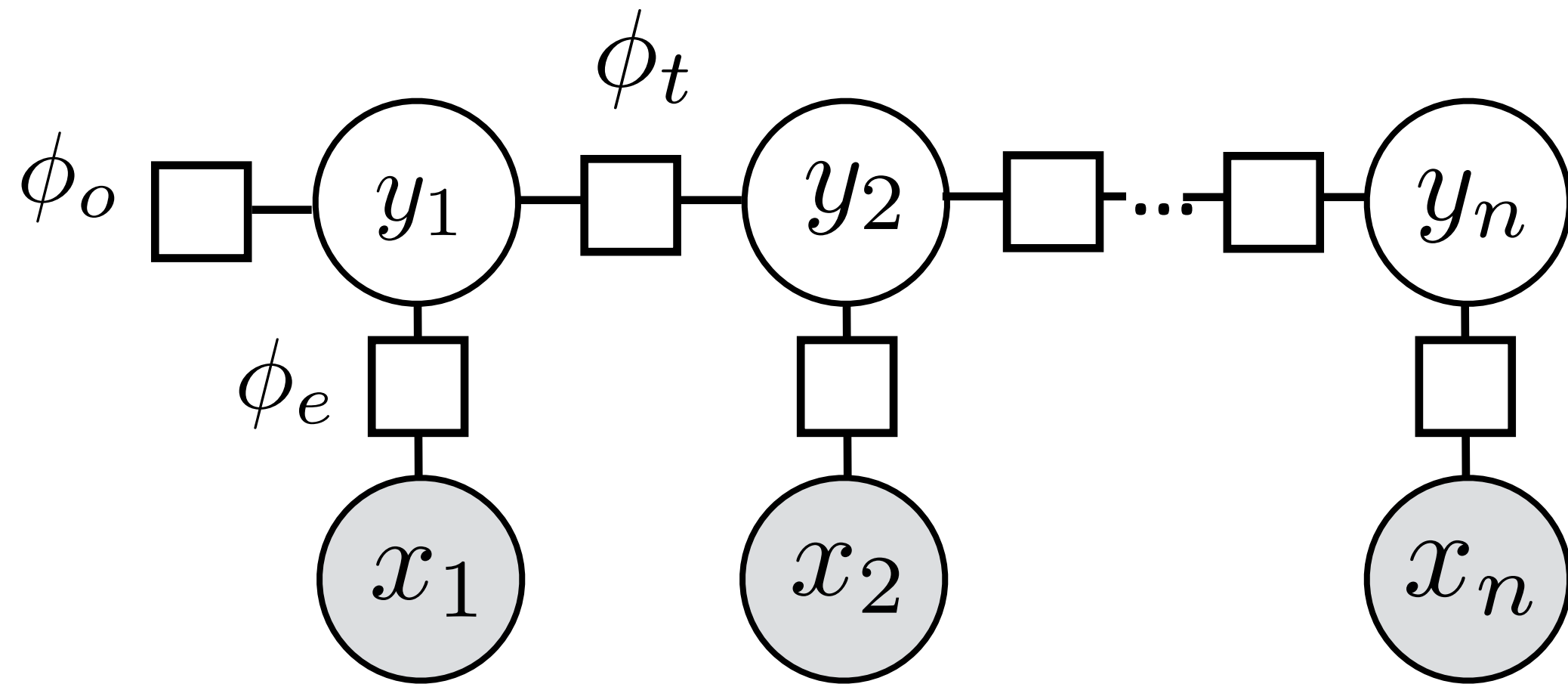
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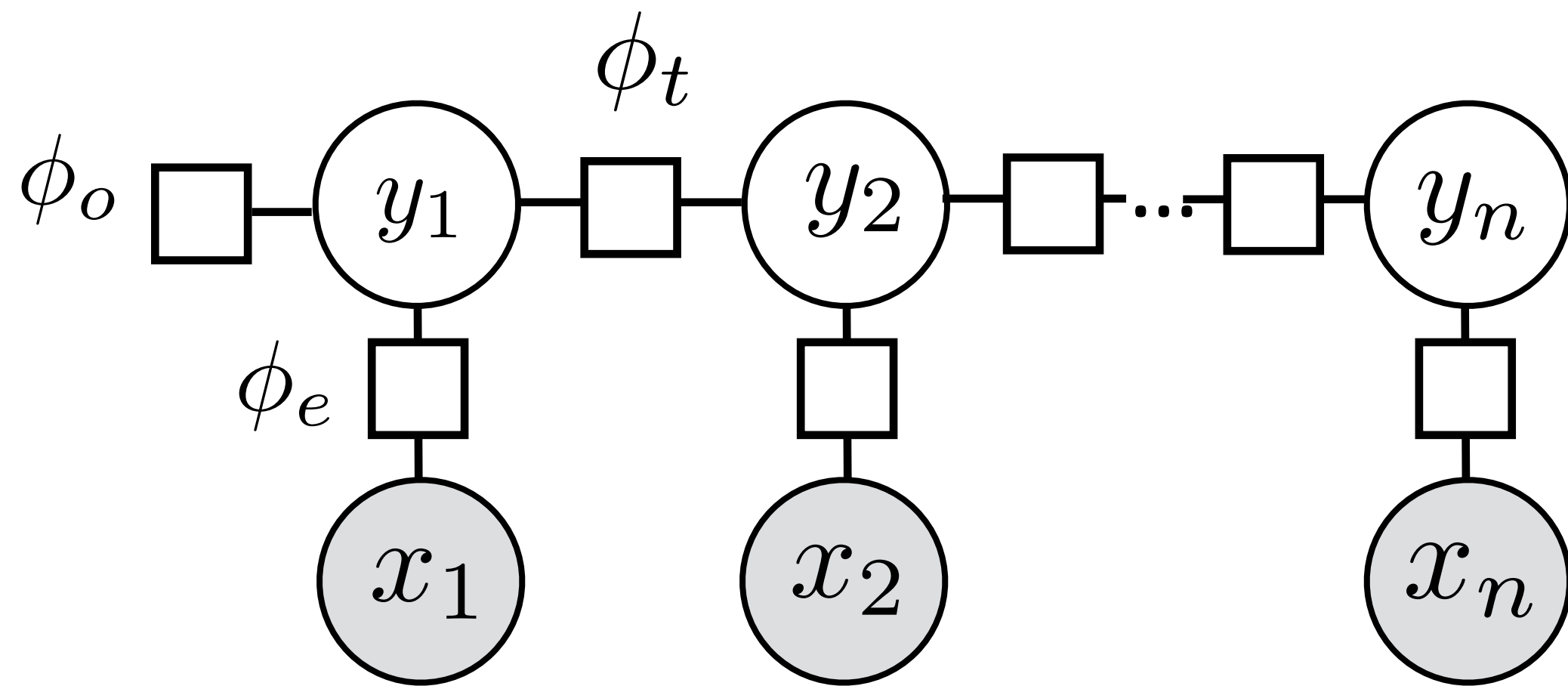


Sequential CRFs



$$P(\mathbf{y}|\mathbf{x}) \propto \exp(\phi_o(y_1)) \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(x_i, y_i))$$

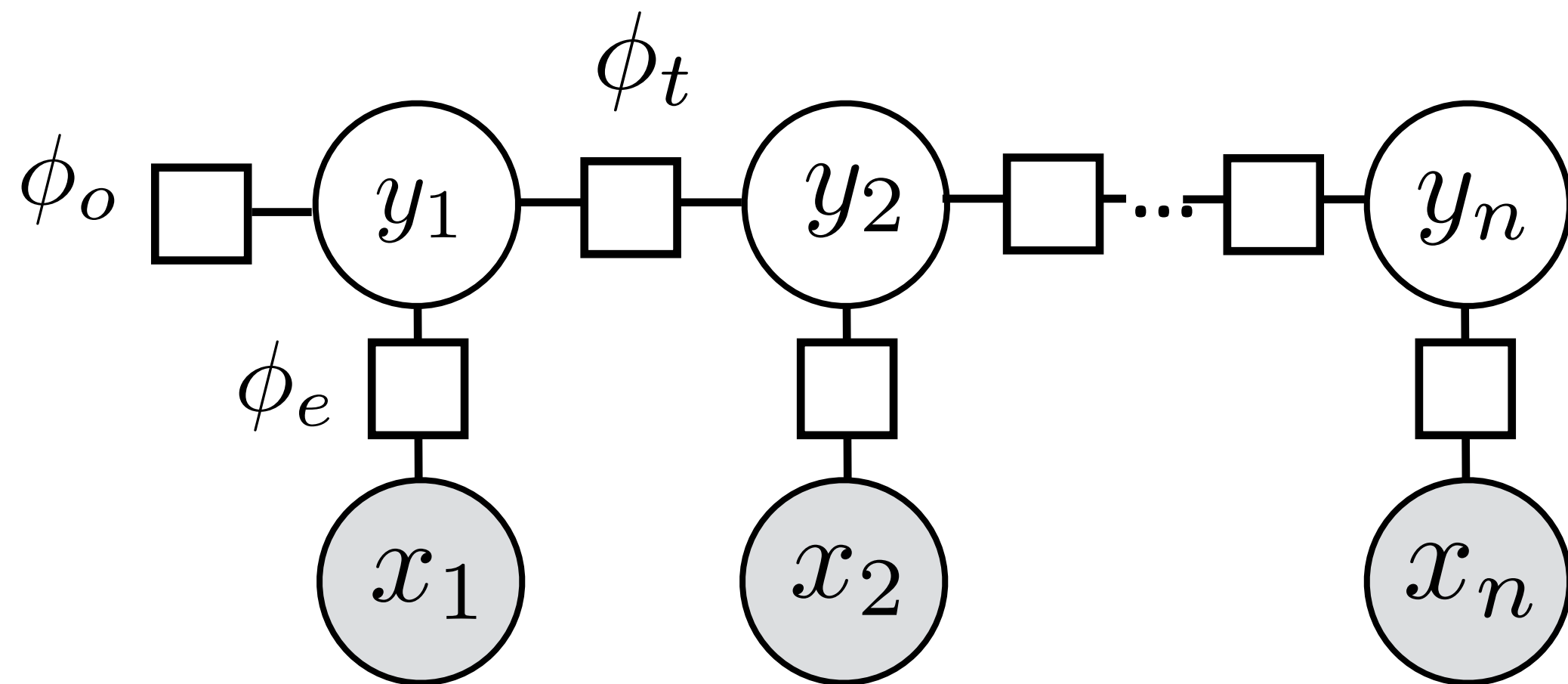
Sequential CRFs



$$P(\mathbf{y}|\mathbf{x}) \propto \exp(\phi_o(y_1)) \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(x_i, y_i))$$

- ▶ We condition on $\mathbf{x}$, so every factor can depend on all of $\mathbf{x}$ (including transitions, but we won't do this)

Sequential CRFs

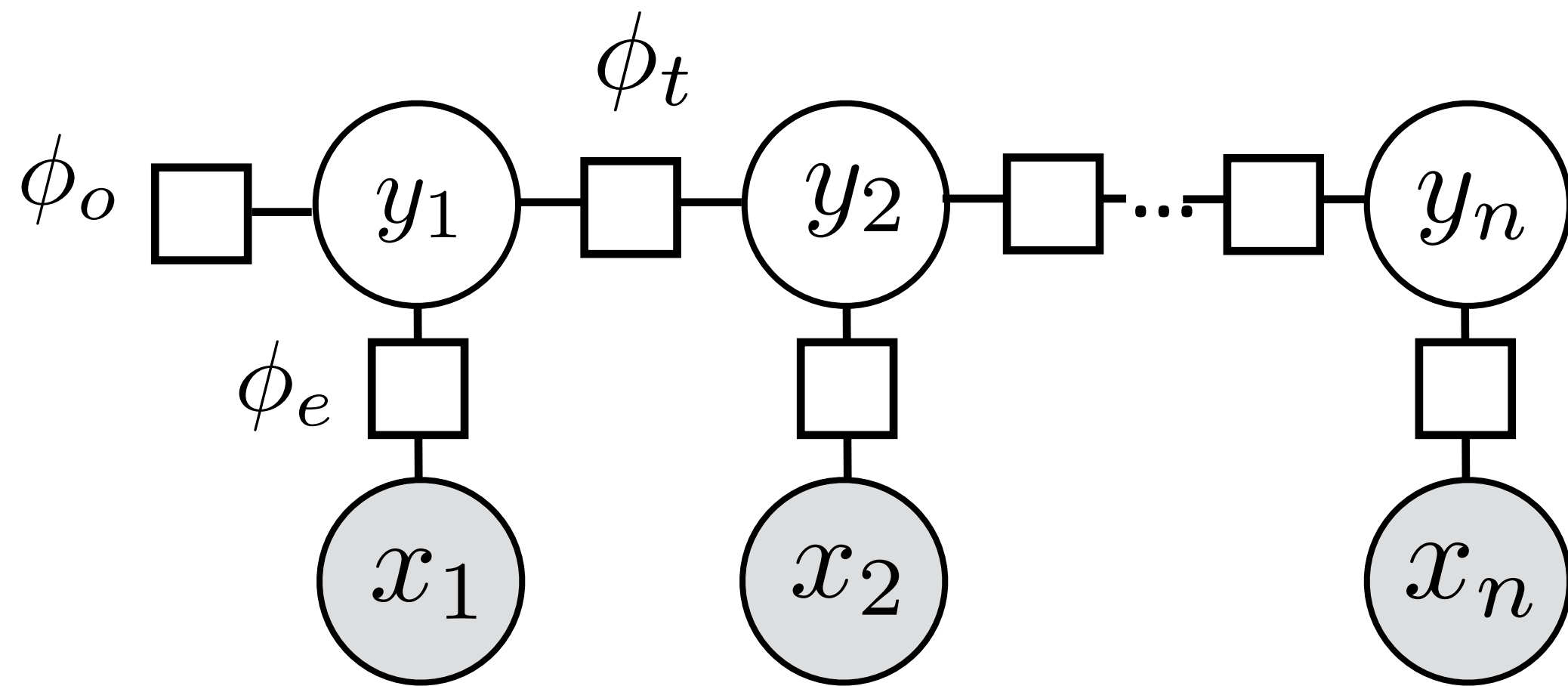


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$\prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$

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Sequential CRFs



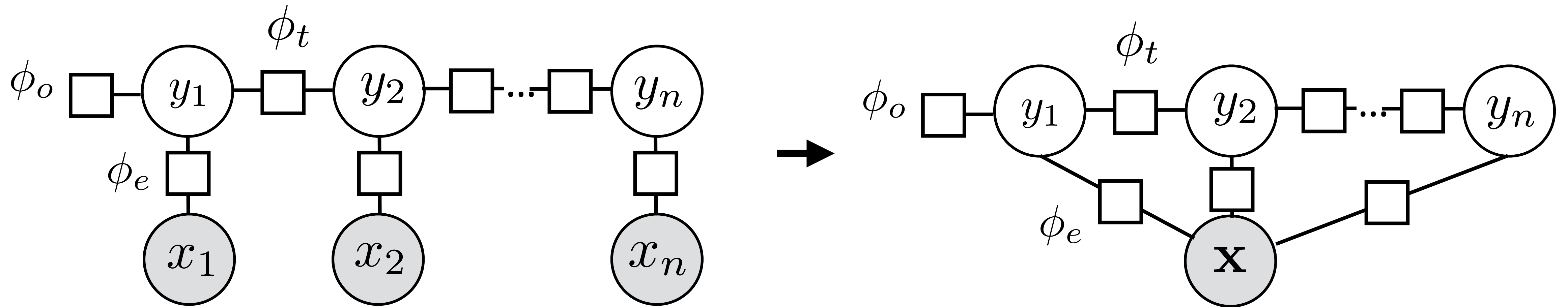
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$$\prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$

token index — lets us look at current word

Sequential CRFs



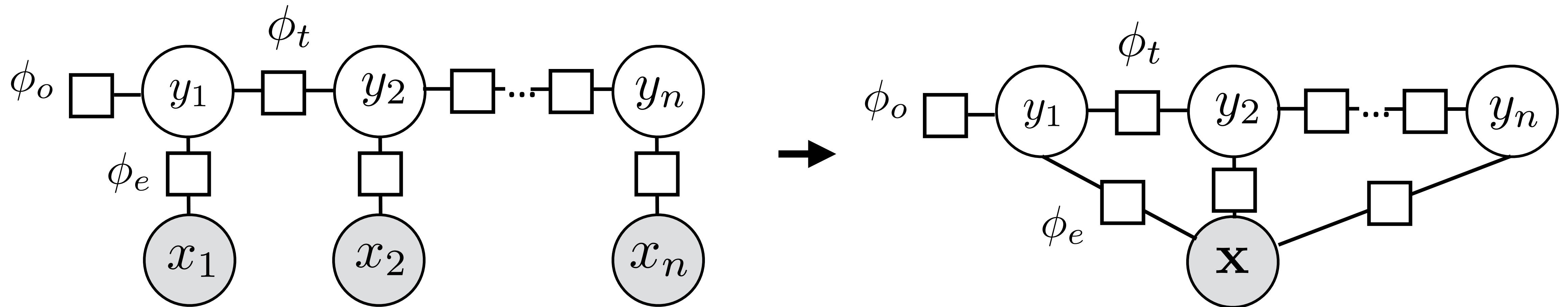
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Sequential CRFs



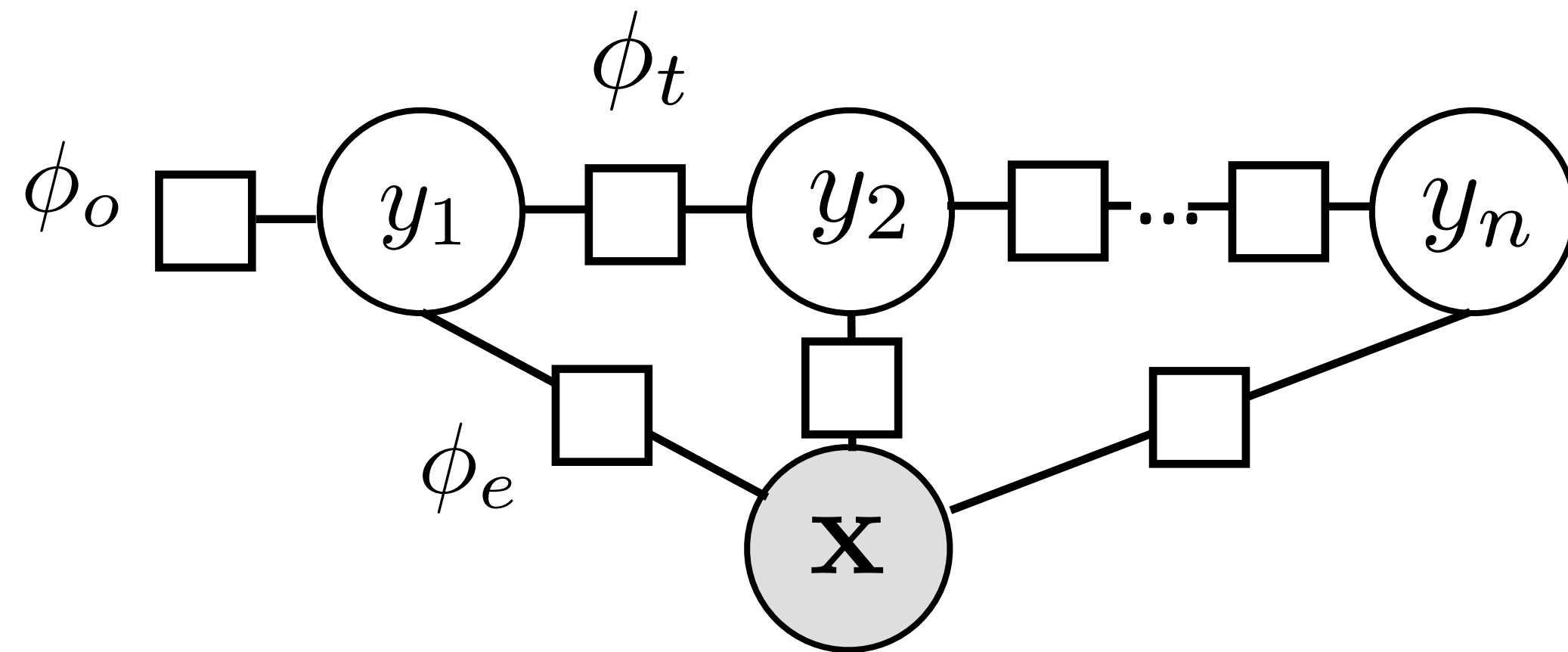
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- ▶ We condition on $\mathbf{x}$, so every factor can depend on all of $\mathbf{x}$ (including transitions, but we won't do this)
- ▶ $\mathbf{y}$ can't depend arbitrarily on $\mathbf{x}$ in a generative model

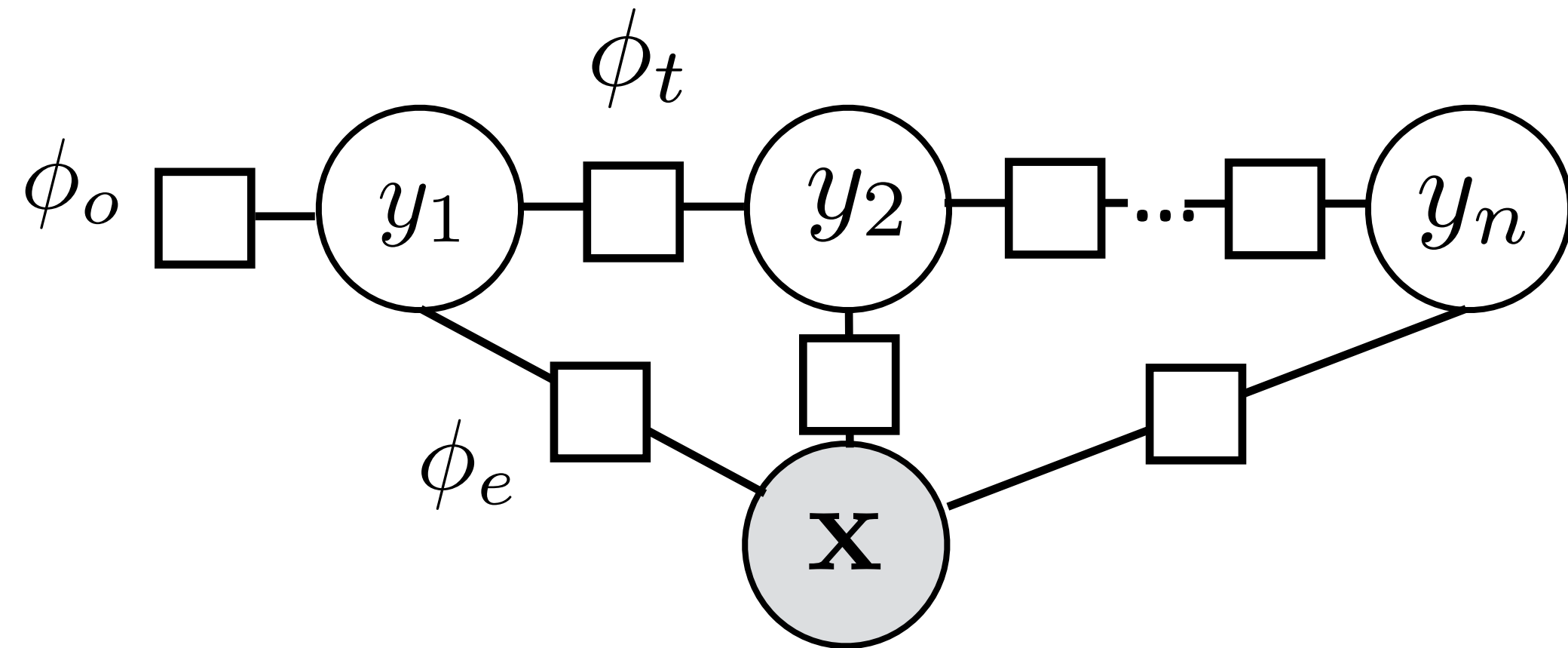
$$\prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$

token index — lets us look at current word

Sequential CRFs

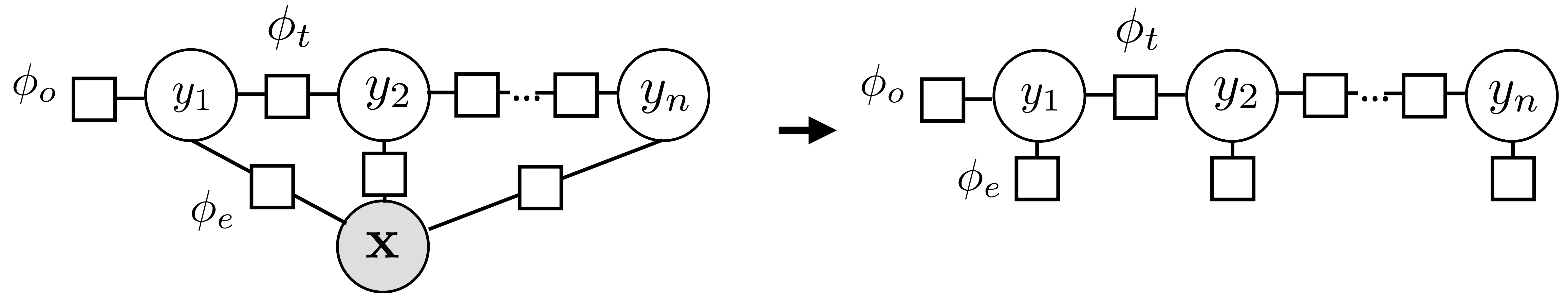


Sequential CRFs



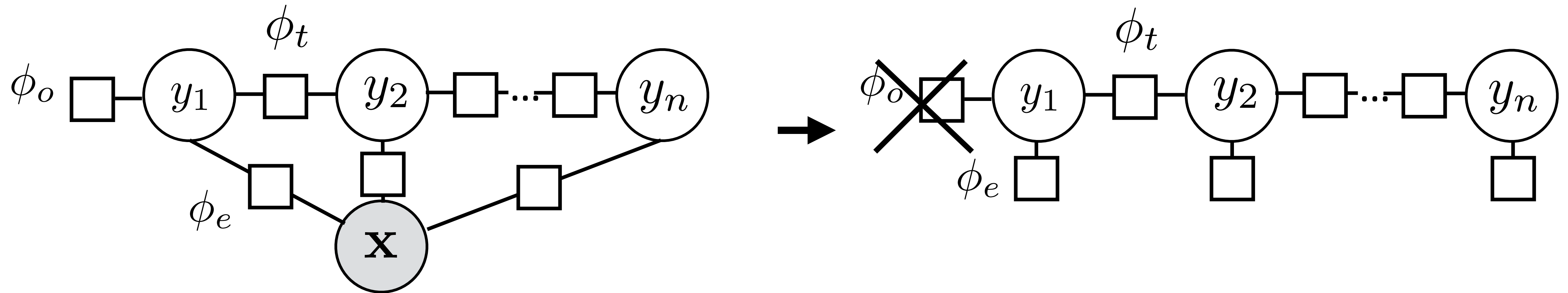
- Notation: omit $\mathbf{x}$ from the factor graph entirely (implicit)

Sequential CRFs



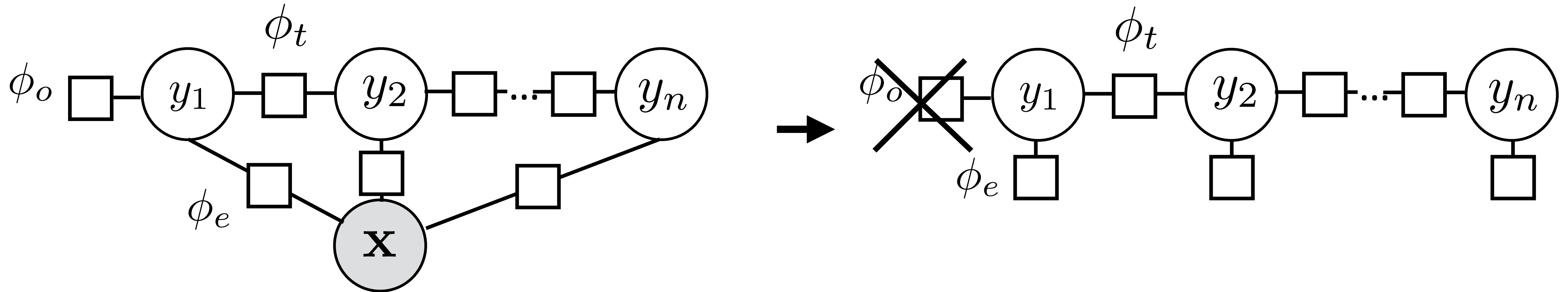
- Notation: omit $\mathbf{x}$ from the factor graph entirely (implicit)

Sequential CRFs



- ▶ Notation: omit $\mathbf{x}$ from the factor graph entirely (implicit)
- ▶ Don't include initial distribution, can bake into other factors

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Sequential CRFs (aka linear-chain CRFs):

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$

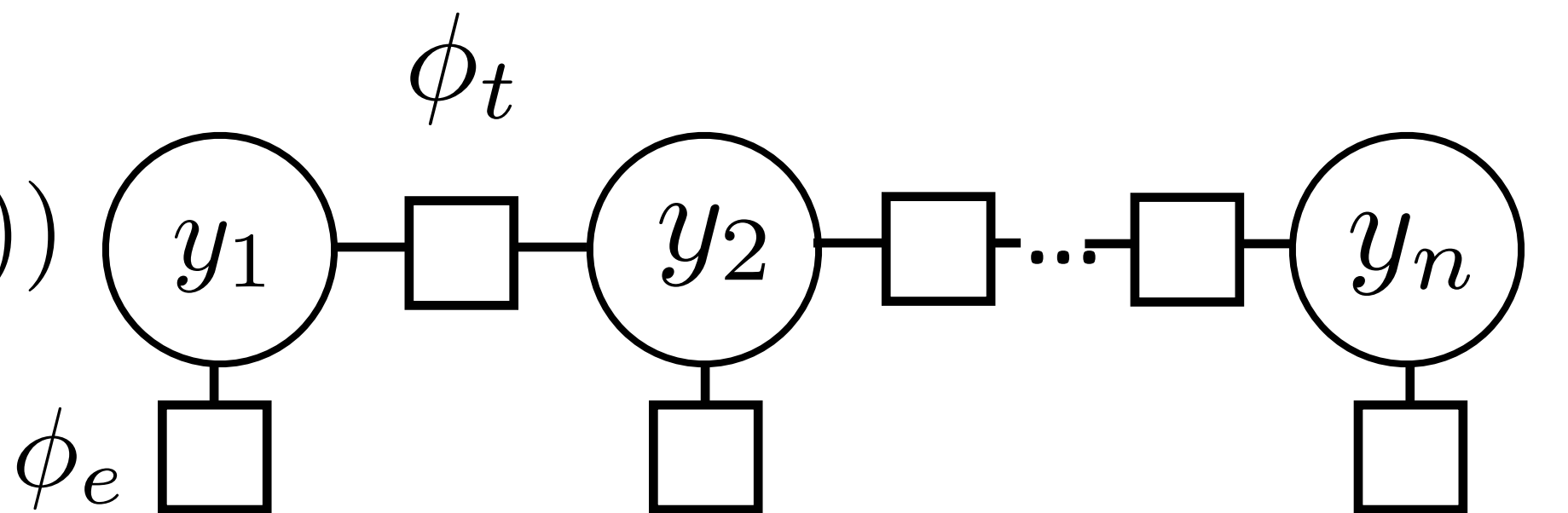
Features for NER

Feature Functions

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$

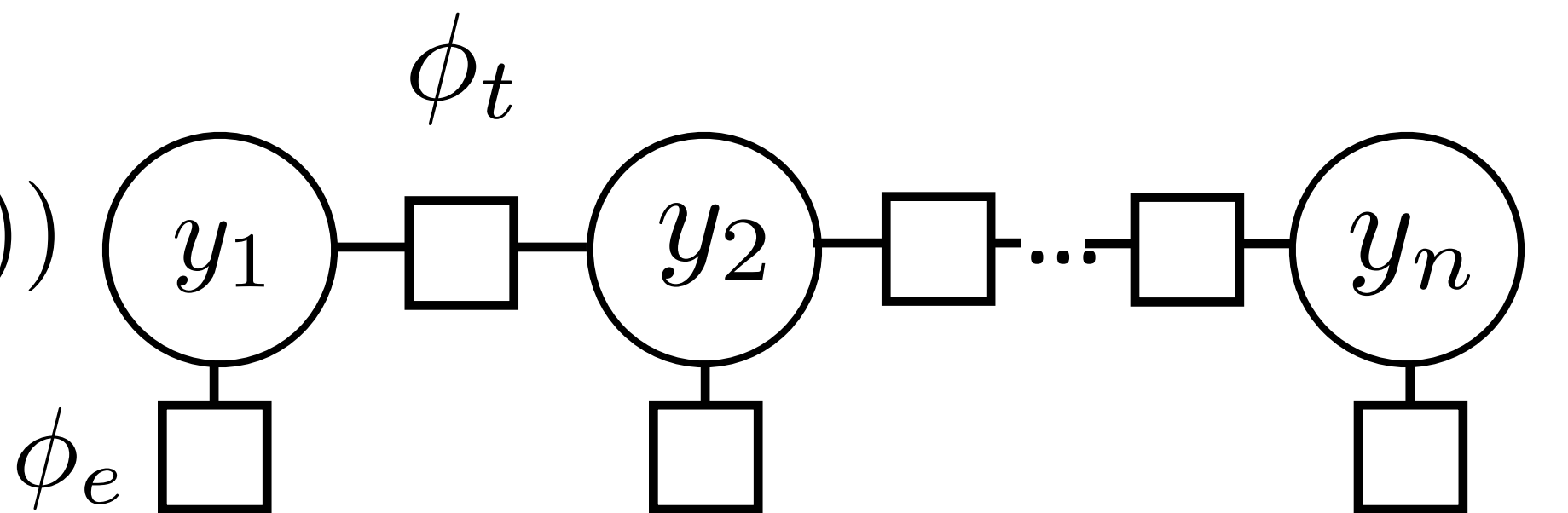
The diagram illustrates a Markov chain structure for a sequence model. It consists of a sequence of nodes $y_1, y_2, \dots, y_n$ represented by circles. Each y_i is connected to y_{i+1} by a horizontal line, with a square node in between. Additionally, each y_i is connected to a square node below it. The transition function ϕ_t is associated with the horizontal edges, and the emission function ϕ_e is associated with the vertical edges.

Feature Functions

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$


- ▶ This can be almost anything! (e.g., use a neural network to compute them)
Here we use linear functions of sparse features:

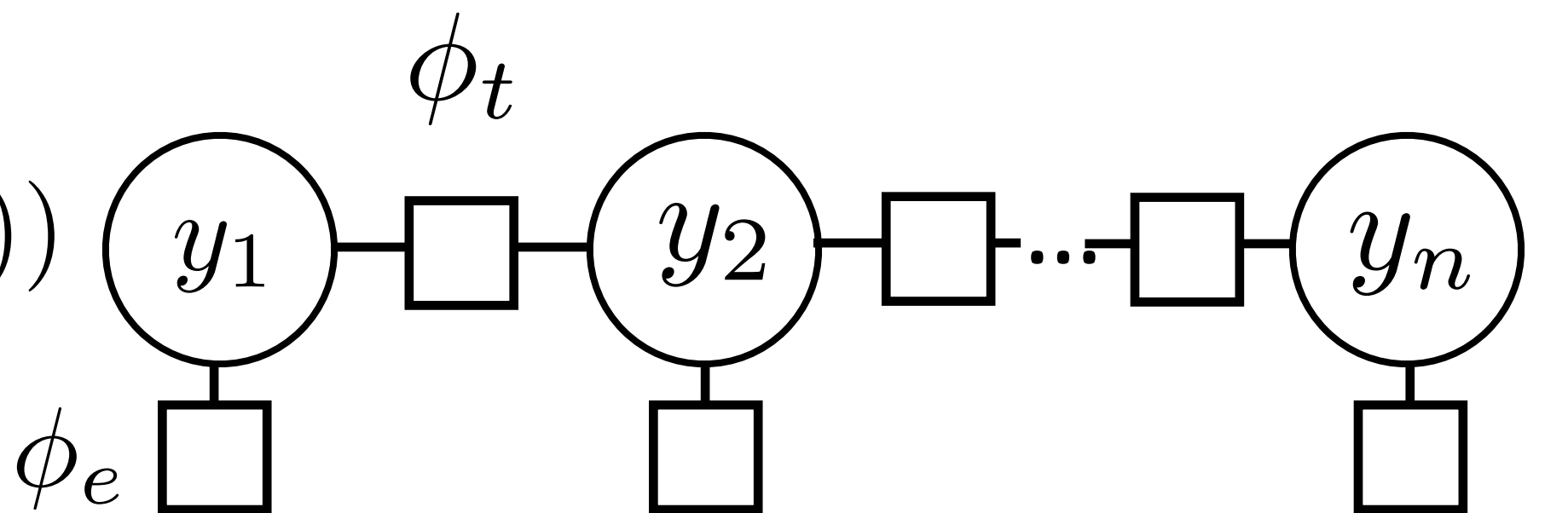
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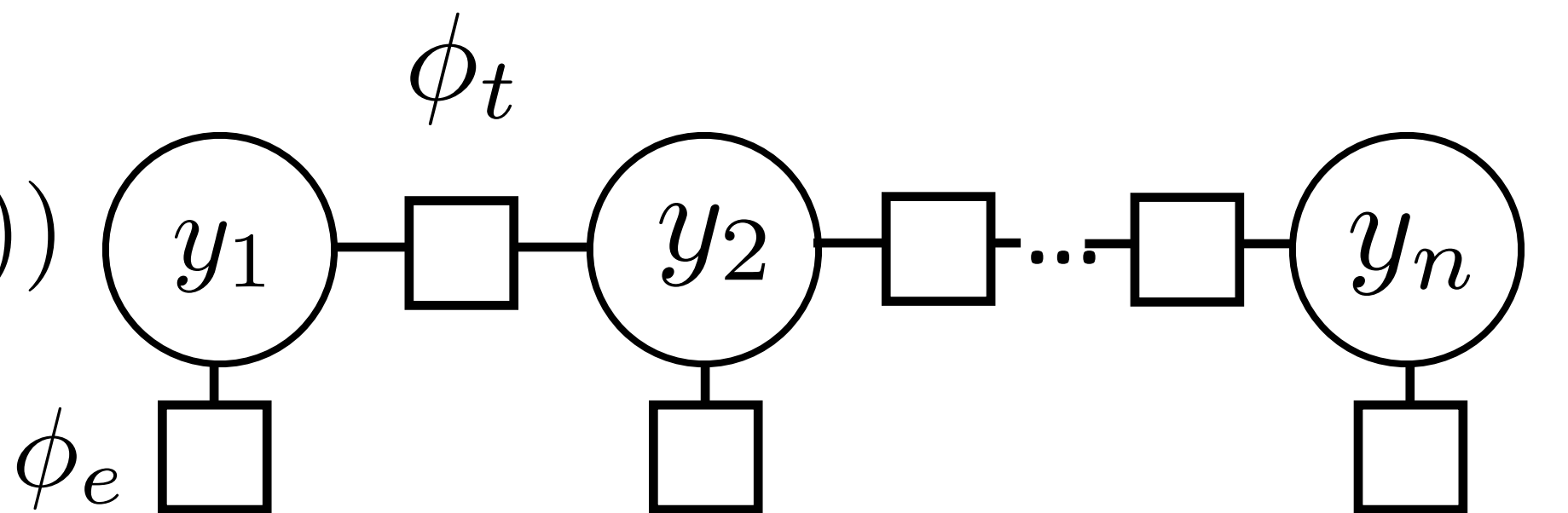
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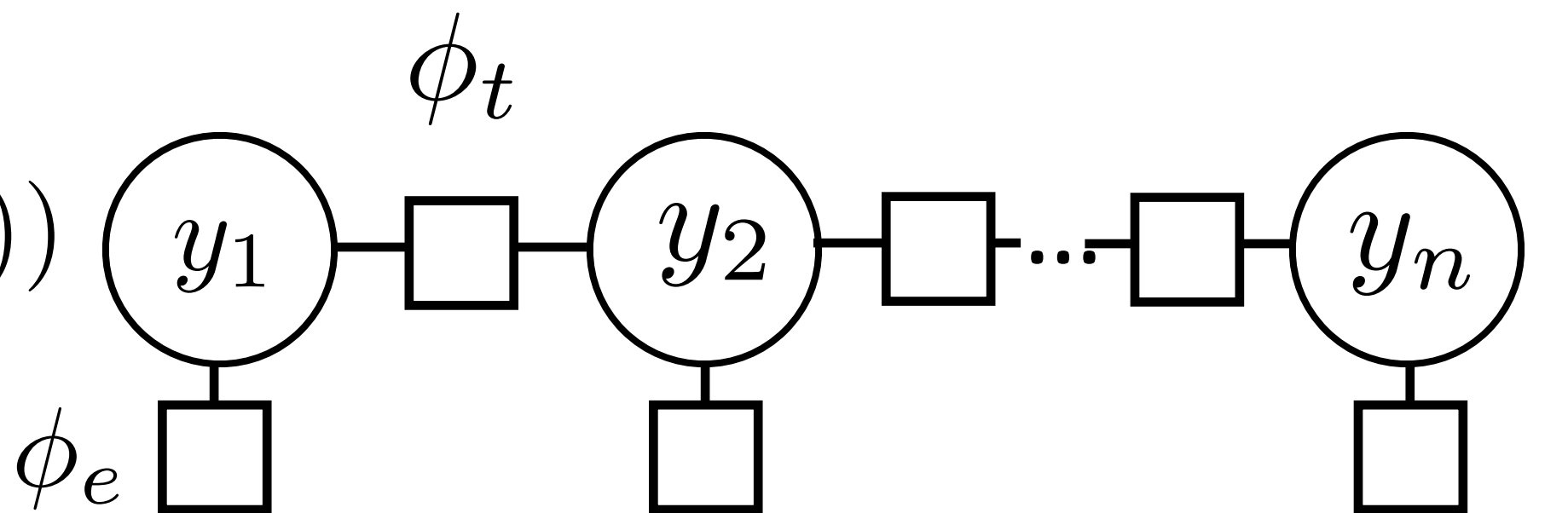
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$$P(\mathbf{y}|\mathbf{x}) \propto \exp w^\top \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$$

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- ▶ Looks like our single weight vector multiclass logistic regression model

Basic Features for NER

$$P(\mathbf{y}|\mathbf{x}) \propto \exp w^\top \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$$

Barack Obama will travel to Hangzhou today for the G20 meeting .

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O B-LOC



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Transitions: $f_t(y_{i-1}, y_i) = \text{Ind}[y_{i-1} \ \& \ y_i] = \text{I}[O - \text{B-LOC}]$

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Emissions: $f_e(y_6, 6, \mathbf{x}) =$

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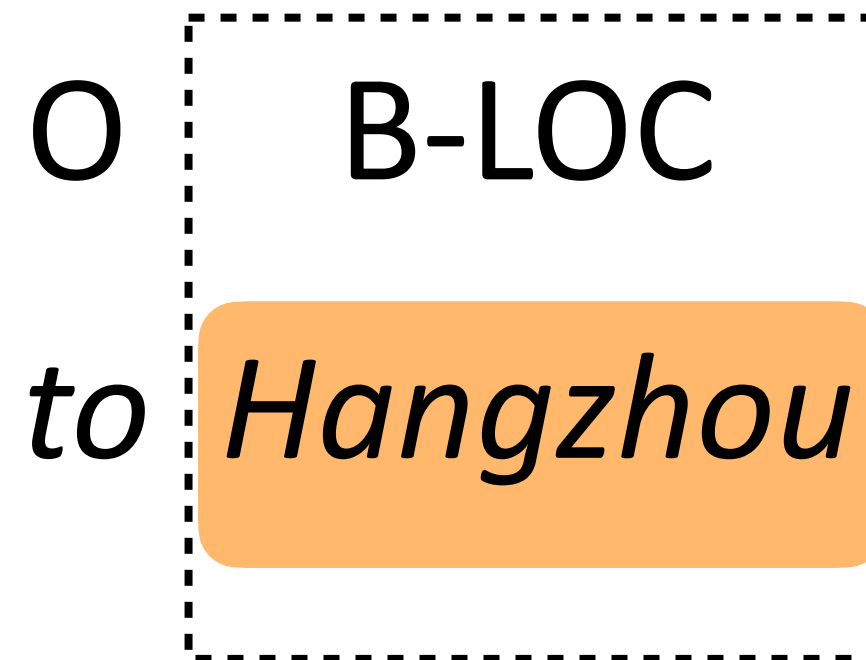
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Emissions: $f_e(y_6, 6, \mathbf{x}) = \text{I}[\text{B-LOC} \ \& \ \text{Current word} = \text{Hangzhou}]$
 $\text{I}[\text{B-LOC} \ \& \ \text{Prev word} = \text{to}]$

Emission Features for NER

LOC

Leicestershire is a nice place to visit...

$$\phi_e(y_i, i, \mathbf{x})$$

PER

Leonardo DiCaprio won an award...

LOC

I took a vacation to Boston

ORG

Apple released a new version...

LOC

Texas governor

PER

Greg Abbott said

ORG

According to the New York Times...

Features for NER

- ▶ Word features (can use in HMM)
 - ▶ Capitalization
 - ▶ Word shape
 - ▶ Prefixes/suffixes
 - ▶ Lexical indicators
- ▶ Context features (can't use in HMM!)
 - ▶ Words before/after
 - ▶ POS Tags before/after (if we run a POS tagger first)
- ▶ Word clusters
- ▶ Gazetteers

Leicestershire

Boston

Apple released a new version...

According to the *New York Times*...

Inference and Learning in CRFs

CRFs in General (recap)

- ▶ CRFs: discriminative model with the following form:

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_k \exp(\phi_k(\mathbf{x}, \mathbf{y}))$$

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↑ any real-valued scoring function of its arguments

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- ▶ Our special case: linear feature-based potentials $\phi_k(\mathbf{x}, \mathbf{y}) = w^\top f_k(\mathbf{x}, \mathbf{y})$

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- ▶ Problem: intractable inference in the general case! Computing Z requires an exponent sum

Linear-chain CRFs Outline

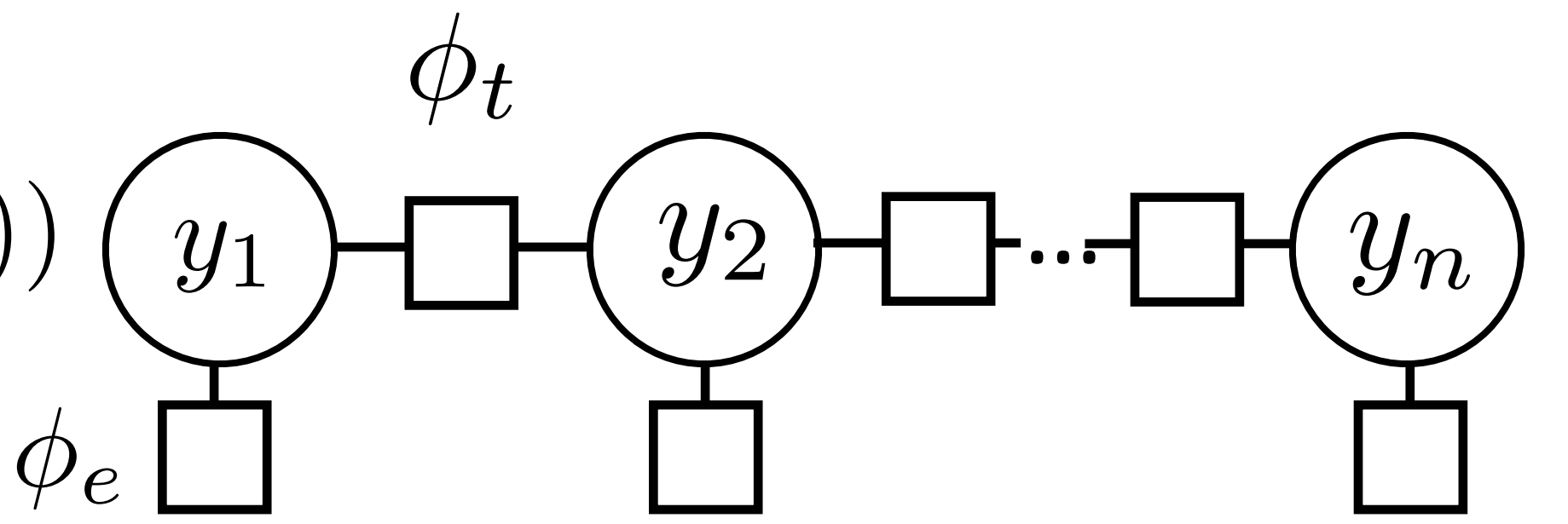
► Model:
$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$

$$P(\mathbf{y}|\mathbf{x}) \propto \exp w^\top \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$$

► Inference

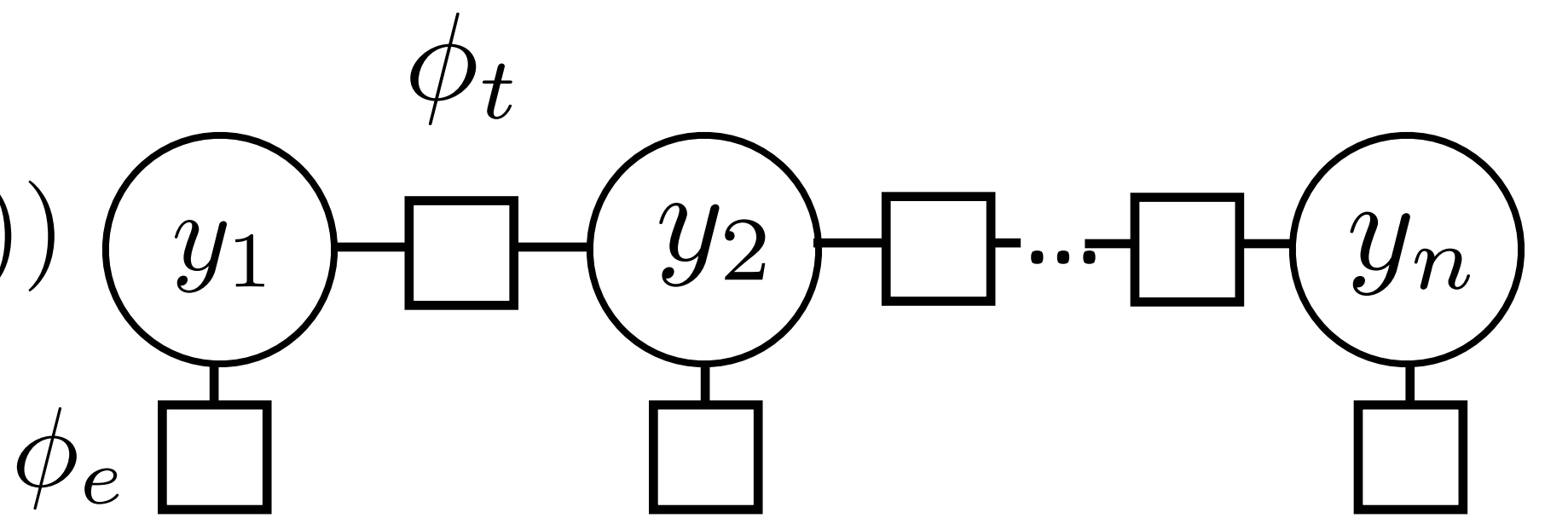
► Learning

Computing (arg)maxes

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$


The diagram illustrates a sequence model structure. It features a horizontal chain of nodes. The first node is a circle labeled y_1 , and the second is a circle labeled y_2 , followed by an ellipsis and then a circle labeled y_n . These circles are connected by horizontal lines to square nodes. A square node is positioned below each circle node (y_1, y_2, y_n), connected by a vertical line. The label ϕ_e is placed to the left of the first square node. Above the horizontal line connecting the first and second circle nodes, the label ϕ_t is placed.

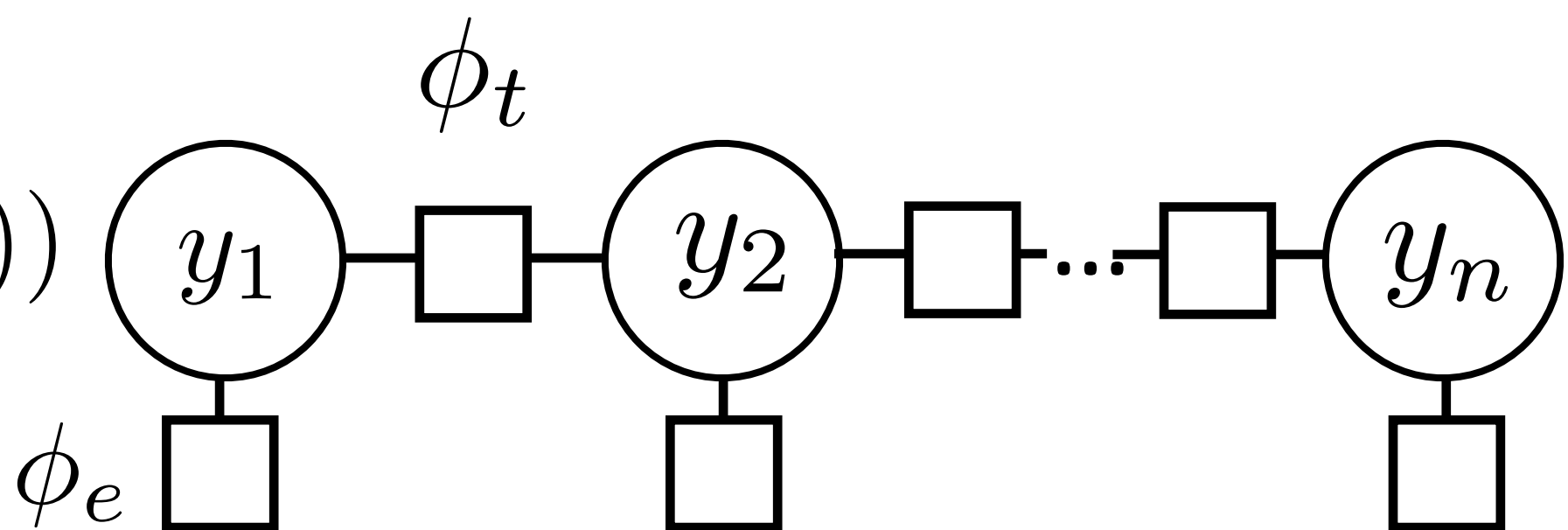
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The diagram illustrates a graphical model for the sequence $y_1, y_2, \dots, y_n$. The nodes $y_1, y_2, \dots, y_n$ are represented by circles. The nodes are connected sequentially by horizontal lines, with square nodes representing the transitions between them. The label ϕ_t is placed above the transition between y_1 and y_2 . The label ϕ_e is placed below the square node connected to y_1 . Ellipses indicate the continuation of the sequence.

- $\operatorname{argmax}_{\mathbf{y}} P(\mathbf{y}|\mathbf{x})$: can use Viterbi exactly as in HMM case

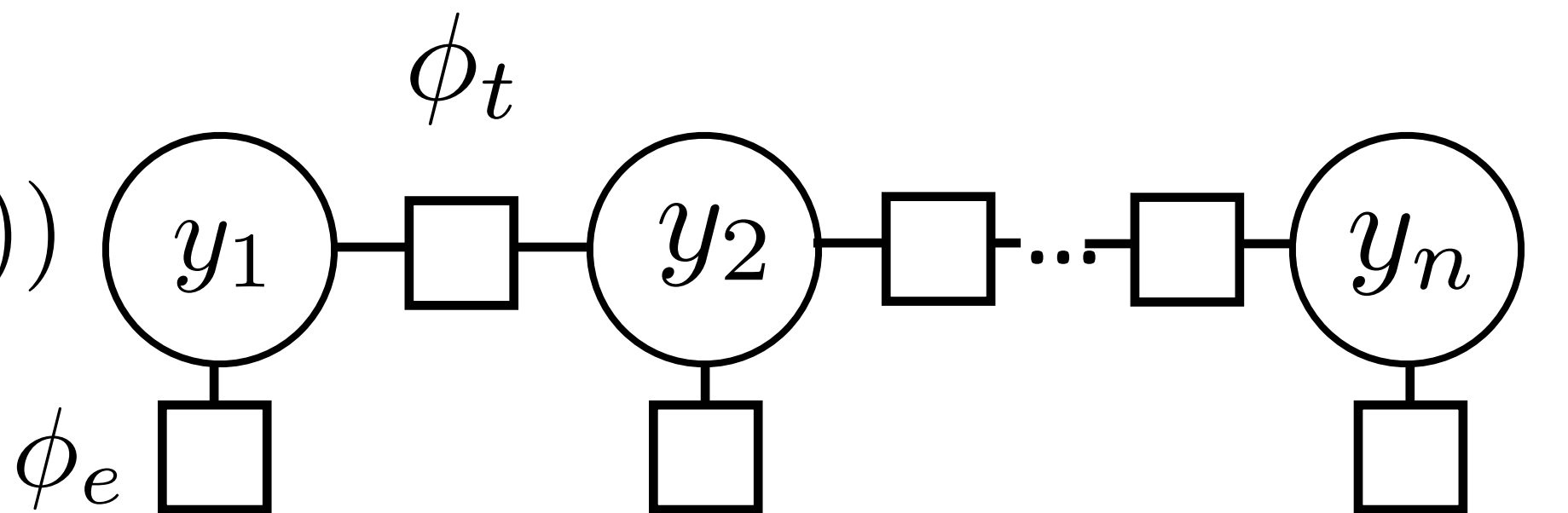
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- $\operatorname{argmax}_{\mathbf{y}} P(\mathbf{y}|\mathbf{x})$: can use Viterbi exactly as in HMM case

$$\max_{y_1, \dots, y_n} e^{\phi_t(y_{n-1}, y_n)} e^{\phi_e(y_n, n, \mathbf{x})} \dots e^{\phi_e(y_2, 2, \mathbf{x})} e^{\phi_t(y_1, y_2)} e^{\phi_e(y_1, 1, \mathbf{x})}$$

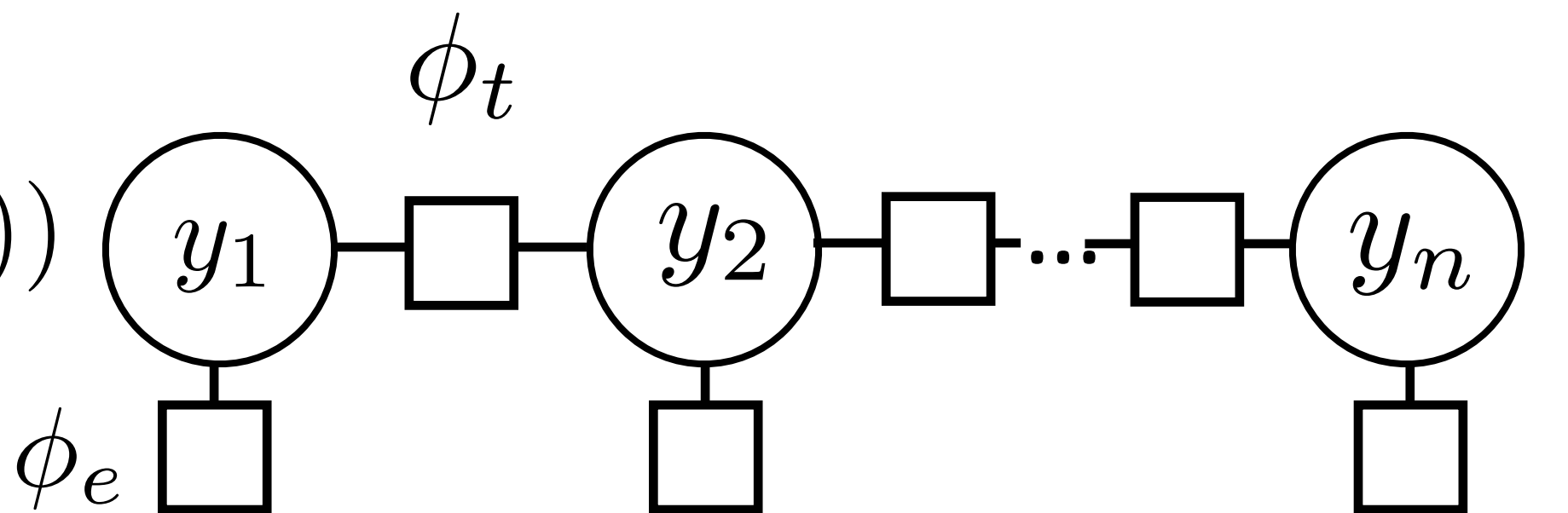
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$$\begin{aligned} & \max_{y_1, \dots, y_n} e^{\phi_t(y_{n-1}, y_n)} e^{\phi_e(y_n, n, \mathbf{x})} \dots e^{\phi_e(y_2, 2, \mathbf{x})} e^{\phi_t(y_1, y_2)} e^{\phi_e(y_1, 1, \mathbf{x})} \\ = & \max_{y_2, \dots, y_n} e^{\phi_t(y_{n-1}, y_n)} e^{\phi_e(y_n, n, \mathbf{x})} \dots e^{\phi_e(y_2, 2, \mathbf{x})} \max_{y_1} e^{\phi_t(y_1, y_2)} e^{\phi_e(y_1, 1, \mathbf{x})} \end{aligned}$$

Computing (arg)maxes

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$


- $\operatorname{argmax}_{\mathbf{y}} P(\mathbf{y}|\mathbf{x})$: can use Viterbi exactly as in HMM case

$$\begin{aligned} & \max_{y_1, \dots, y_n} e^{\phi_t(y_{n-1}, y_n)} e^{\phi_e(y_n, n, \mathbf{x})} \dots e^{\phi_e(y_2, 2, \mathbf{x})} e^{\phi_t(y_1, y_2)} e^{\phi_e(y_1, 1, \mathbf{x})} \\ = & \max_{y_2, \dots, y_n} e^{\phi_t(y_{n-1}, y_n)} e^{\phi_e(y_n, n, \mathbf{x})} \dots e^{\phi_e(y_2, 2, \mathbf{x})} \boxed{\max_{y_1} e^{\phi_t(y_1, y_2)} e^{\phi_e(y_1, 1, \mathbf{x})}} \end{aligned}$$

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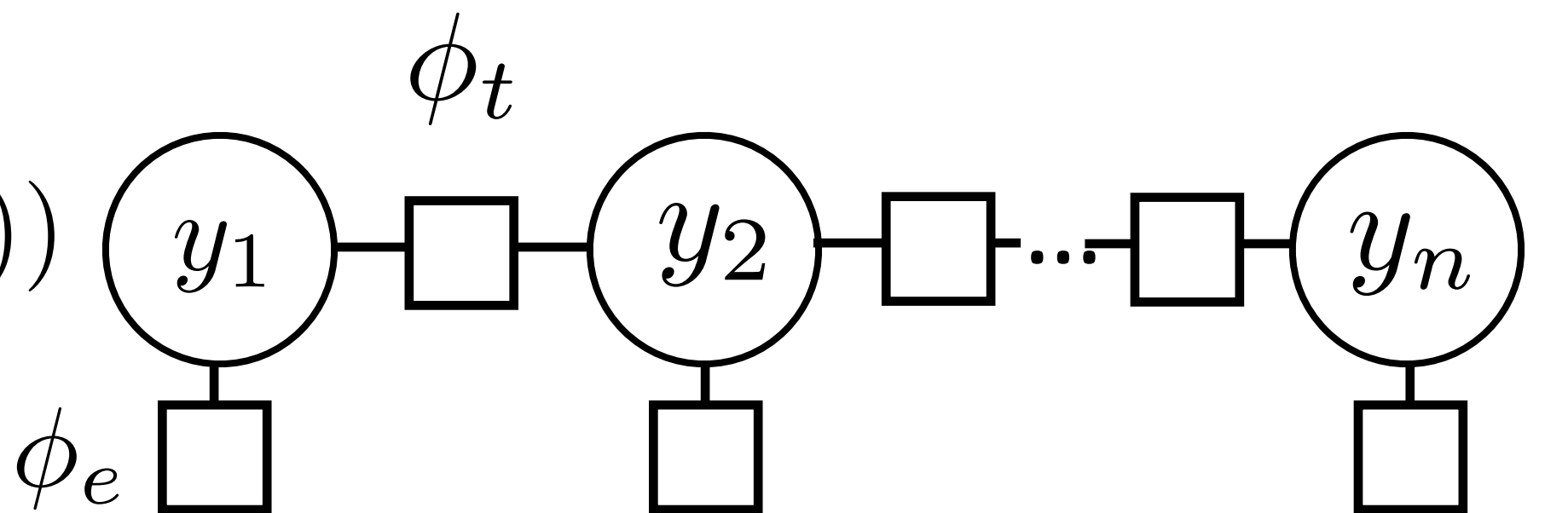
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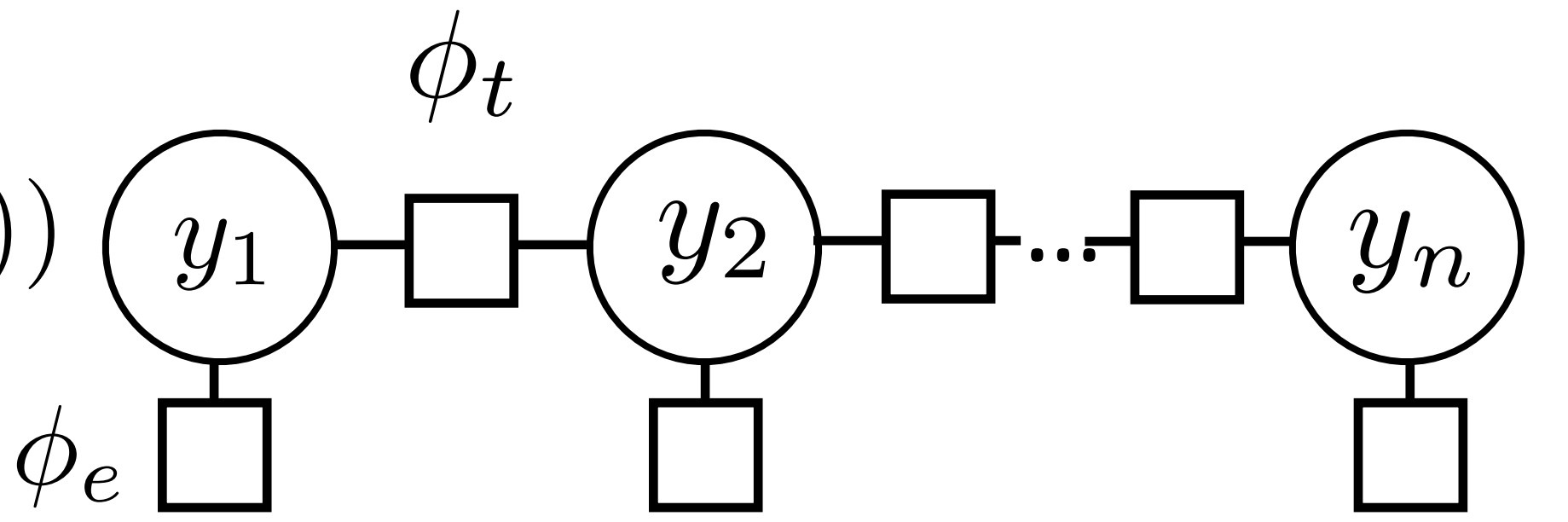
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Computing (arg)maxes

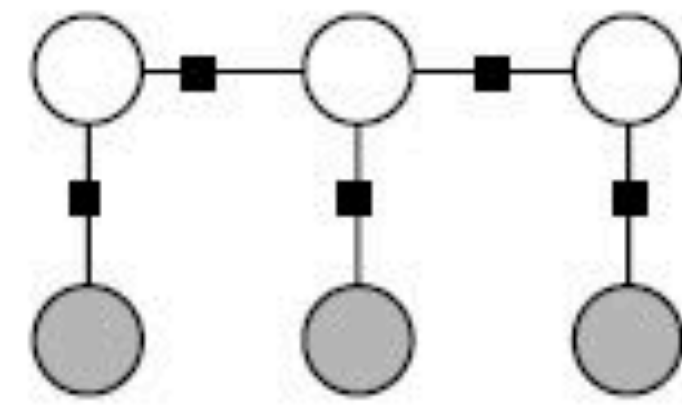
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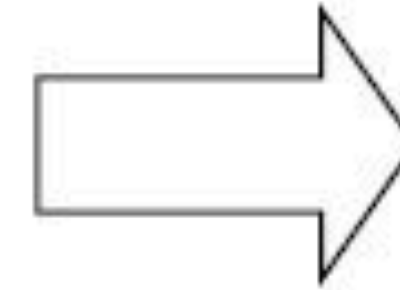
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- $\exp(\phi_t(y_{i-1}, y_i))$ and $\exp(\phi_e(y_i, i, \mathbf{x}))$ play the role of the Ps now, same dynamic program

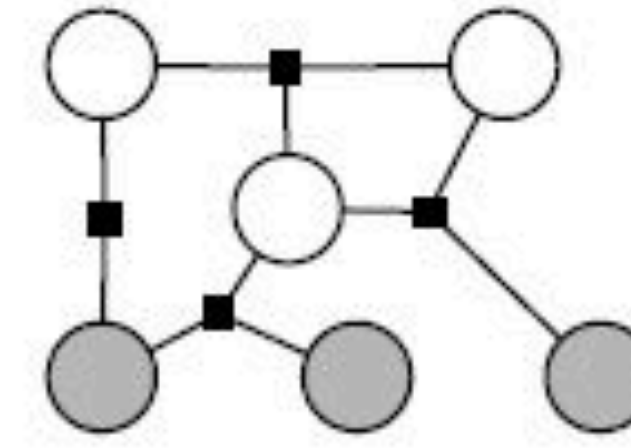
Inference in General CRFs



Linear-chain CRFs

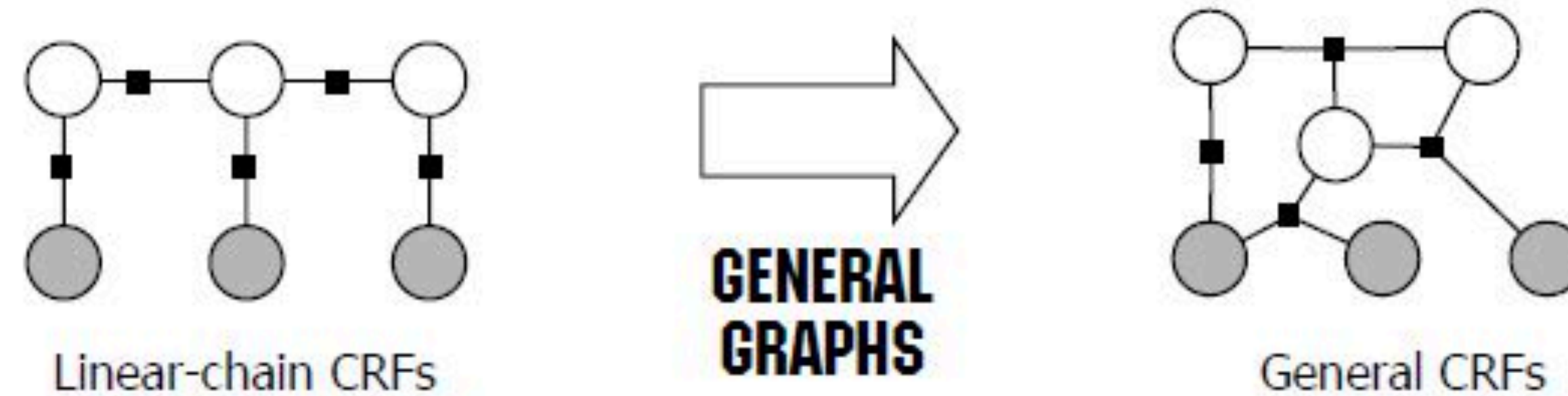


**GENERAL
GRAPHS**



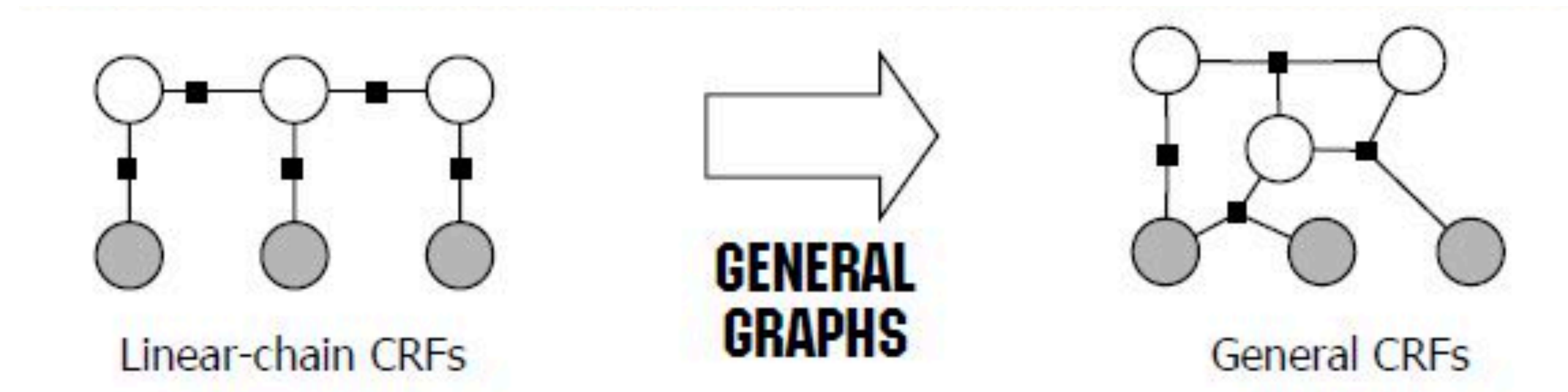
General CRFs

Inference in General CRFs



- ▶ Can do inference in any tree-structured CRF

Inference in General CRFs



- ▶ Can do inference in any tree-structured CRF
- ▶ Max-product algorithm: generalization of Viterbi to arbitrary tree-structured graphs (sum-product is generalization of forward-backward)

CRFs Outline

► Model:
$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$

$$P(\mathbf{y}|\mathbf{x}) \propto \exp w^\top \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$$

- Inference: $\operatorname{argmax} P(\mathbf{y}|\mathbf{x})$ from Viterbi
- Learning

Training CRFs

$$P(\mathbf{y}|\mathbf{x}) \propto \exp w^\top \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$$

Training CRFs

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$$\begin{aligned} \frac{\partial}{\partial w} \mathcal{L}(\mathbf{y}^*, \mathbf{x}) = & \sum_{i=2}^n f_t(y_{i-1}^*, y_i^*) + \sum_{i=1}^n f_e(y_i^*, i, \mathbf{x}) \\ & - \mathbb{E}_{\mathbf{y}} \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right] \end{aligned}$$

Training CRFs

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intractable! $\nearrow -\mathbb{E}_{\mathbf{y}} \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$

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- ▶ Let's focus on emission feature expectation

Training CRFs

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$$\mathbb{E}_{\mathbf{y}} \left[\sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right] = \sum_{\mathbf{y} \in \mathcal{Y}} P(\mathbf{y} | \mathbf{x}) \left[\sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$$

Training CRFs

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Training CRFs

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Forward-Backward Algorithm

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- ▶ How do we compute these marginals $P(y_i = s | \mathbf{x})$?

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- ▶ What did Viterbi compute? $P(\mathbf{y}_{\max} | \mathbf{x}) = \max_{y_1, \dots, y_n} P(\mathbf{y} | \mathbf{x})$

Forward-Backward Algorithm

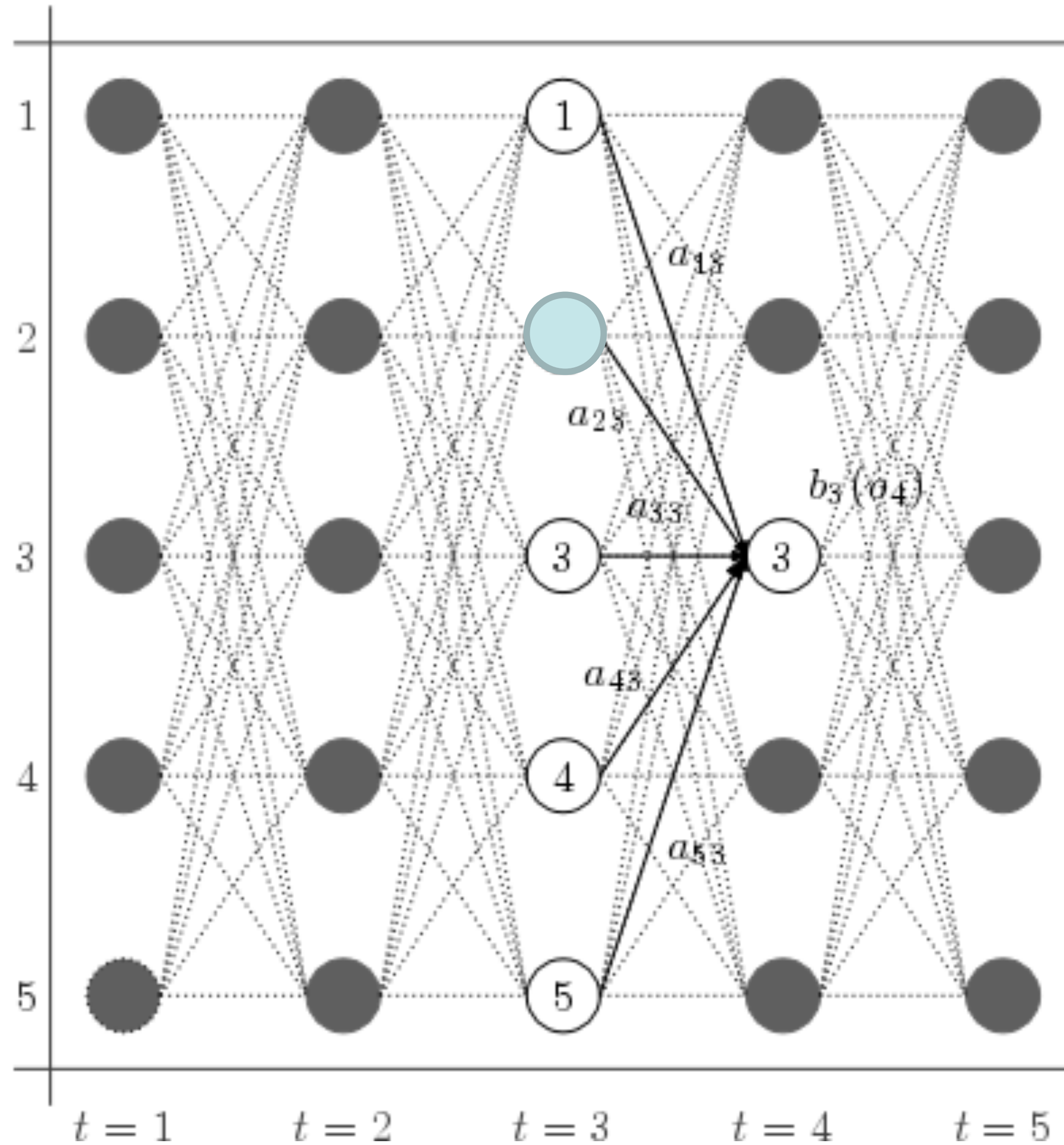
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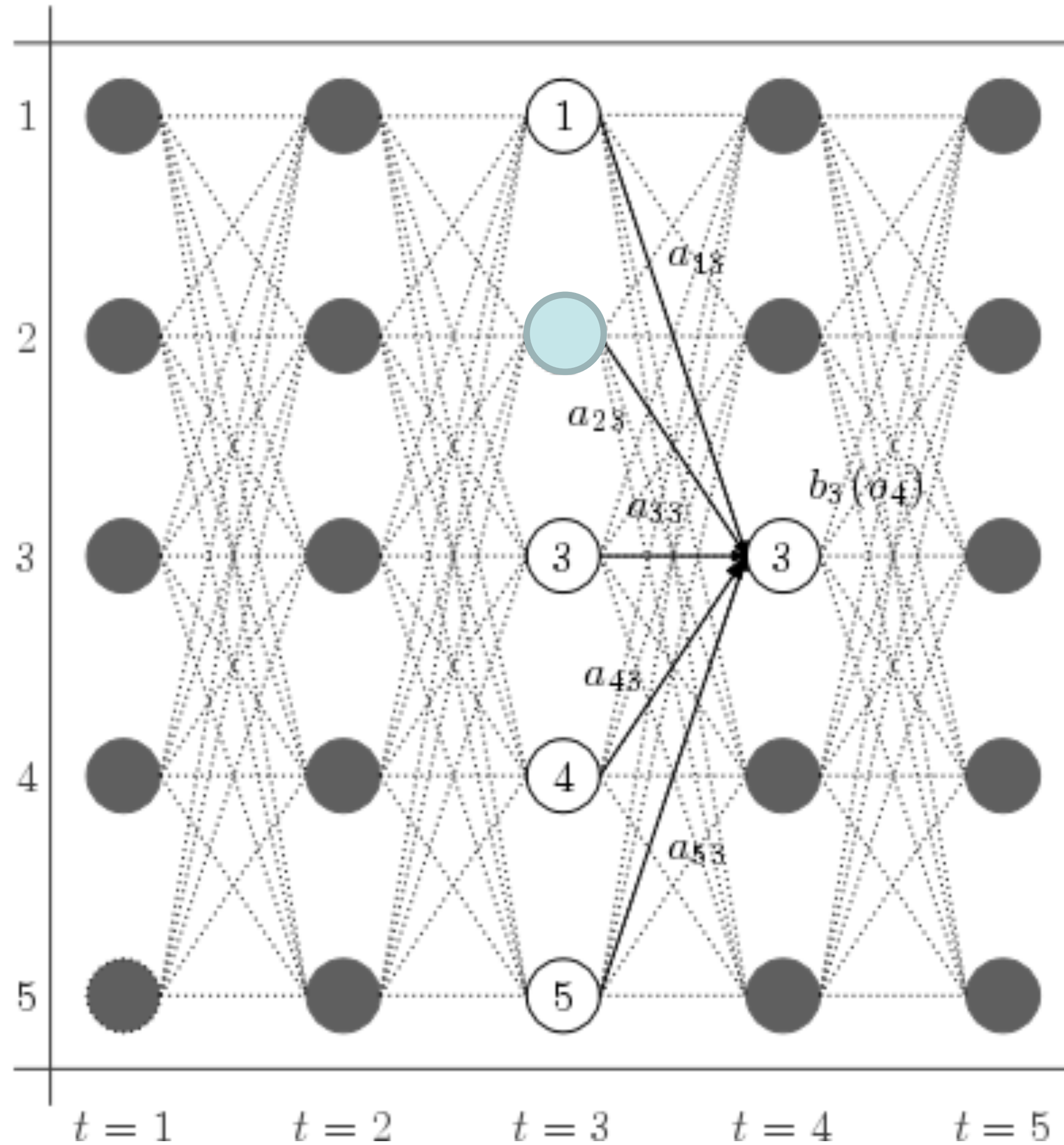
- ▶ What did Viterbi compute? $P(\mathbf{y}_{\max} | \mathbf{x}) = \max_{y_1, \dots, y_n} P(\mathbf{y} | \mathbf{x})$

- ▶ Can compute marginals with dynamic programming as well using the forward-backward algorithm

Forward-Backward Algorithm

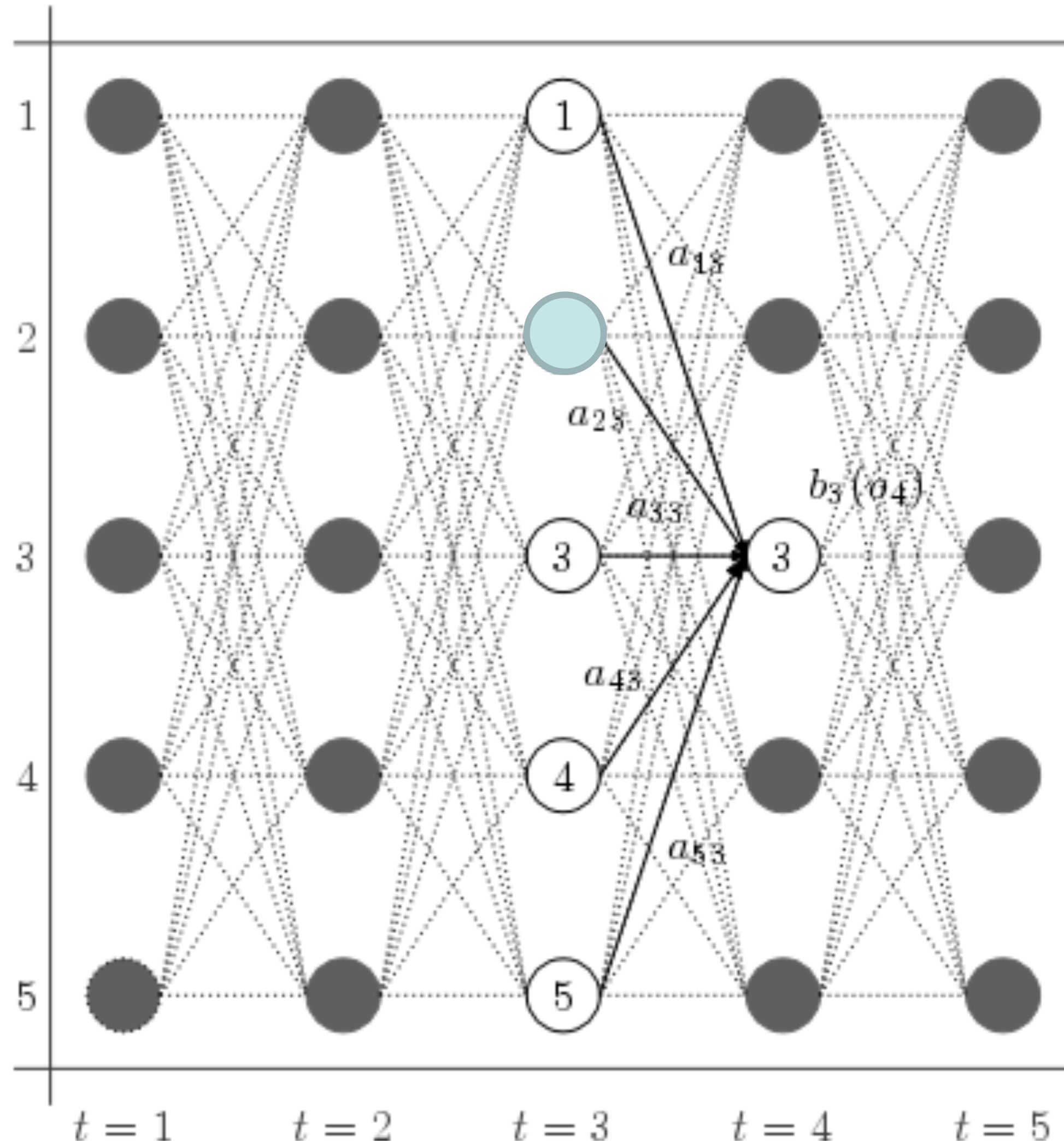


Forward-Backward Algorithm



$$P(y_3 = 2 | \mathbf{x}) =$$

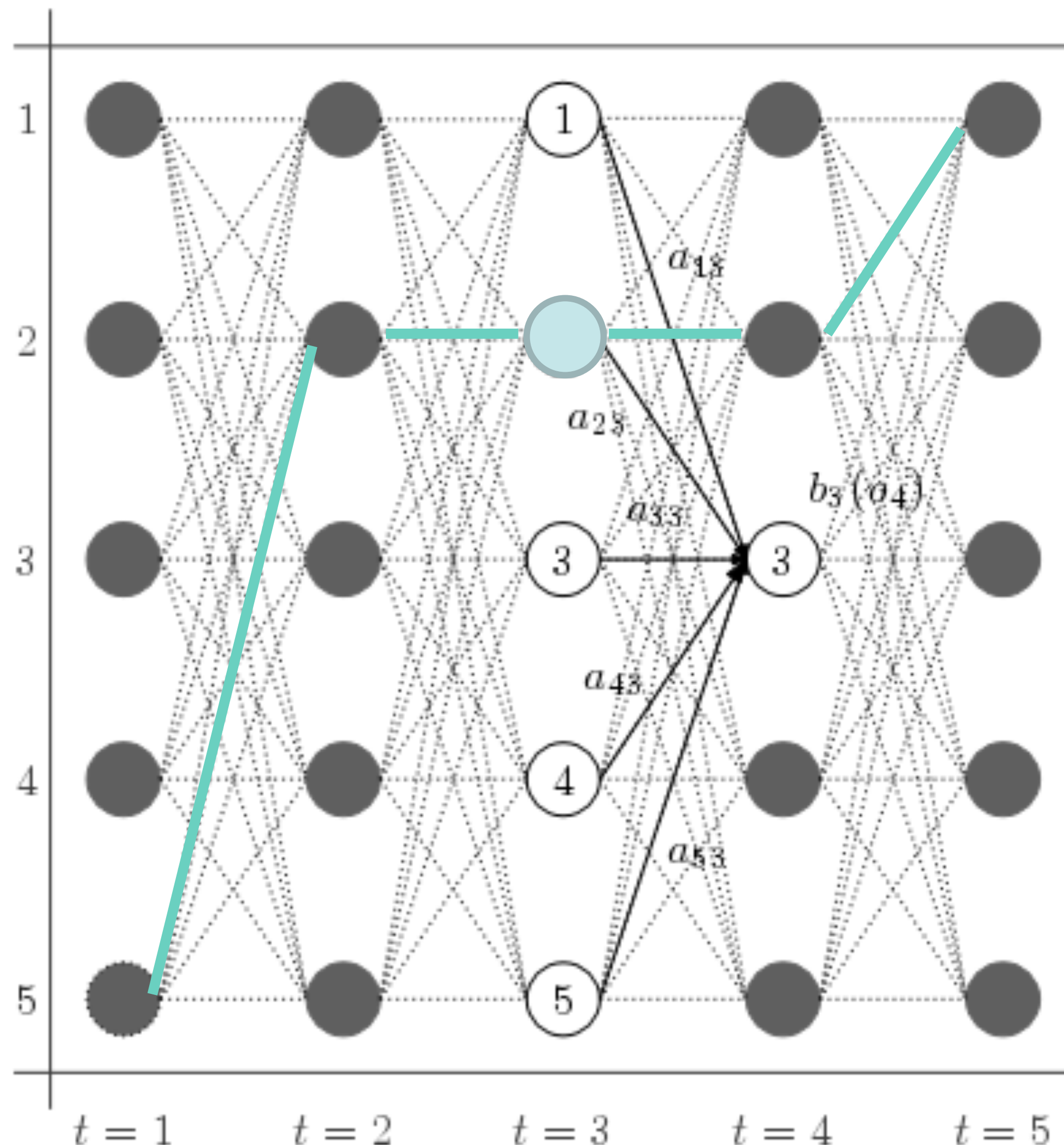
Forward-Backward Algorithm



$$P(y_3 = 2 | \mathbf{x}) =$$

$$\frac{\text{sum of all paths through state 2 at time 3}}{\text{sum of all paths}}$$

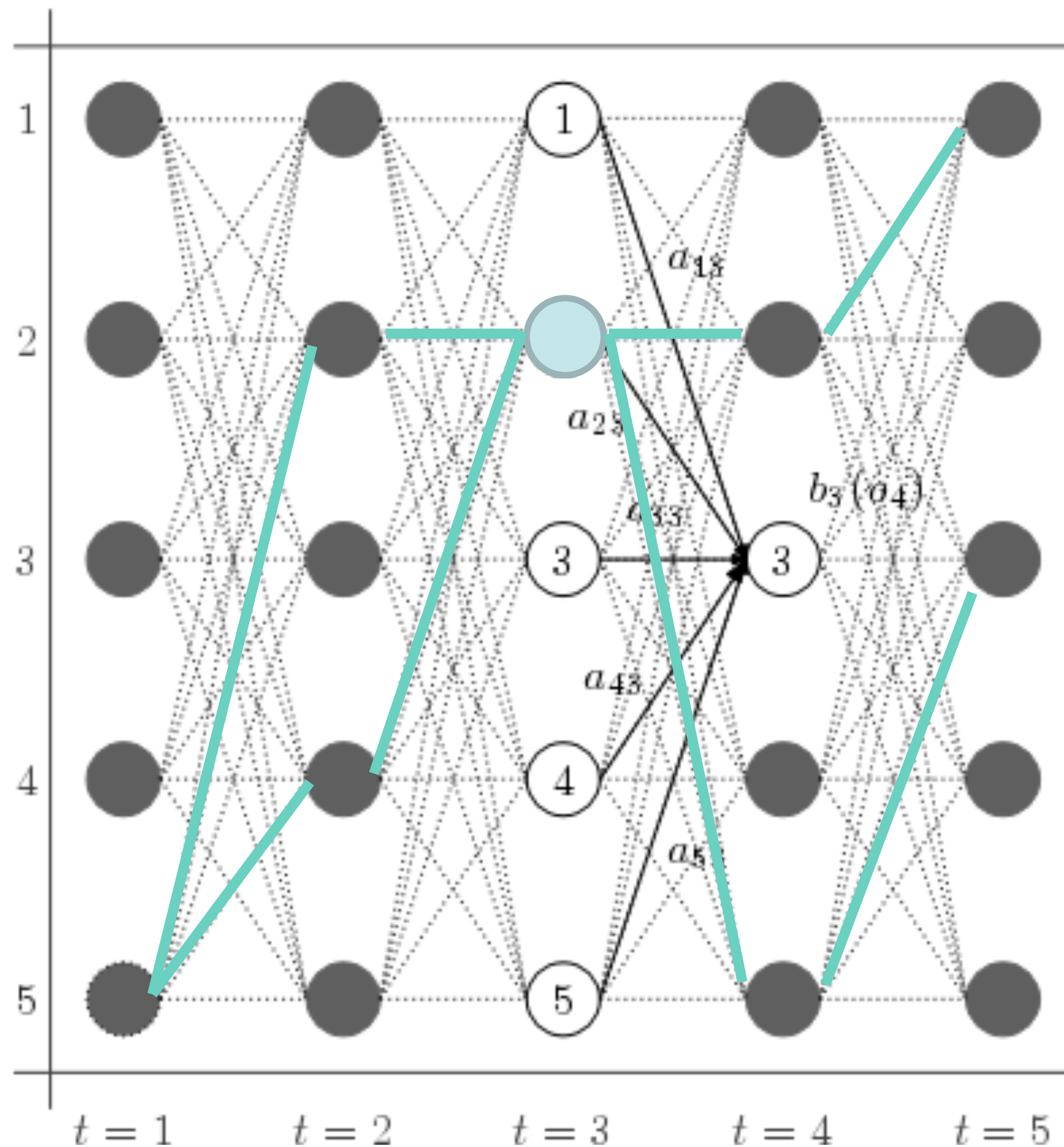
Forward-Backward Algorithm



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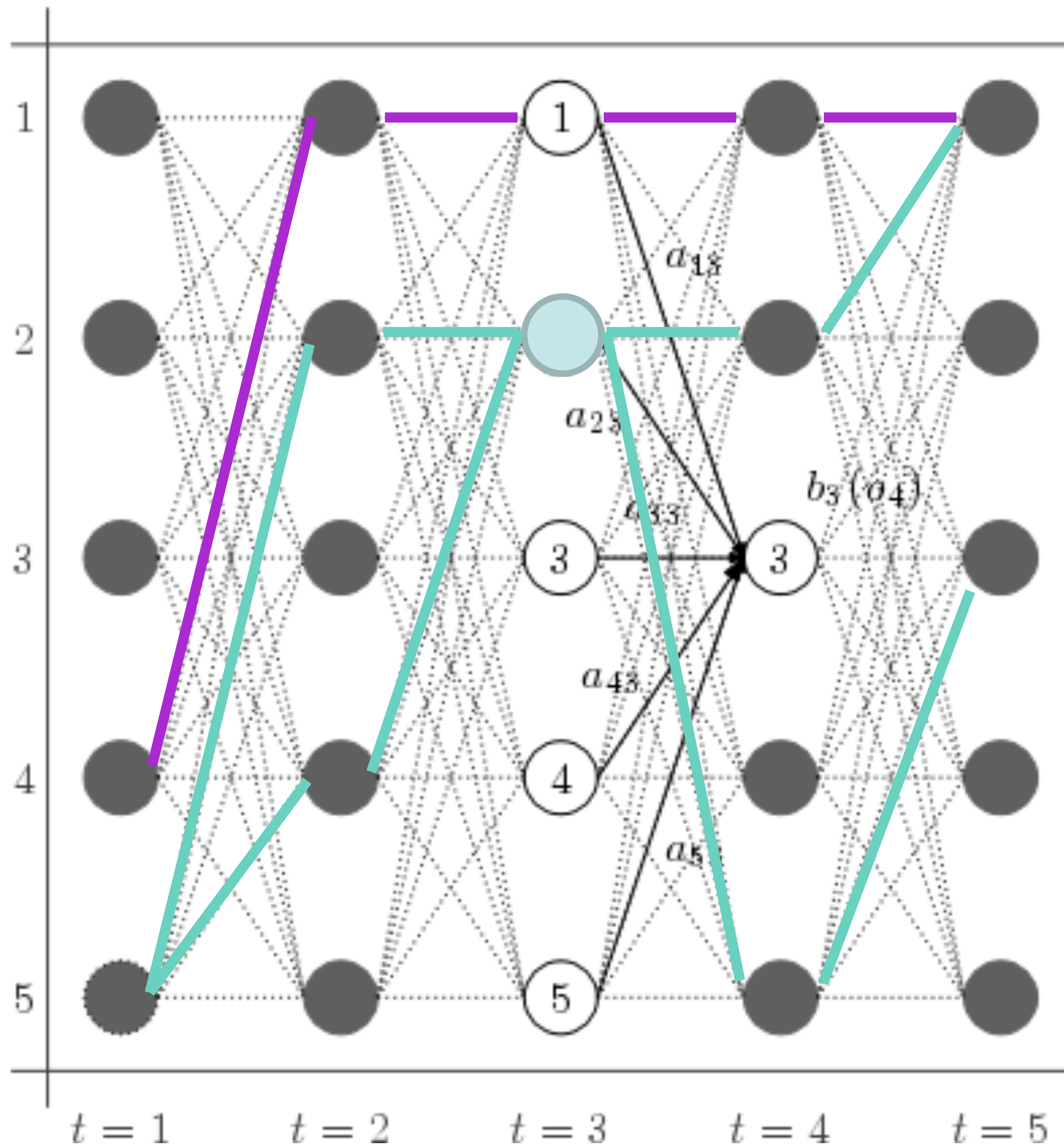
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Forward-Backward Algorithm



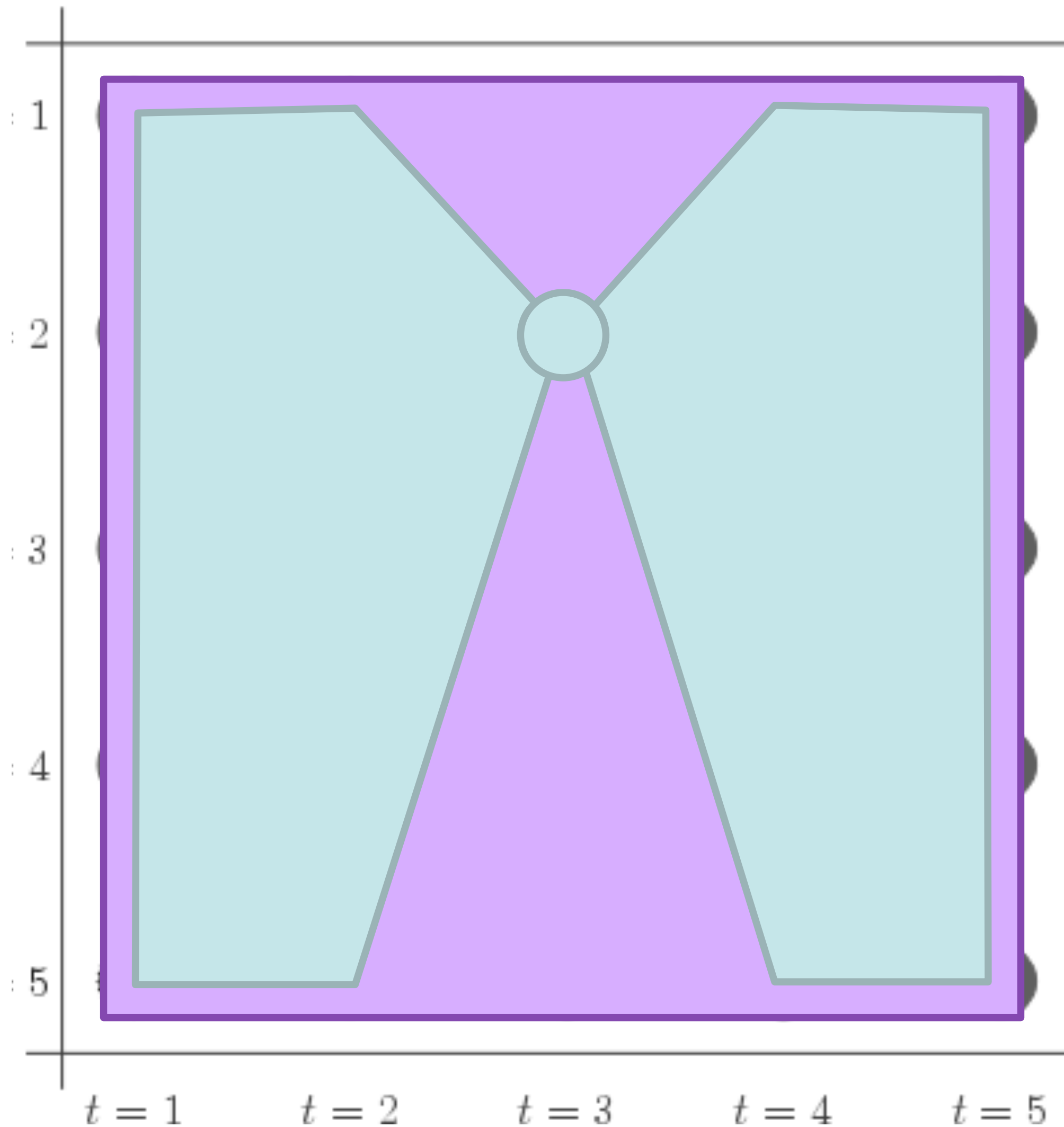
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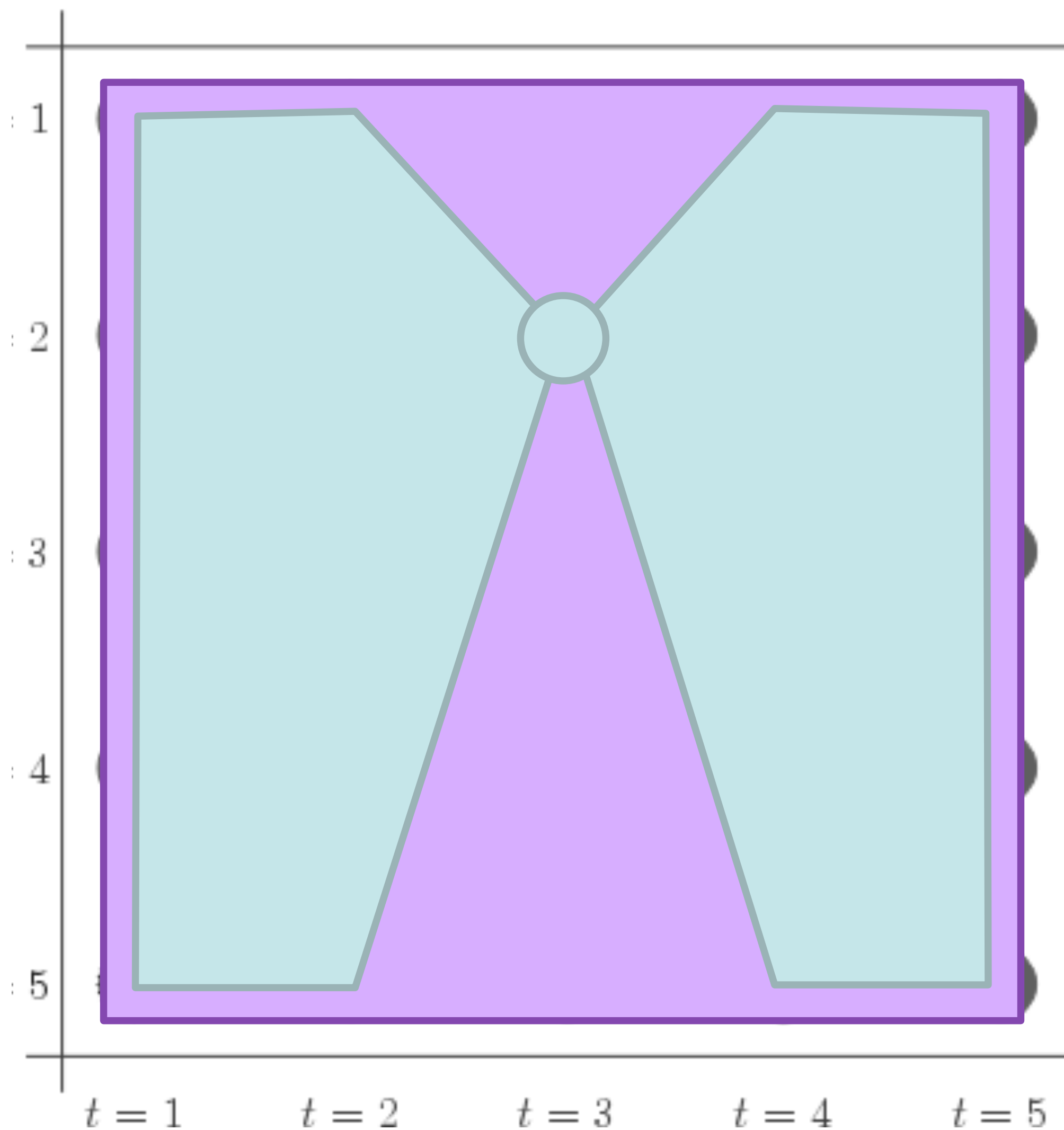
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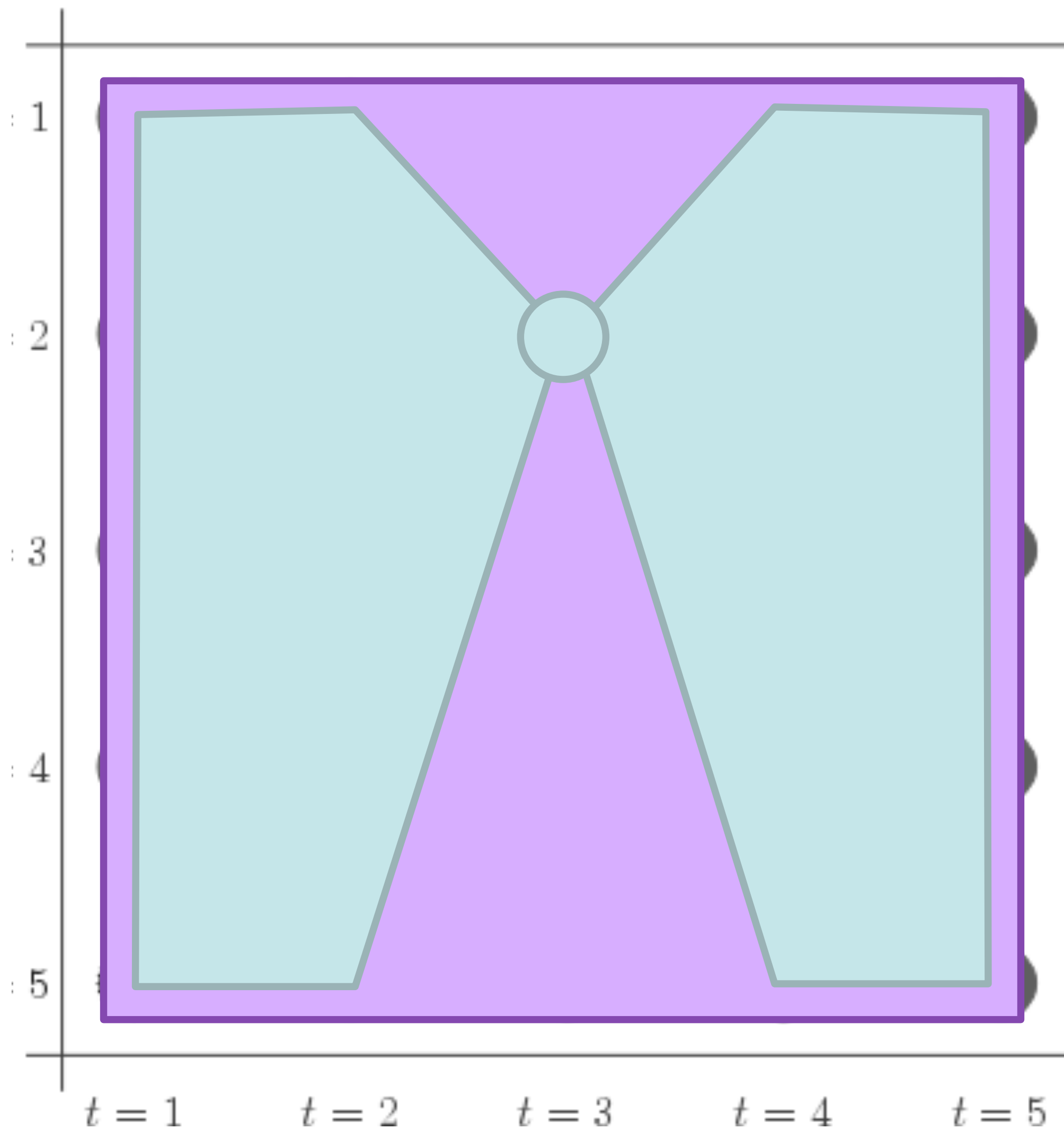


$$P(y_3 = 2 | \mathbf{x}) =$$

$$\frac{\text{sum of all paths through state 2 at time 3}}{\text{sum of all paths}}$$

$$= \frac{\text{light blue path diagram}}{\text{purple square diagram}}$$

Forward-Backward Algorithm



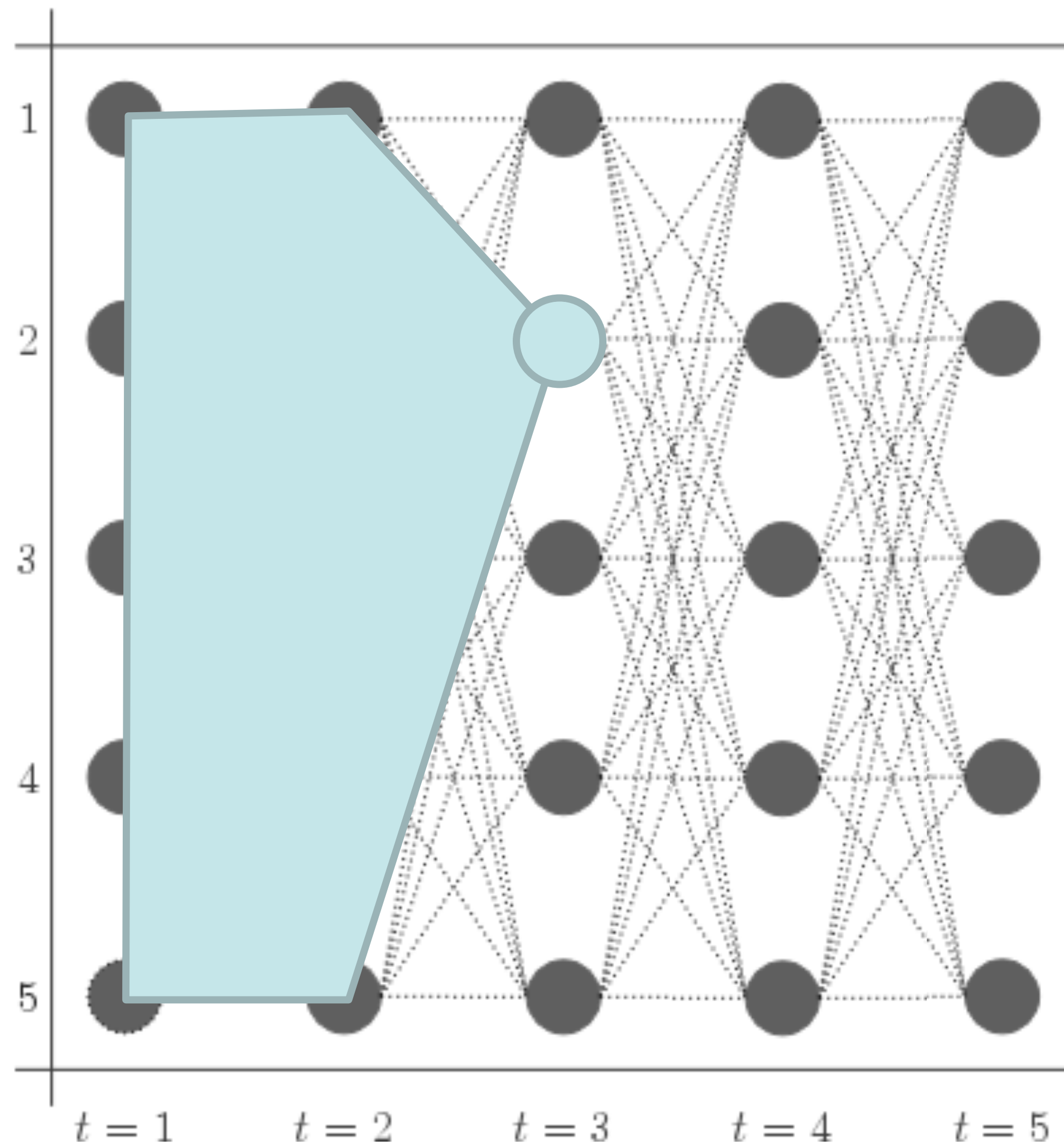
$$P(y_3 = 2 | \mathbf{x}) =$$

sum of all paths through state 2 at time 3
sum of all paths

$$= \frac{\text{light blue shape}}{\text{purple shape}}$$

- ▶ Easiest and most flexible to do one pass to compute and one to compute

Forward-Backward Algorithm



- ▶ Initial:

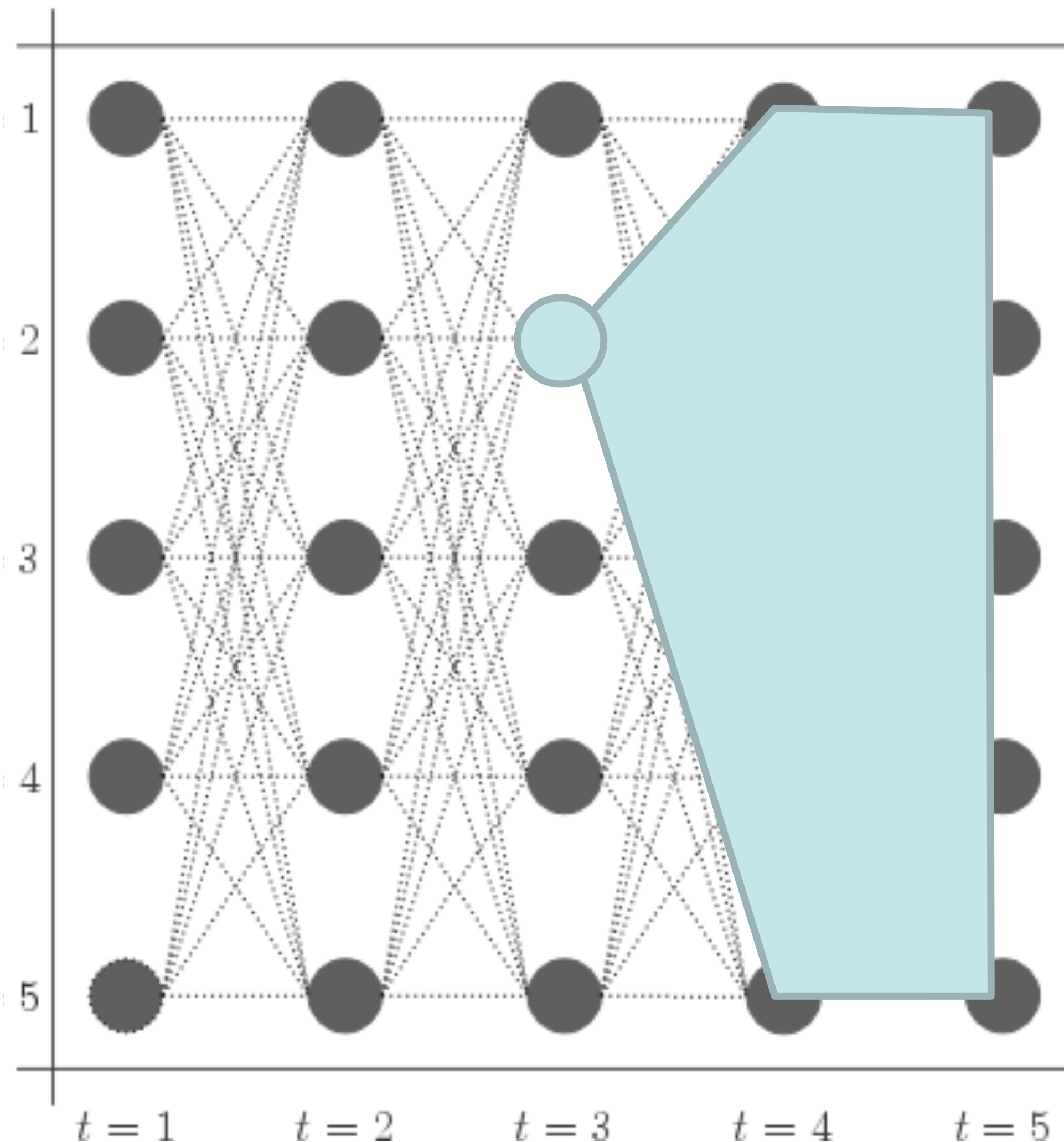
$$\alpha_1(s) = \exp(\phi_e(s, 1, \mathbf{x}))$$

- ▶ Recurrence:

$$\alpha_t(s_t) = \sum_{s_{t-1}} \alpha_{t-1}(s_{t-1}) \exp(\phi_e(s_t, t, \mathbf{x})) \exp(\phi_t(s_{t-1}, s_t))$$

- ▶ Same as Viterbi but summing instead of maxing!
- ▶ These quantities get very small!
Store everything as log probabilities

Forward-Backward Algorithm



- ▶ Initial:

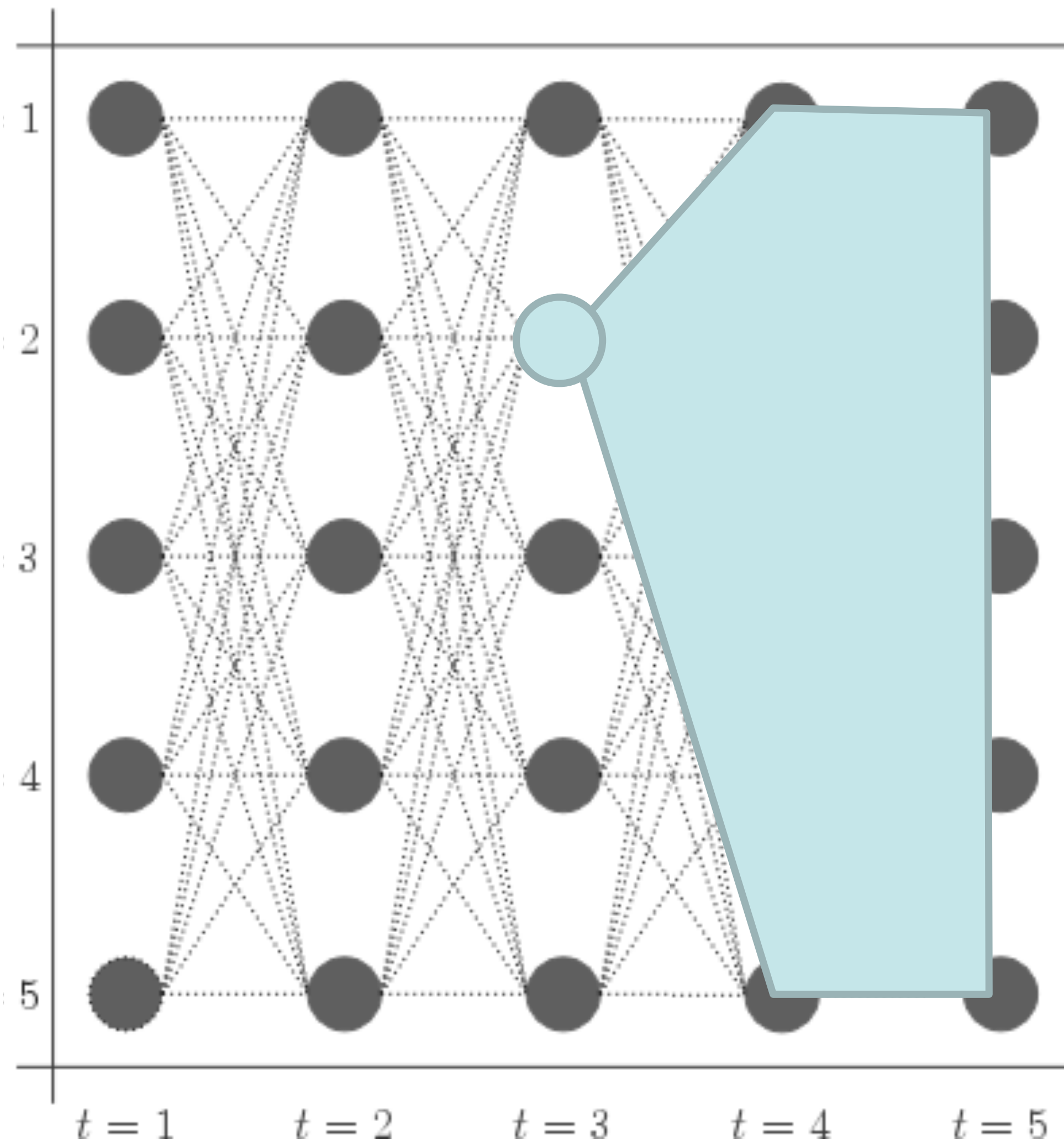
$$\beta_n(s) = 1$$

- ▶ Recurrence:

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- ▶ Big differences: count emission for the *next* timestep (not current one)

Forward-Backward Algorithm



- Initial:

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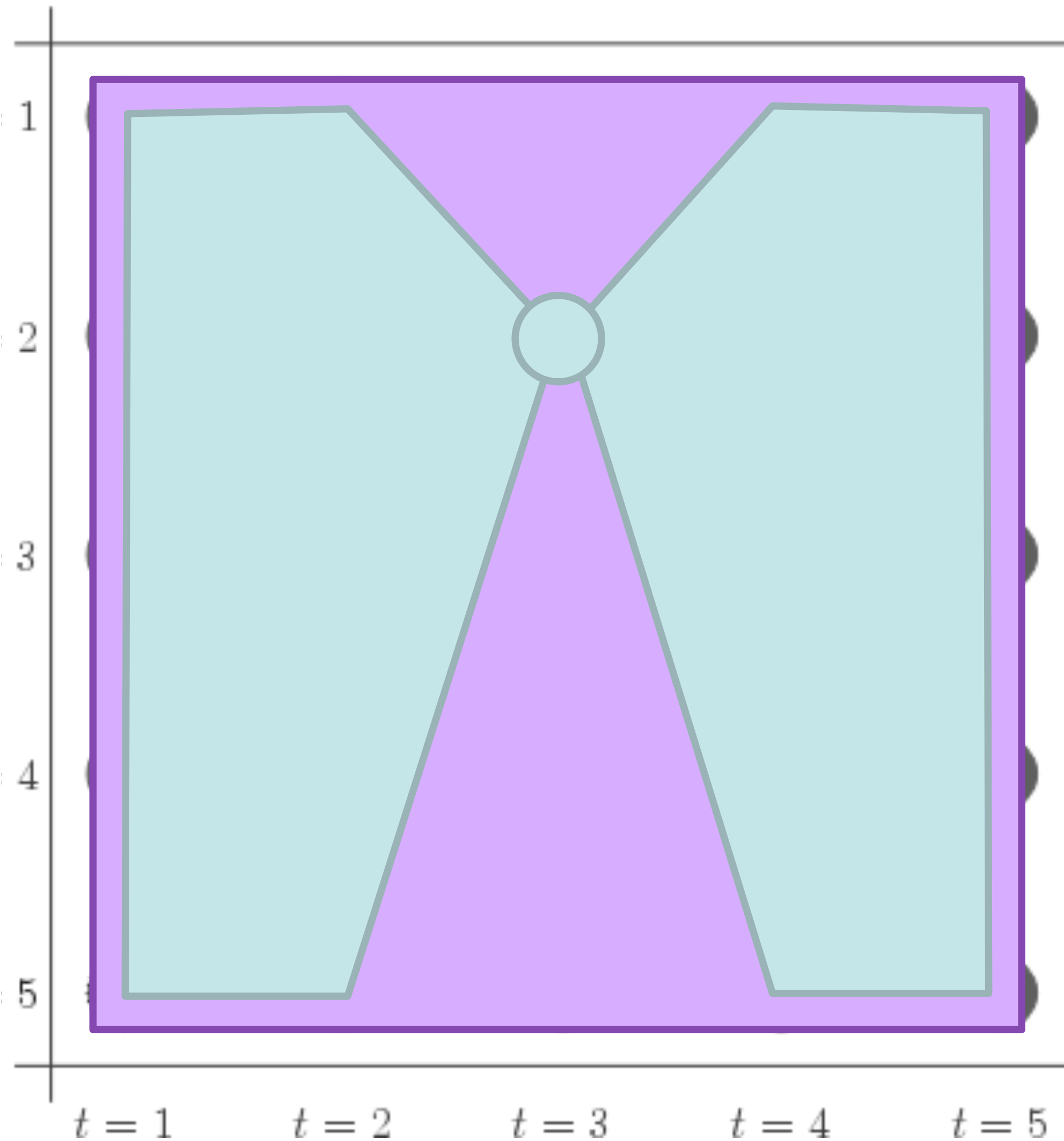
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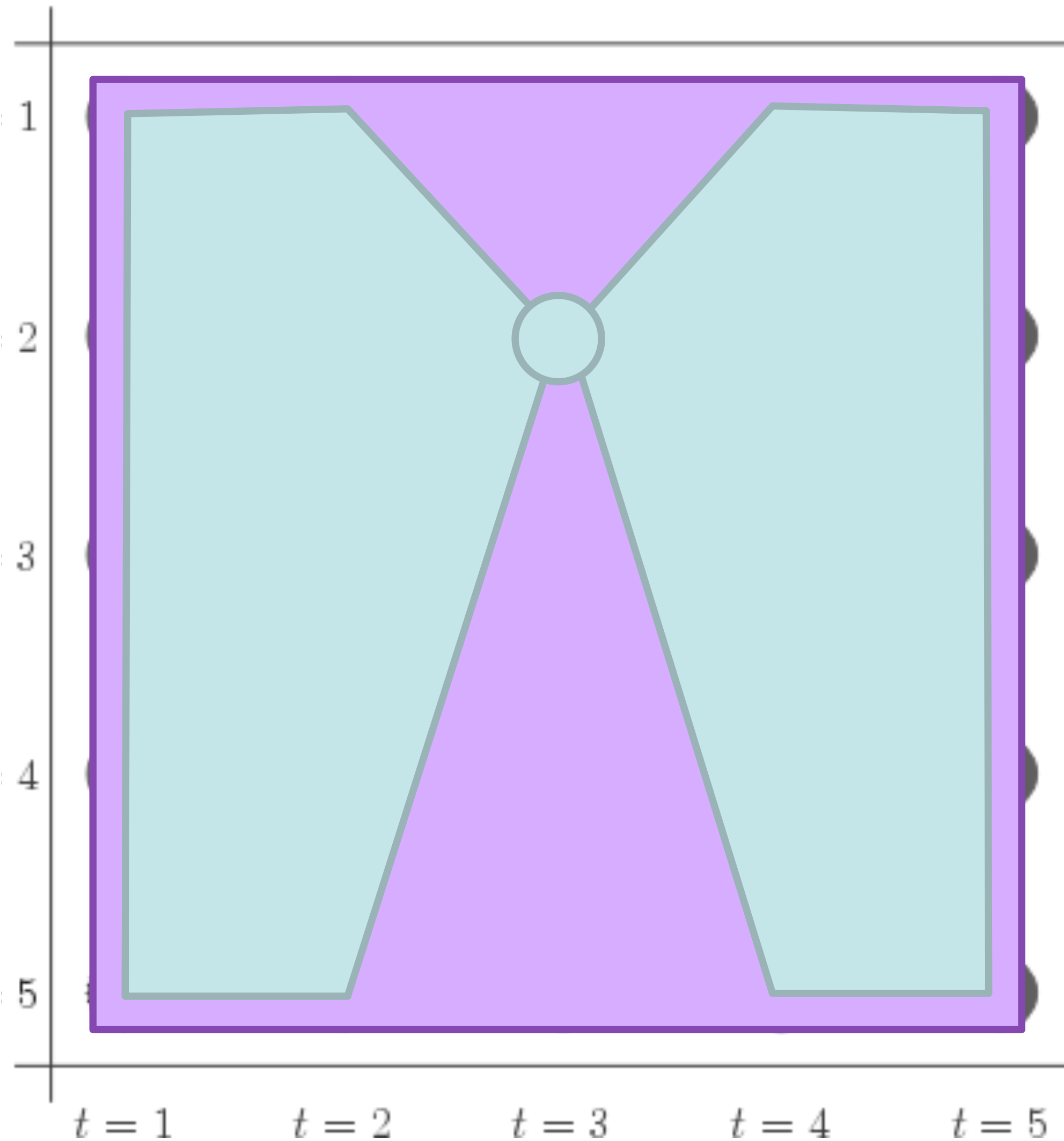
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$$\beta_n(s) = 1$$

$$\beta_t(s_t) = \sum_{s_{t+1}} \beta_{t+1}(s_{t+1}) \exp(\phi_e(s_{t+1}, t+1, \mathbf{x})) \exp(\phi_t(s_t, s_{t+1}))$$

$$P(s_3 = 2 | \mathbf{x}) = \frac{\alpha_3(2)\beta_3(2)}{\sum_i \alpha_3(i)\beta_3(i)} = \frac{\text{Diagram of state 2 at } t=3}{\text{Diagram of all states at } t=3}$$



Forward-Backward Algorithm

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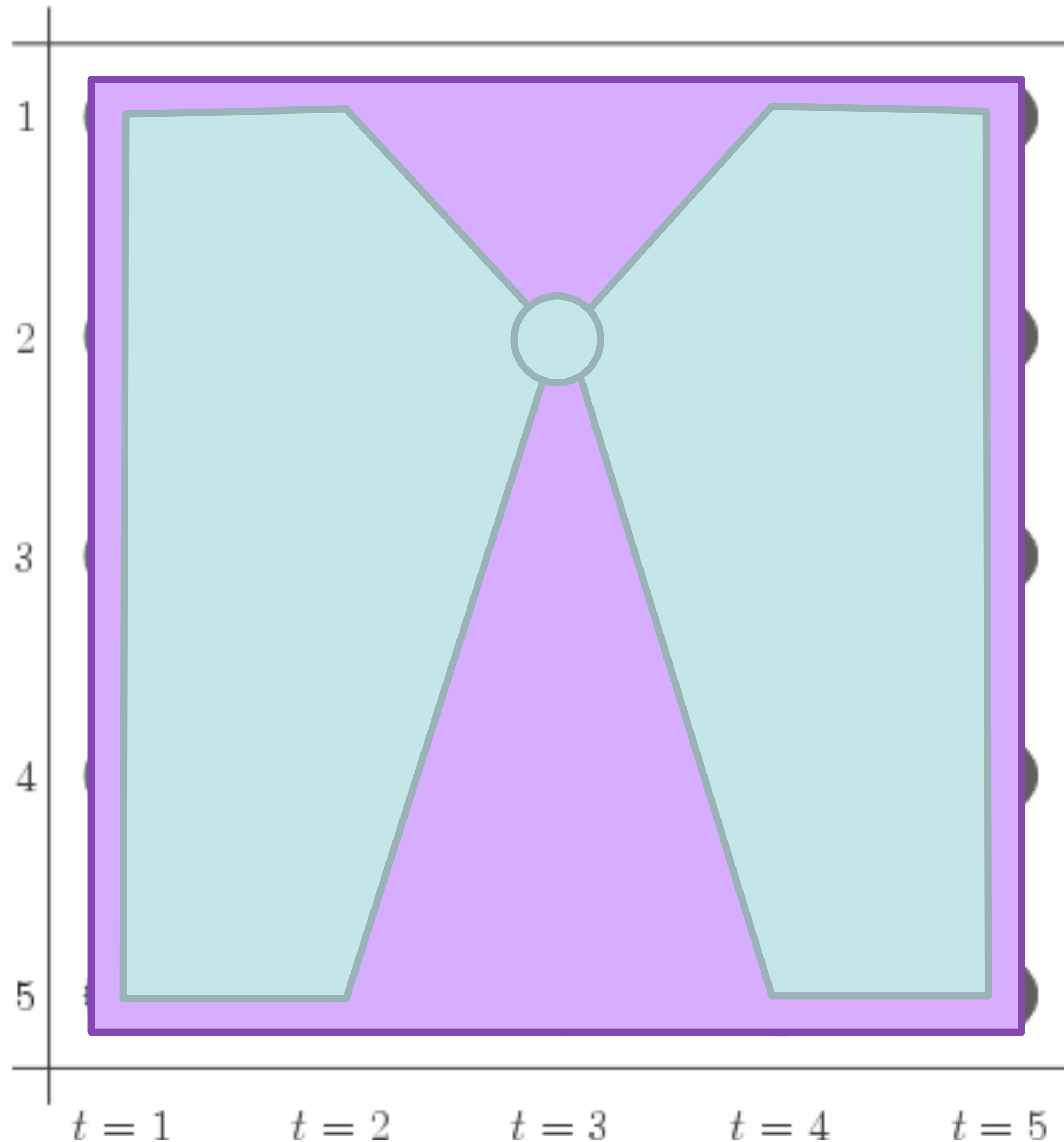
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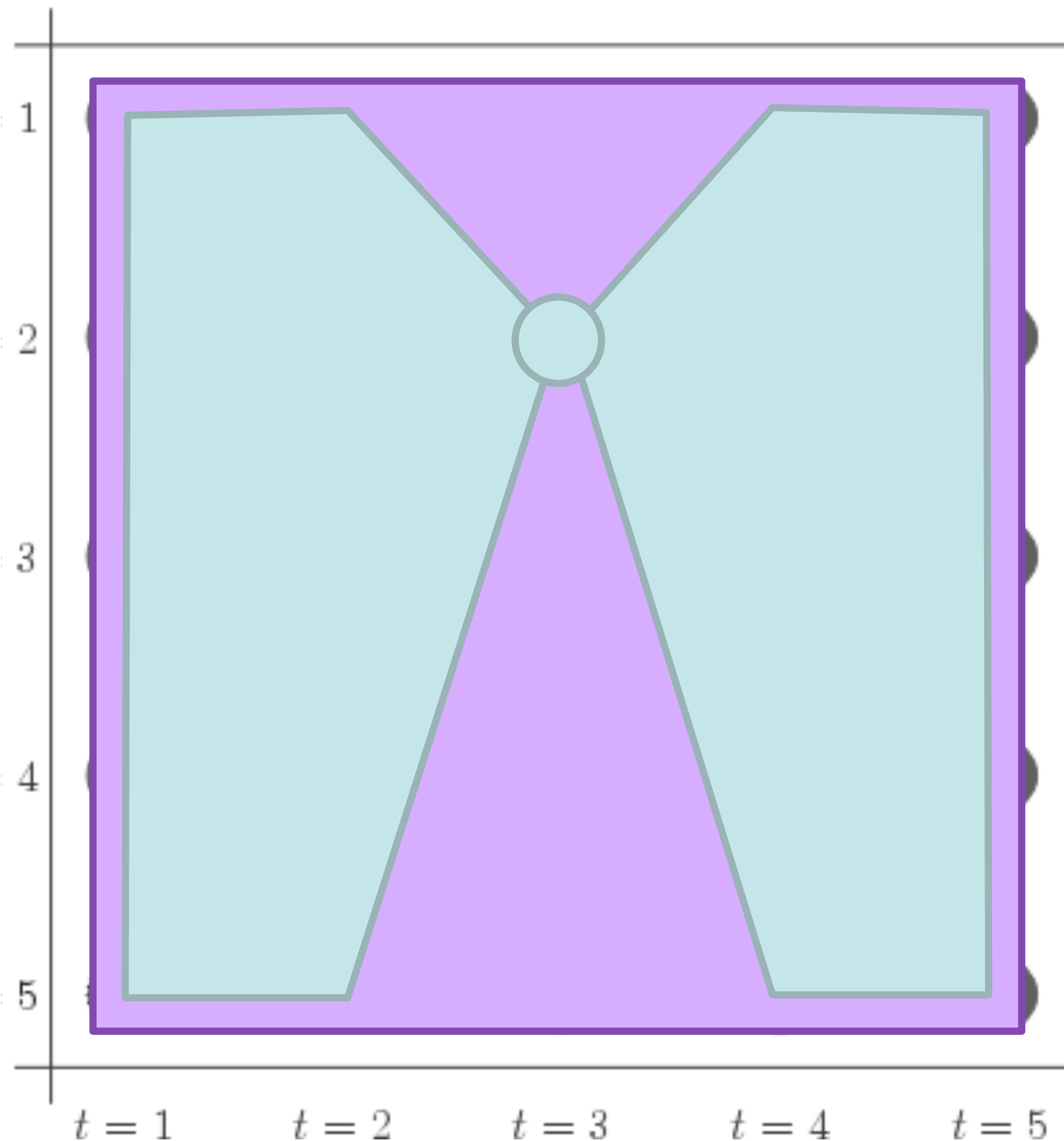
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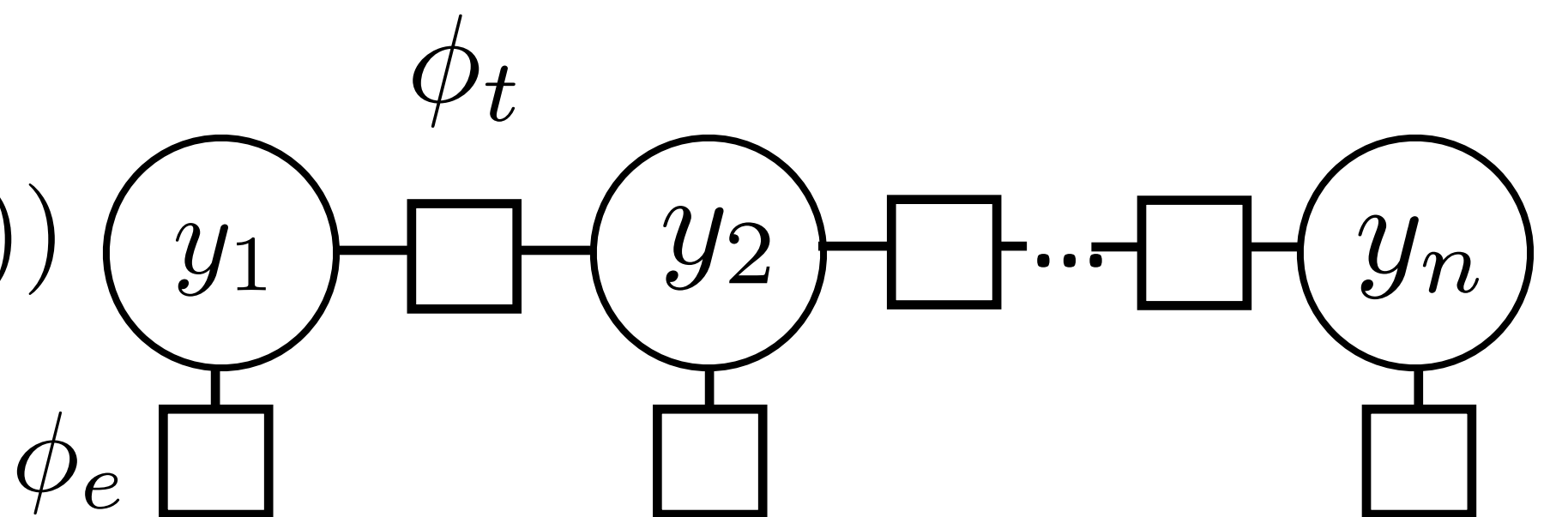
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$$P(s_3 = 2 | \mathbf{x}) = \frac{\alpha_3(2)\beta_3(2)}{\sum_i \alpha_3(i)\beta_3(i)} = \frac{\text{light blue shape}}{\text{purple shape}}$$

► What is the denominator here? Z



Computing Marginals

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$


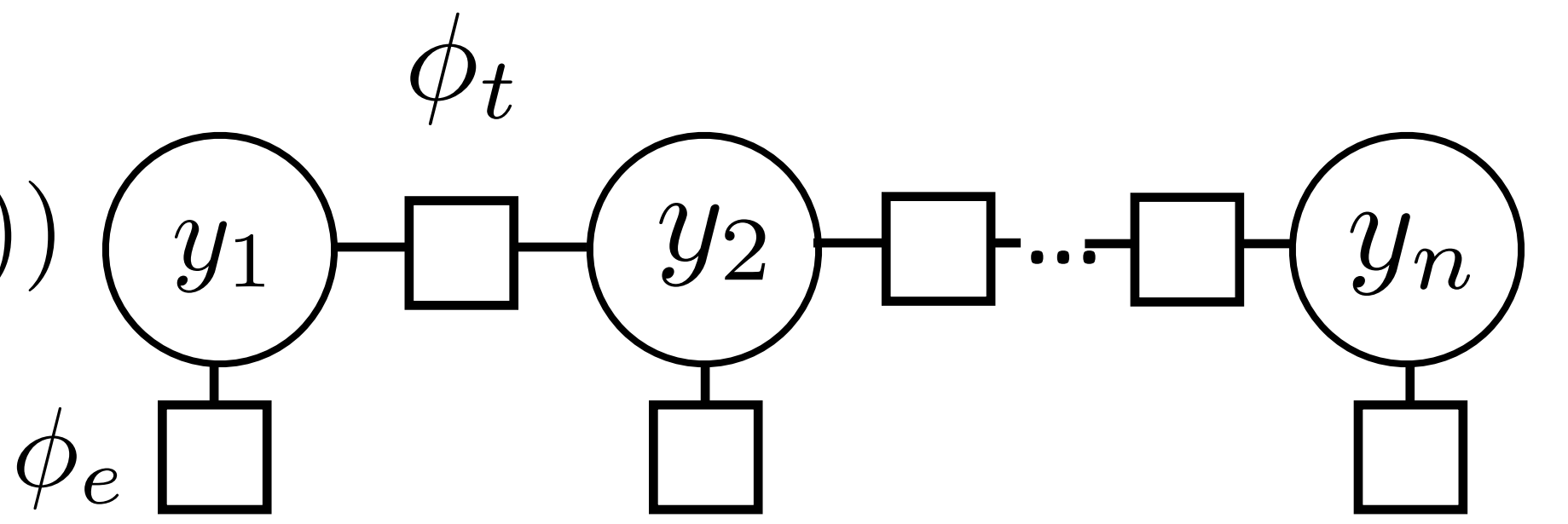
The diagram illustrates a graphical model for a sequence of variables $y_1, y_2, \dots, y_n$. The variables are represented by circles. The first two variables, y_1 and y_2 , are explicitly labeled, and the last variable, y_n , is also labeled. Between y_1 and y_2 , and between y_2 and the next node, there are square nodes representing parameters. The label ϕ_t is placed above the first square node, and ϕ_e is placed below the first square node. The diagram shows a chain of nodes connected by horizontal lines, with vertical lines connecting the circles to the squares below them. Ellipses indicate the continuation of the sequence.

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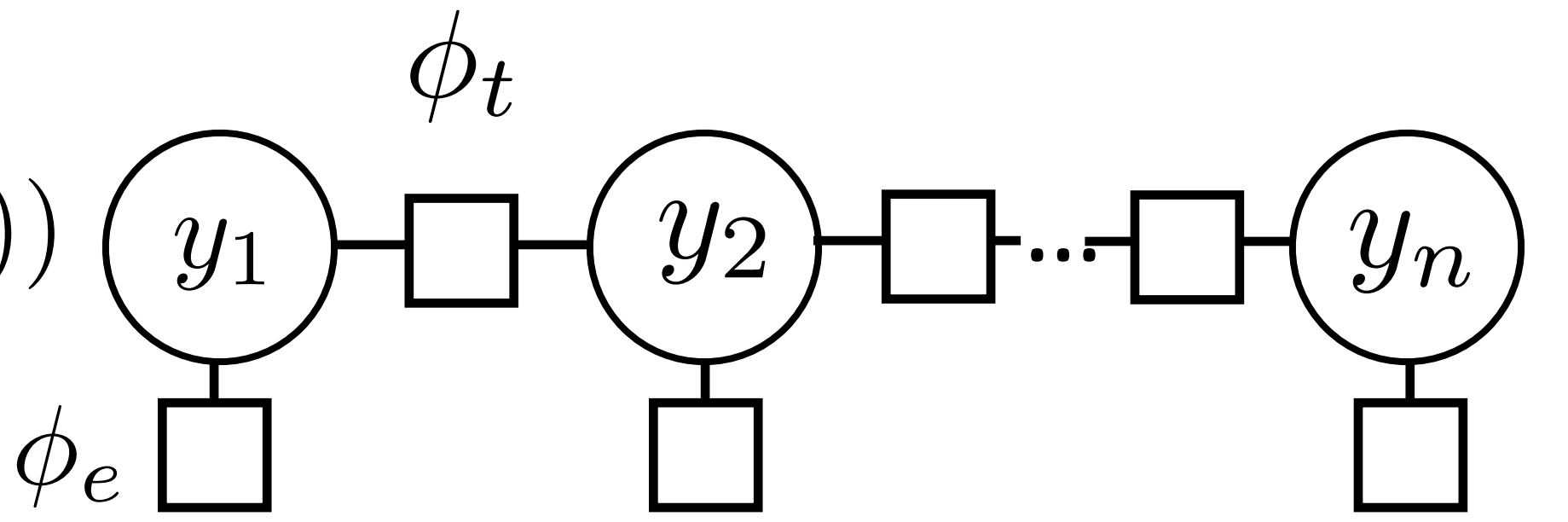
► Normalizing constant $Z = \sum_{\mathbf{y}} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$

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- ▶ For both HMMs and CRFs:

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Z for CRFs,
P(**x**) for HMMs

Training CRFs

- For emission features:

$$\frac{\partial}{\partial w} \mathcal{L}(\mathbf{y}^*, \mathbf{x}) = \sum_{i=1}^n f_e(y_i^*, i, \mathbf{x}) - \sum_{i=1}^n \sum_s P(y_i = s | \mathbf{x}) f_e(s, i, \mathbf{x})$$

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- ▶ ... but, you can build a pretty good system without learned transition features (e.g., use heuristic weights, or just enforce constraints like B-PER → I-ORG is illegal)

CRFs Outline

► Model:
$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$

$$P(\mathbf{y}|\mathbf{x}) \propto \exp w^\top \left[\sum_{i=2}^n f_t(y_{i-1}, y_i) + \sum_{i=1}^n f_e(y_i, i, \mathbf{x}) \right]$$

- Inference: $\operatorname{argmax} P(\mathbf{y}|\mathbf{x})$ from Viterbi
- Learning: run forward-backward to compute posterior probabilities; then

$$\frac{\partial}{\partial w} \mathcal{L}(\mathbf{y}^*, \mathbf{x}) = \sum_{i=1}^n f_e(y_i^*, i, \mathbf{x}) - \sum_{i=1}^n \sum_s P(y_i = s | \mathbf{x}) f_e(s, i, \mathbf{x})$$

Pseudocode

for each epoch

 for each example

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- apply the gradient update

Implementation Tips for CRFs

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- ▶ If things are too slow, run a profiler and see where time is being spent. Forward-backward should take most of the time

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Application in NER

NER Results

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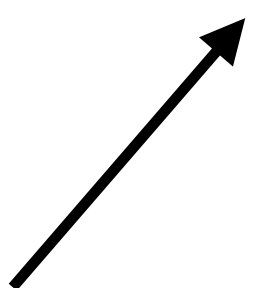
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Tanjug

From Wikipedia, the free encyclopedia

Tanjug (/ˈtʌnjʊɡ/) (Serbian Cyrillic: Танјуг) is a Serbian state news agency based in Belgrade.^[2]

Nonlocal Features

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Nonlocal Features

ORG?

PER?

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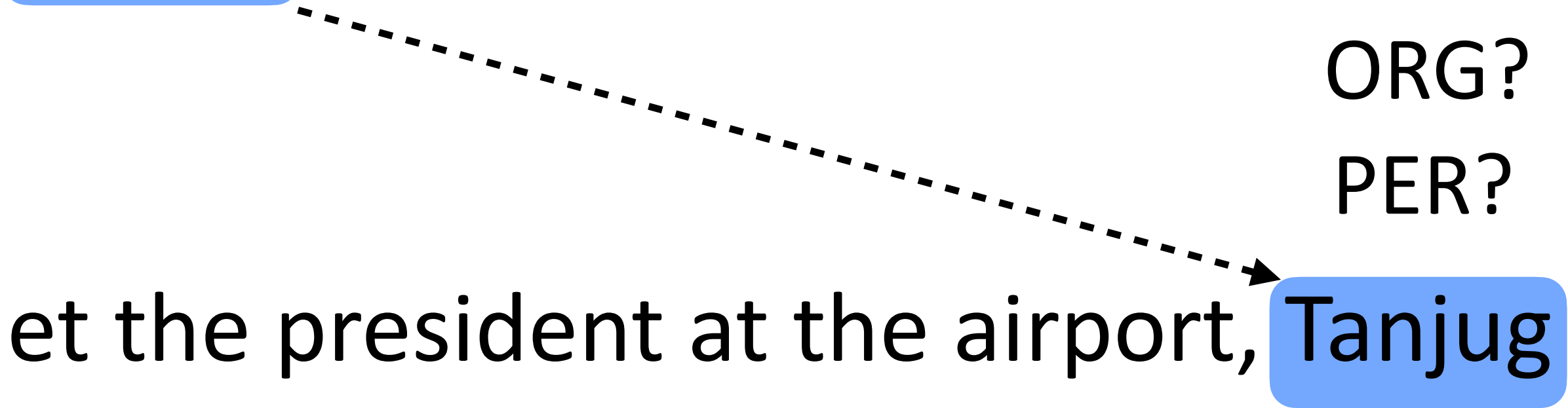
PER?

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- ▶ More complex factor graph structures can let you capture this, or just decode sentences in order and use features on previous sentences

Semi-Markov CRF Models

Barack Obama will travel to Hangzhou today for the G20 meeting .

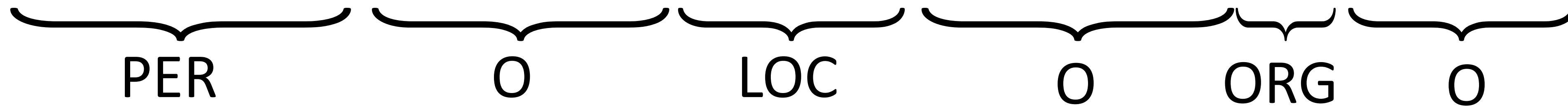
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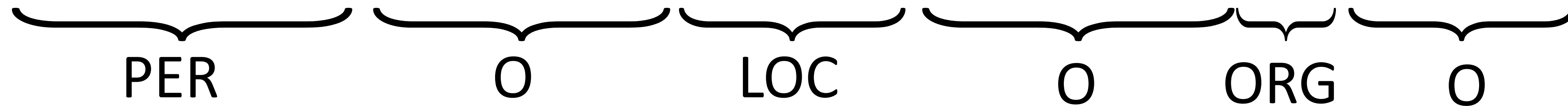
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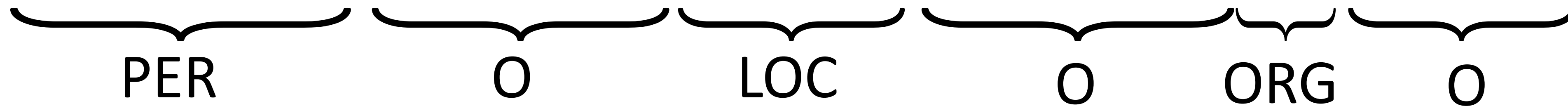
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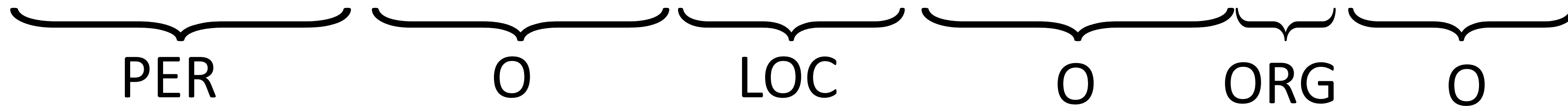
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How well do NER systems do?

	System	Resources Used	F_1
+	LBJ-NER	Wikipedia, Nonlocal Features, Word-class Model	90.80
-	(Suzuki and Isozaki, 2008)	Semi-supervised on 1G-word unlabeled data	89.92
-	(Ando and Zhang, 2005)	Semi-supervised on 27M-word unlabeled data	89.31
-	(Kazama and Torisawa, 2007a)	Wikipedia	88.02
-	(Krishnan and Manning, 2006)	Non-local Features	87.24
-	(Kazama and Torisawa, 2007b)	Non-local Features	87.17
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Ratinov and Roth (2009)

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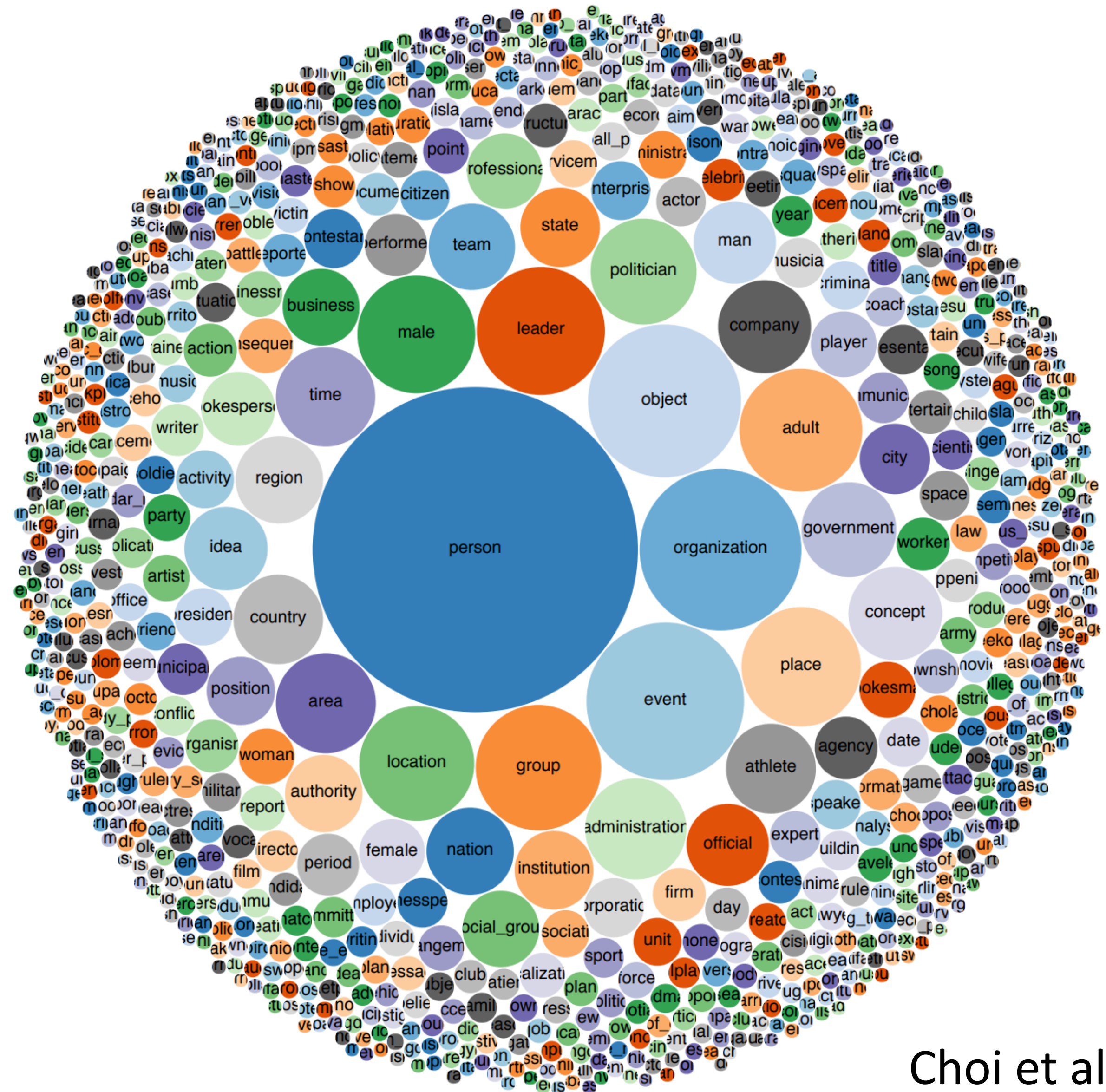
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Fine-tuning approach	Dev F1	Test F1
BERT _{LARGE}	96.6	92.8
BERT _{BASE}	96.4	92.4
Devlin et al. (2019)		

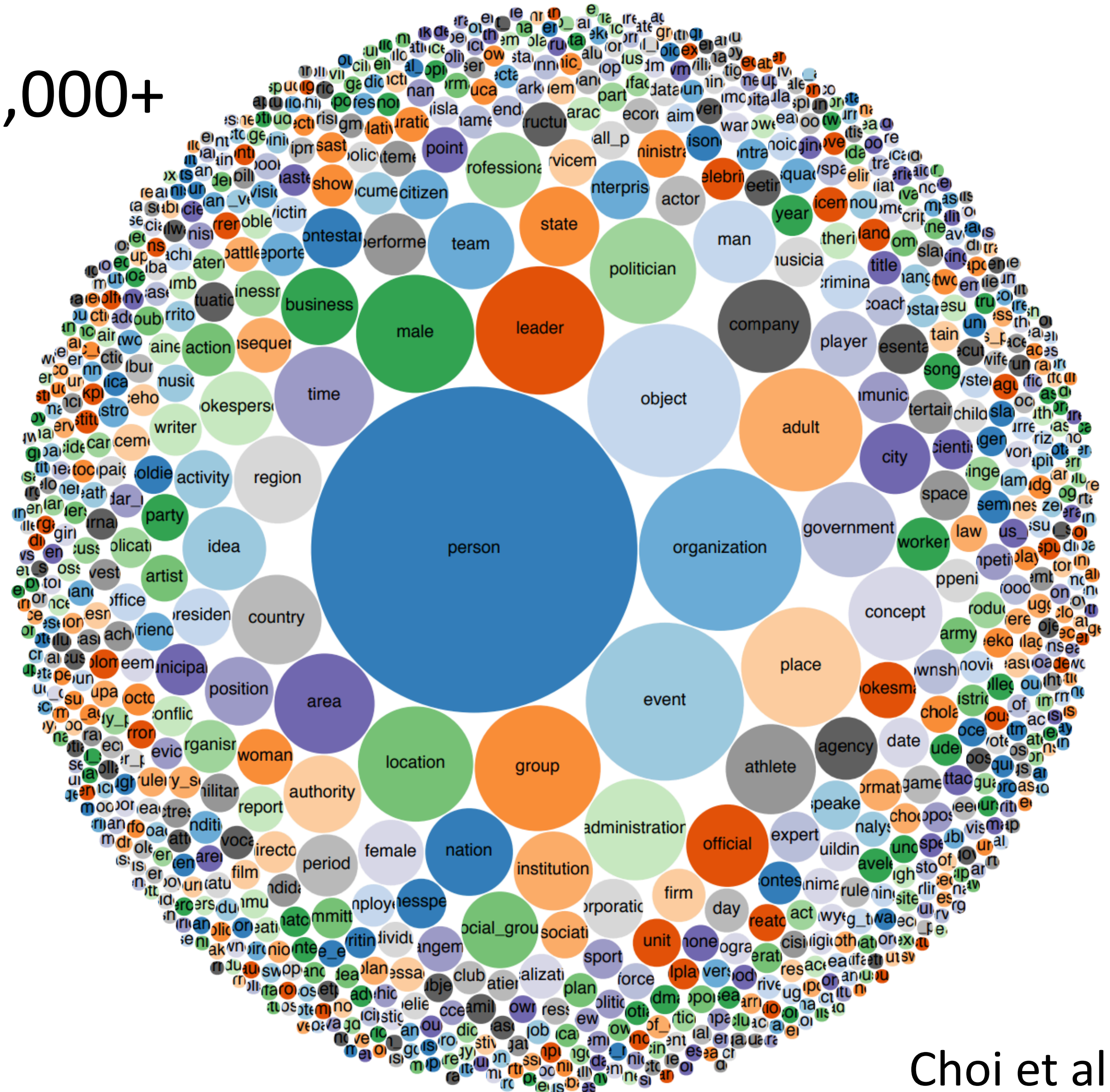
Ultra-fine Entity Typing



Choi et al. (2018)

Ultra-fine Entity Typing

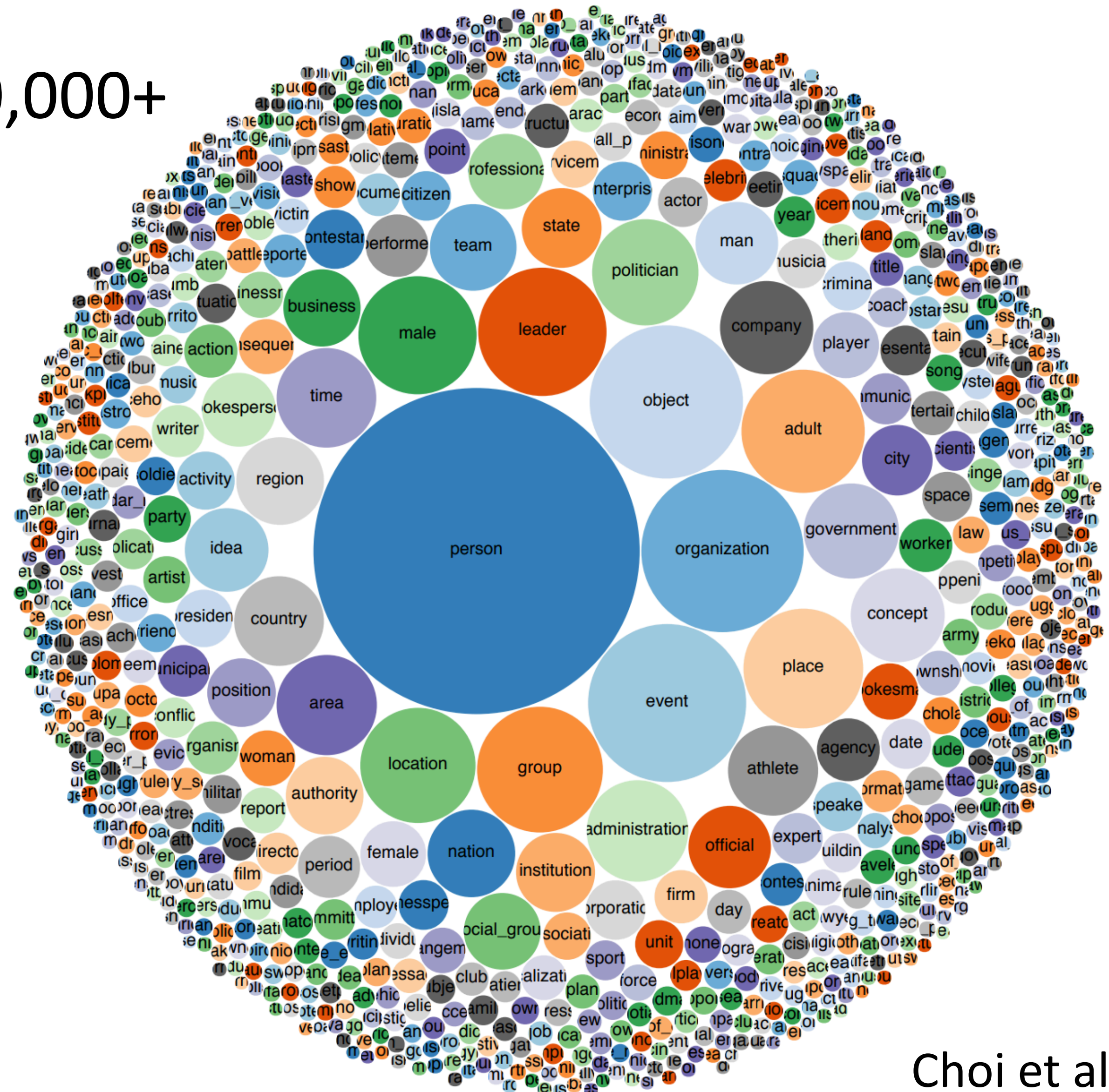
- ▶ Scaling up to larger typesets 10,000+



Choi et al. (2018)

Ultra-fine Entity Typing

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- ▶ Person/official/president ...



Choi et al. (2018)

Sentence Alignment

- Neural CRF for Sentence Alignment

NEWSLA

WAR & PEACE SCIENCE KIDS MONEY HEALTH

SCIENCE 1738 SHARE

Archaeologist may have found remains of ancient Egyptian Queen Nefertiti

By Robert Gebelhoff, Washington Post. 08.17.15



The 3,330-year-old bust of Nefertiti sits in an exhibition in the Kulturforum in Berlin, Germany, March 1, 2005. Photo: AP/Herbert Knosowski

Nefertiti — she's an ancient Egyptian queen and the source of a fantastic mystery regarding the iconic remnants of long-lost royalty.

For decades, archaeologists have speculated on the location of the queen's remains, the last royal mummy missing from the dynasty of the famous King Tutankhamun, better known as King Tut. But now, an archaeologist claims that he has found her

MAX
1140L
960L
720L
420L
WRITE
QUIZ

NEWSLA

WAR & PEACE SCIENCE KIDS MONEY LAW HEALTH

SCIENCE 1738 SHARE

Mystery of ancient Egypt solved? Tomb of queen may be hidden near King Tut

By Washington Post, adapted by Newsela staff. 08.17.15



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The ancient Egyptian Queen Nefertiti has long been at the center of a mystery. For years, archaeologists have wondered where her tomb might be hidden. Nefertiti belonged to the family line of the famous King Tutankhamun, better known as King Tut. Indeed, some believe she was Tut's mother. While the other royals in her line are

1140L
960L
720L
420L
WRITE
QUIZ

Professional editors rewrite news articles into 4 different readability levels for grade 3-12 students.

Sentence Alignment

► Neural CRF for Sentence Alignment

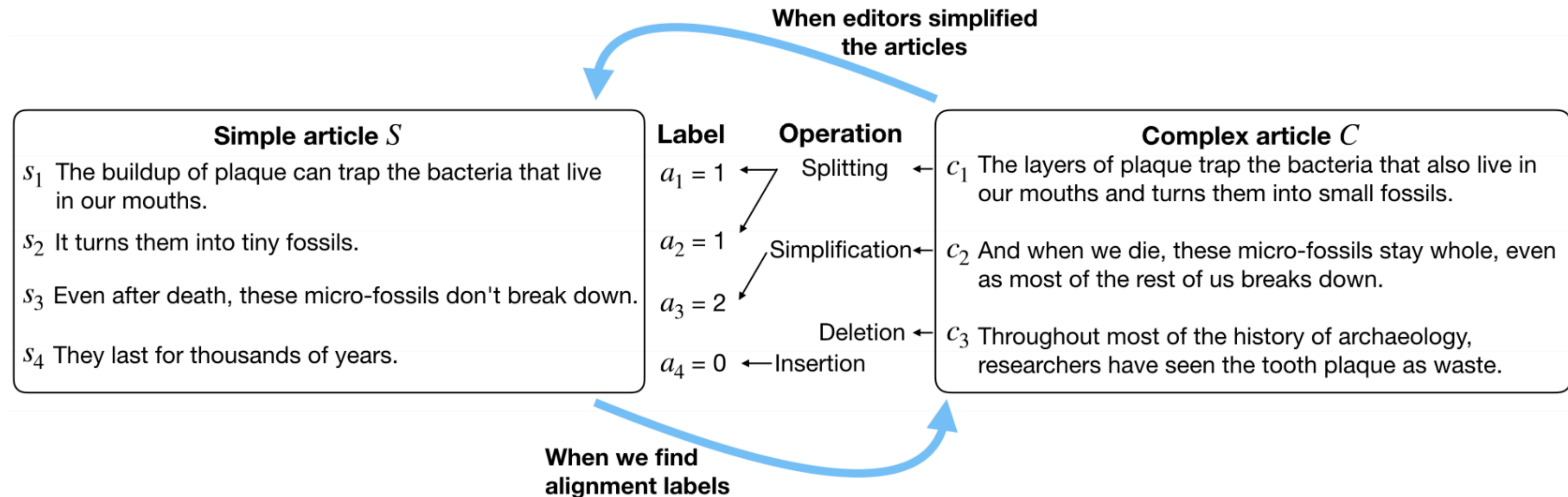
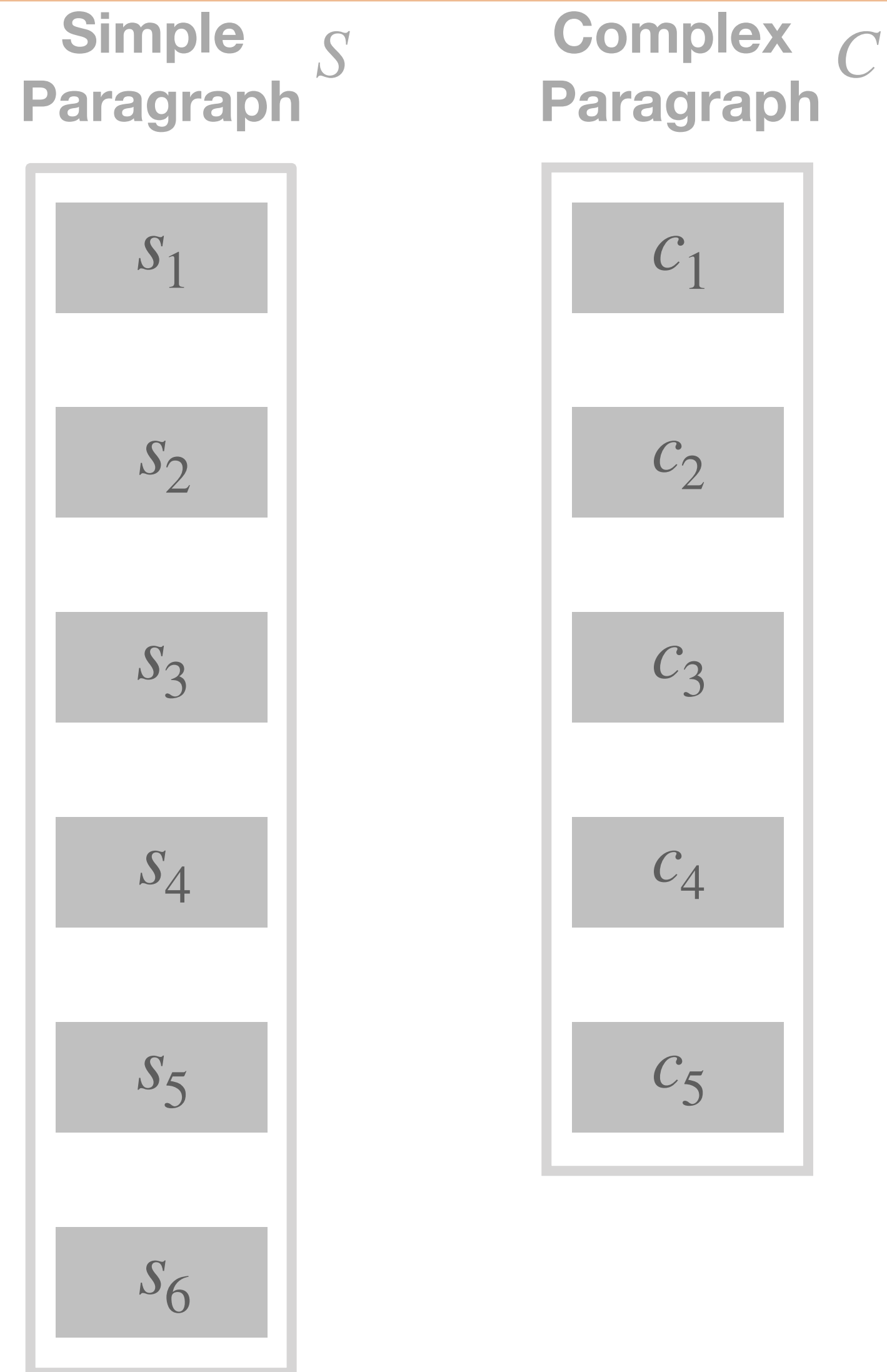


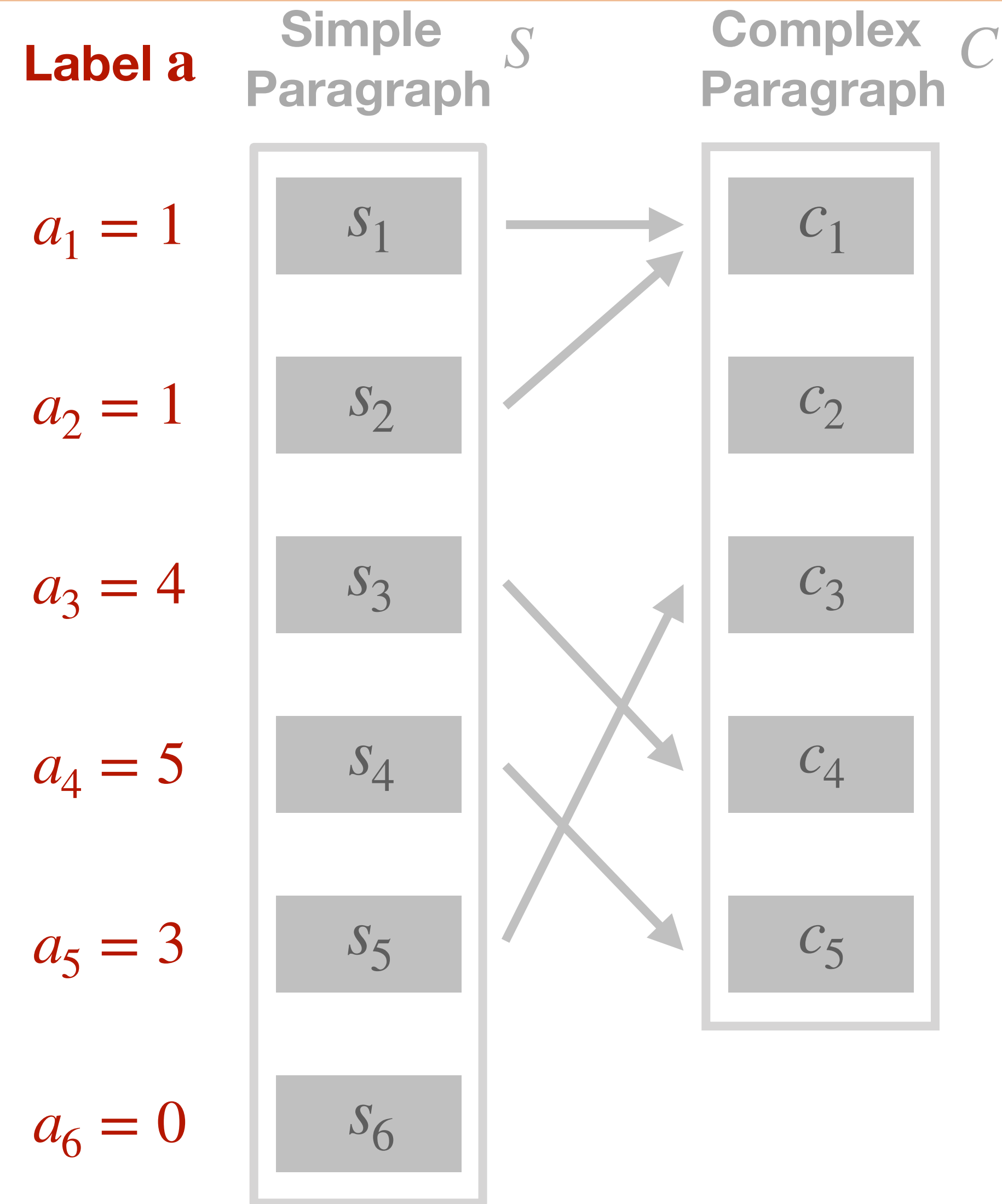
Figure 1: An example of sentence alignment between an original news article (right) and its simplified version (left) in Newsela. The label a_i for each simple sentence s_i is the index of complex sentence c_{a_i} it aligns to.

Sentence Alignment

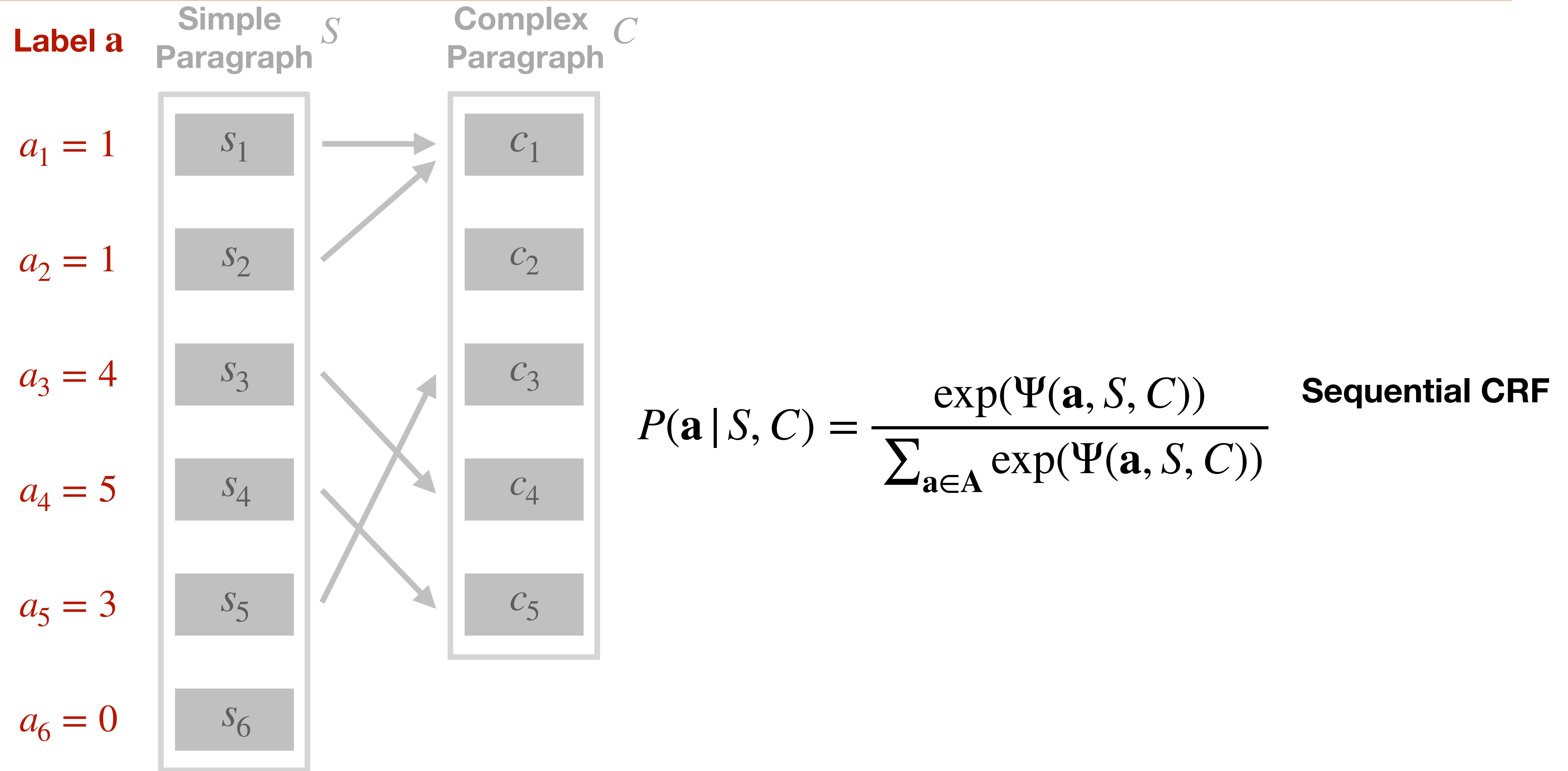
Sentence Alignment



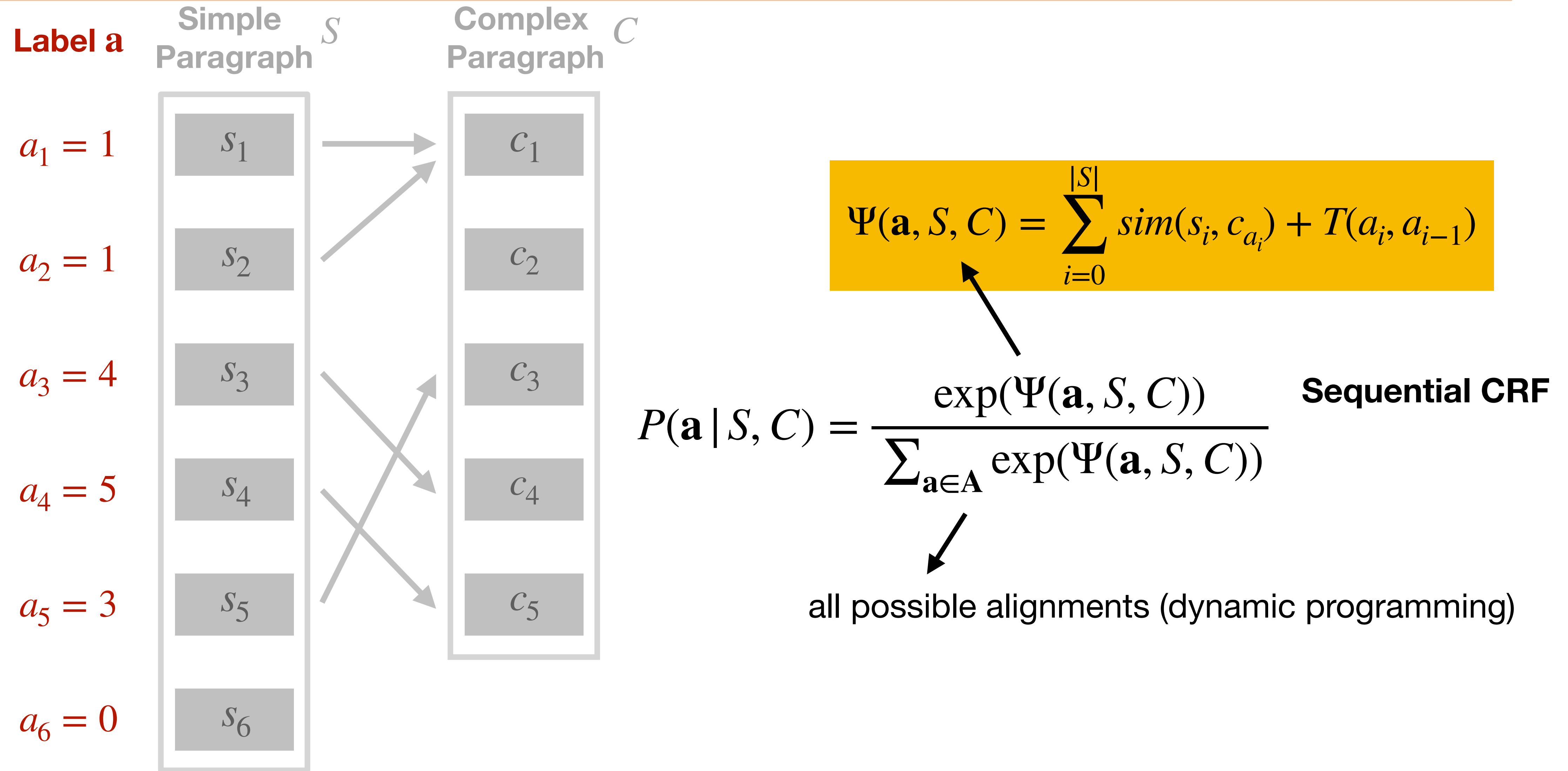
Sentence Alignment



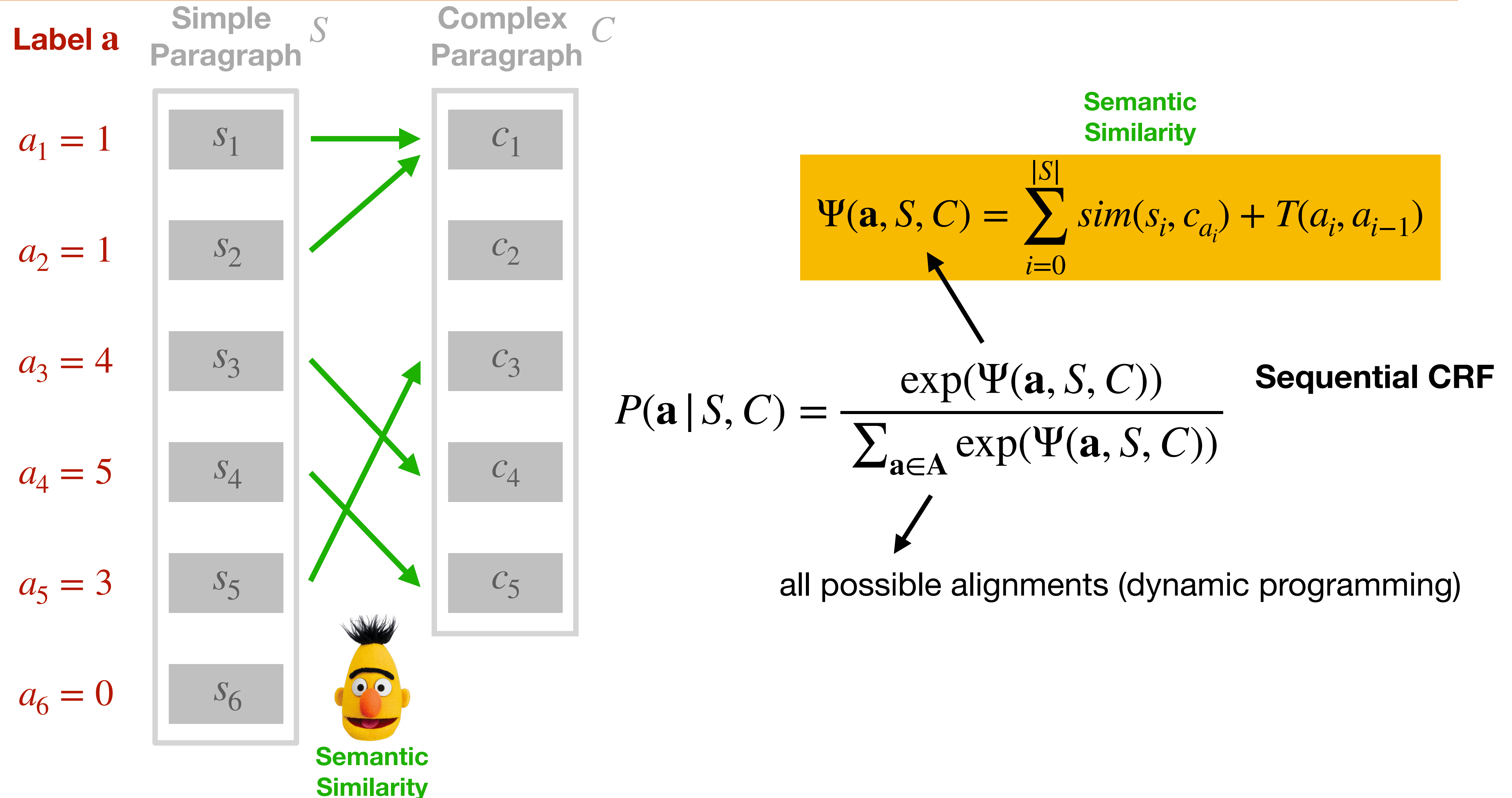
Sentence Alignment



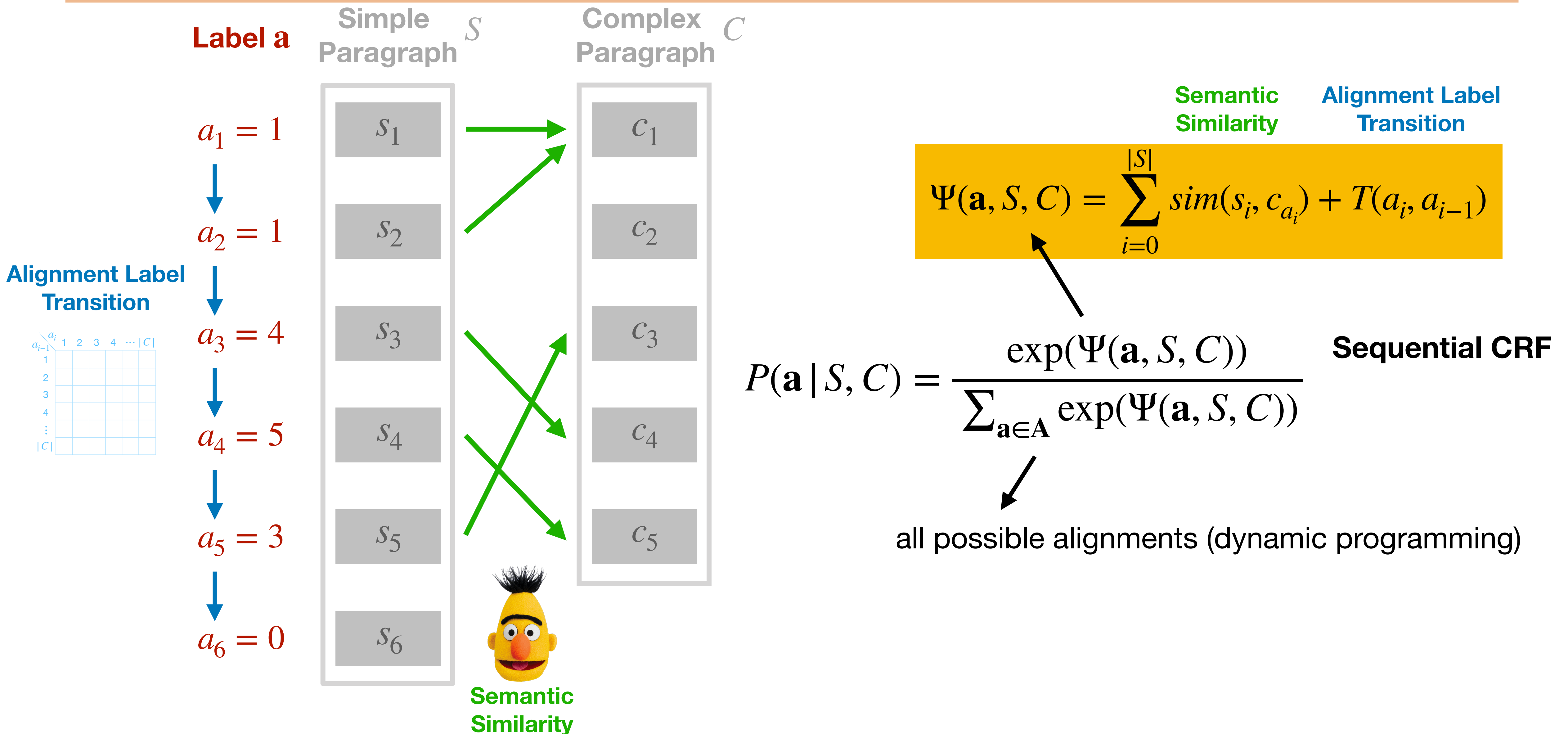
Sentence Alignment



Sentence Alignment



Sentence Alignment



Sentence Alignment

- Structure prediction + BERT_{finetune} → A neural CRF alignment model.

		aligned + partial vs. others*		
		Precision	Recall	F1
Greedy	JaccardAlign (Xu et al., 2015)	98.66	67.58	80.22
Dynamic Programming	MASSAlign (Paetzold et al., 2017)	95.49	82.27	88.39
Greedy	CATS (Štajner et al., 2018)	88.56	91.31	89.92
Threshold	BERT _{finetune}	94.99	89.62	92.22
Threshold	BERT _{finetune} + paragraph alignment	98.05	88.63	93.10
CRF	Our CRF aligner	97.86	91.31	95.59

+5.7

* Results are on the manually annotated Newsela dataset.

Next Up

Next Up

- ▶ Recurrent Neural Network

Next Up

- ▶ More sequential models
 - ▶ Recurrent Neural Network

Next Up

- ▶ More sequential models
 - ▶ Recurrent Neural Network
 - ▶ Neural CRF model